Middleton Beach Artificial Surf Reef Environmental Impact Assessment

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Middleton Beach Artificial Surf Reef Environmental Impact Assessment

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City of Albany

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BMT Western Australia Pty Ltd

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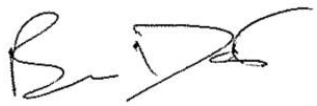
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Acronyms

Acronym	Definition
ANZECC/ARMCANZ	Australian and New Zealand Environment and Conservation Council and Agriculture and Resource Management Council of Australia and New Zealand
APA	Albany Port Authority
ARI	Assessment on Referral Information
ASLSC	Albany Surf Life Saving Club
ASR	Artificial Surf Reef
BCH	Benthic Communities and Habitat
BPPH	Benthic Primary Producer Habitat
ºC	Degrees Celsius
CoA	City of Albany
DBCA	Department of Biodiversity, Conservation and Attractions
DoL	Department of Lands
DoT	Department of Transport
DPIRD	Department of Primary Industries and Regional Development
DPLH	Department of Planning, Lands and Heritage
DWER	Department of Water and Environmental Regulation
EAG	Environment Assessment Guideline
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
EP Act	Environmental Protection Act
EPA	Environmental Protection Authority
EPBC	Environment Protection and Biodiversity Conservation
GPS	Global Positioning System
ha	Hectare
HSE	Health Safety Environment
ILUA	Indigenous Land Use Agreement
IMS	Introduced Marine Species
ISQG	Interim Sediment Quality Guidelines
LAA	Land Administration Act
LAU	Local Assessment Unit
m^3	Cubic metre
mm	millimetre
MNES	Matters of National Environmental Significance
NE	North East
NPER	Non-Public Environmental Review
NSHA	Noongar Standard Heritage Agreement
NSW	New South Wales

NTM	Notice to Mariners
NW	North West
PER	Public Environmental Review
RHDHV	Royal Haskoning DHV
SCUBA	Self Contained Underwater Breathing Apparatus
SD Act	Sea Dumping Act
SE	South East
SG	Steering Group
SW	South West
SWALSC	South West Aboriginal Land and Sea Council
TBT	Tributyltin
WA	Western Australia
WC	Waterways Conservation

1. Introduction

1.1 The proposal

The City of Albany (CoA) proposes to construct an artificial surfing reef (ASR) at Middleton Beach, southern Western Australia (WA) (Figure 1.1) (the 'Project'). The proposed ASR will be built at the southern end of the beach with the aim of enhancing the recreational surfing conditions for local residents and tourists. The ASR will be designed to target beginner-intermediate surfers.

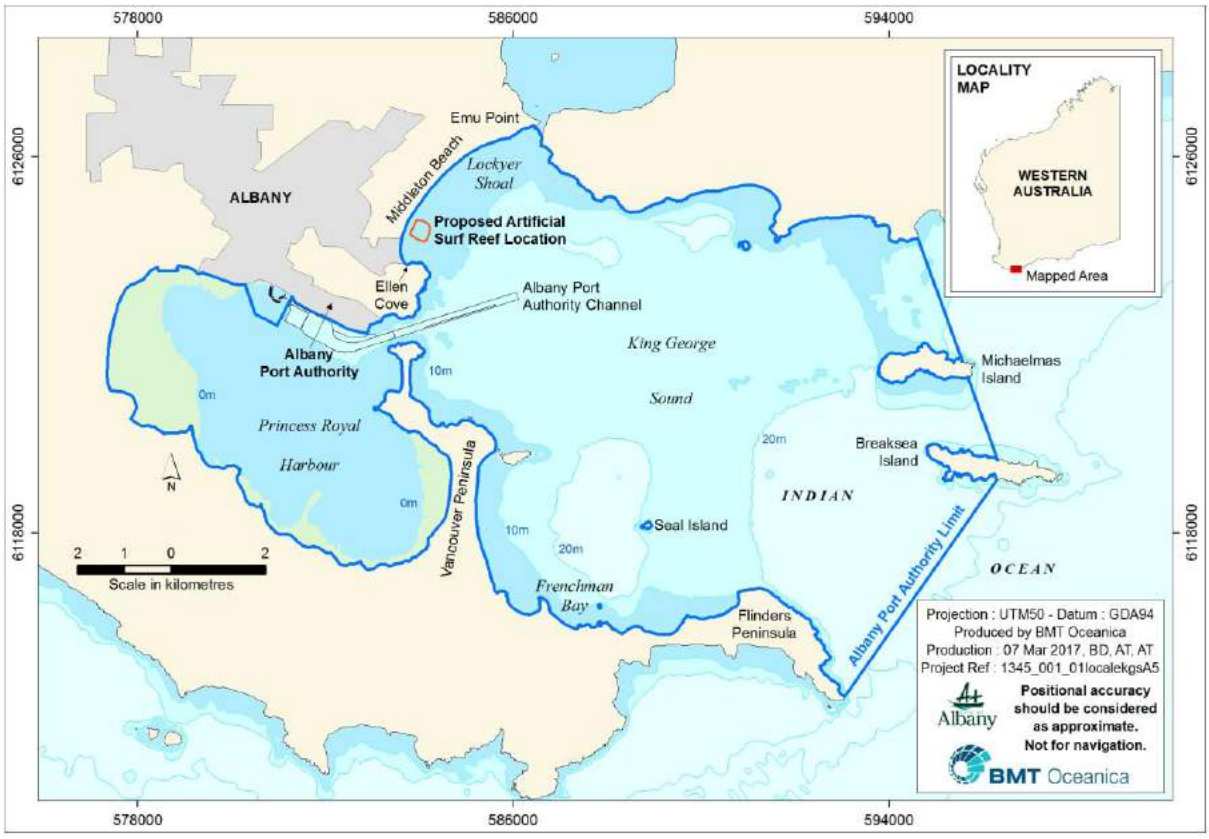


Figure 1.1 Project location, Middleton Beach, Albany, Western Australia

1.2 Project justification and benefits

Currently, the closest suitable surfing locations from Albany are around 40 minutes' drive, and are generally disregarded by beginner and junior surfers (CoA 2016). With a lack of public transport available to reach appropriate locations (e.g. Mutton Bird Beach or Nanarup), the opportunities to surf on a regular basis are limited, particularly for young people (CoA 2016). Aside from the safety aspects associated with the current need to drive distances to find surfable waves, the current locations are isolated and unmonitored. Enabling these activities to be undertaken at Middleton Beach will improve safety through increased monitoring and proximity to the Albany Surf Life Saving Club (ASLSC) and medical and emergency facilities in the city (CoA 2016).

The CoA is hoping to attract and retain a younger generation, who currently tend to be drawn away to metropolitan areas where a wider variety of recreational facilities exist. It is hoped that the ASR will provide a significant attractor for retaining this demographic, as well as expanding the recreational amenity for older residents who currently need to travel to surf, either to isolated beaches some way from the city or to other locations such as Margaret River (CoA 2016).

The intention is to design the ASR to create a consistent, quality wave appropriate for holding events at state, national and international levels (CoA 2016). Surfing WA have stated that they would foresee holding 3-4 events per year in Albany that are not currently possible due to the poor quality of surf on Albany's central beaches (CoA 2016).

The Project therefore responds to the need to diversify and grow the regional economy (CoA 2016). The potential tourism benefits from improved surf are clear, but a more general uplift in visitation and length of stay would also be expected (CoA 2016). The Project will complement other initiatives in the region to further develop adventure tourism assets, such as for the 'Snake Run' skate park, mountain biking and bush walking (CoA 2016). It is possible, through the construction of the ASR, for Albany to develop a reputation as a surfing "hub". With existing infrastructure in retail and hospitality, the facilitation of a recognised hub in Albany would provide substantial benefit both economically and socially (CoA 2016).

1.3 Proponent details

The Project proponent is the City of Albany. The name and legal address of the proponent and key Project contacts are given in Table 1.1.

Table 1.1 Name and contact details of the Project proponent and other key contacts

Role	Name and contact details
Proponent	City of Albany 102 North Road, Yakamia Albany WA 6331
Principal	City of Albany 102 North Road, Yakamia Albany WA 6331
Environmental Consultants	BMT WA Pty Ltd 4/20 Parkland Road Osborne Park WA 6017 Email: ben.davis@bmtglobal.com

1.4 This document

This document presents an environmental impact assessment (EIA) of the Project, providing detailed information on the Project proposal, its potential environmental impacts and the proposed management of these impacts. This document has been prepared to support future regulatory assessment and approvals of the Project.

2. Project Description

2.1 Proposed activity

CoA proposes to build an ASR ~300 m offshore of the southern Middleton Beach, Albany (Figure 1.1, Figure 2.1). The contract for the detailed design phase of the ASR will be determined through a competitive tender process.

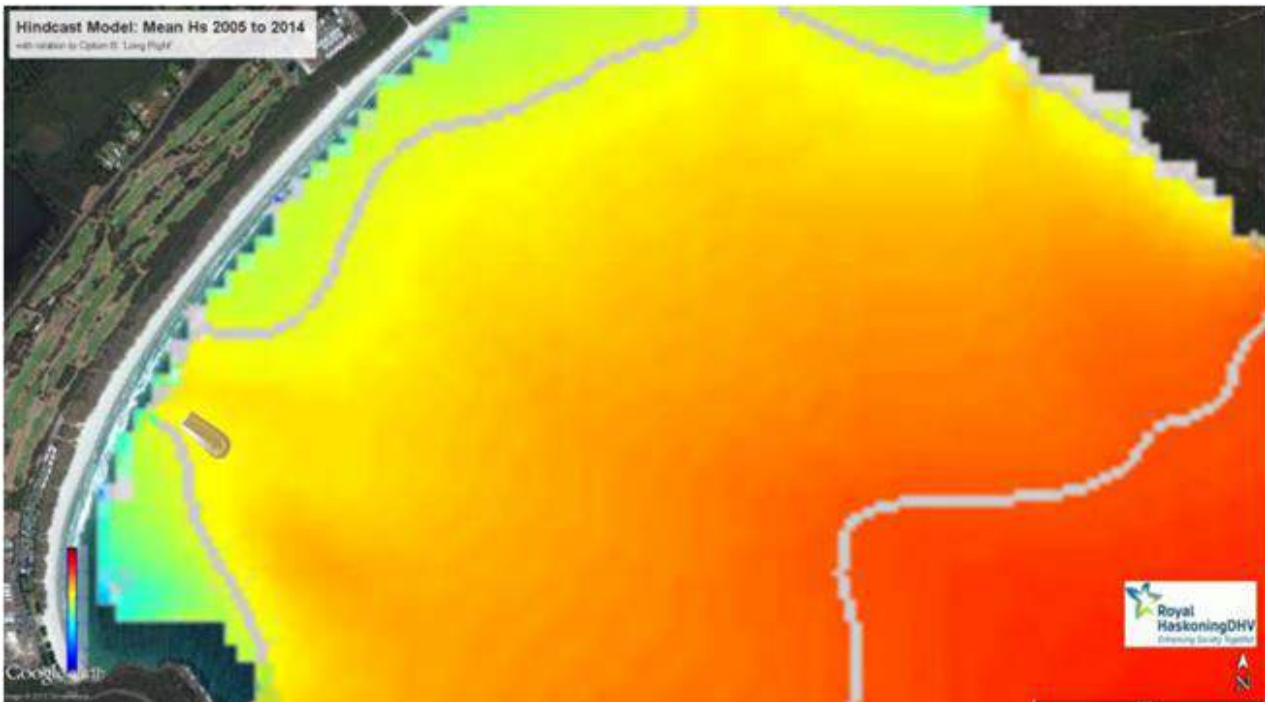


Source: RHDHV (2015a)

Figure 2.1 Artists impression of the proposed Artificial Surf Reef

Studies by an expert coastal engineer concluded that Middleton Beach is a suitable location for an ASR, owing to a unidirectional wave climate, long swell periods and average wave height of around 0.65 m (RHDHV 2015a).

Concurrent baseline monitoring without the ASR over a 13 month period showed that only 6 surfing days were rated better than 'Average' (RHDHV 2015a). With the ASR, it is predicted that better-than-average surfing days will be increased at least 30 times (to 180+ days), with waves breaking 50% of the time at a -0.75 m crest level (RHDHV 2015a). The current proposed ASR location is at the southern end of the southern wave-focusing zone of Middleton Beach (Figure 2.2). This location was chosen because of its proximity to the surfer's car park, other amenities, and distance from nearby seagrass beds (RHDHV 2015b). Another wave focussing zone, at Emu Point, north of the proposed ASR location, was not considered feasible as it was not close to amenities and was nearby to seagrass beds (Figure 2.2, RHDHV 2015b)



Source: RHDHV (2015b)

Figure 2.2 Wave focussing zones of Middleton Beach, Albany, with the proposed Albany Artificial Surfing Reef design overlayed

2.2 Construction methods

It is proposed to construct the ASR using barge-mounted equipment (RHDHV 2015b). The use of barge-mounted machinery may expose the Project to weather-related delays; however, it means that the construction will not impact on the public use of Middleton Beach (i.e. unlike shore-based construction methods). Dumb barges, towed using local tugboats, will be used to transport rock material from nearby Albany Port (RHDHV 2015b). Construction material is likely to be locally sourced rock, ~800-1200 mm in diameter. Approximately 50,000 m^3 of rock material will be transported to Albany Port by road haulage to construct the proposed ASR.

The intended construction method is as follows (RHDHV 2015b):

1. Place the toe rocks around the perimeter of the footprint of the structure to form the structure boundary. This would be done by placing the rocks individually from a barge equipped with a pin jib crane and a rock grab equipped with global positioning system (GPS) technology and specialised software.
2. Fill the area within the perimeter toe rocks with bulk place core material with a pin jib crane and a bucket grab and/or excavator bucket (Figure 2.3), monitoring the levels using side scan survey and GPS technology to get approximate finished levels.
3. Level/smooth the top of core material. This could be done using a barge mounted excavator equipped with GPS technology working at the same time as the above operation.
4. Placement of the armour materials using the barge mounted pin jib crane and rock grab equipped with GPS technology.
5. Final checking and repositioning of the final layer to achieve construction tolerances. Again, this could be done using a barge-mounted excavator equipped with GPS technology working at the same time as the above operation.

Multi-beam echo sounder surveys and GPS logs will be used throughout the construction period to ensure that the required line, levels, slopes and construction tolerances have been met as this will be pivotal to the success of the project.



Source: RHDHV (2015b)

Figure 2.3 Placement of rock material using a pin jib crane and bucket grab

2.2.1 Timing

Construction is anticipated to take a total of 12 months, and will commence following completion of the detailed design phase. The preferable period for the majority of material placement and construction activities to occur is in May-October (Table 2.1) to take advantage of the prevalent offshore winds, which will provide for better working conditions (RHDHV 2015b). Wind conditions outside of this period are generally onshore between December-March, which would make wave conditions at site unsafe for the placement of rock material. The construction period over winter will also reduce the impact on public users of the beach (see Section 6.8) and impact to seagrasses (see Section 4.1).

Table 2.1 Proposed construction timeline for the artificial surf reef

Task	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Pre-mobilisation activities, including delivery of quarried material to site, pre-construction surveys												
Mobilisation and launching of barges and construction equipment												
Construction of ASR												
Finalise ASR, as-built surveys and final construction activities												
Demobilisation of all plant and materials												

Source: RHDHV (2015b)

2.3 Alternatives considered

2.3.1 Not building the artificial surf reef

The CoA wishes to build an ASR for the reasons outlined in Section 1.2. While any associated marine and terrestrial environmental impacts are obviously negated by not constructing the ASR, there will be no positive impact on the economy or social environment.

2.3.2 Use of alternative building materials

The ASR Project engineers (RHDHV) have prepared a summary table to assess the suitability of differing materials for the construction of the ASR (Table 2.2). Geotextile containers have been used to construct the majority of ASRs globally. However, in recent years there has been a trend away from the use of geotextile containers and toward more traditional construction materials, such as rock boulders (RHDHV 2015a). A review of artificial reefs in NSW (WRL 2013) recommended against the use of alternative construction methods, over more traditional construction methods, until the technology was better developed. For the ASR Project it is recommended that granite rock is used, due to its durability, limited environmental impacts and local availability (RHDHV 2017, Table 2.2).

Table 2.2 Alternative materials considered for the construction of the Albany Artificial Surf Reef

Material	Issues
Granite rock	• Potential for turbidity to be generated during construction
Limestone rock	• Some turbidity issues during initial construction phase • Continued, slow breakdown of material may cause turbidity and individual stone degradation, impacting structural stability • Larger stones will be required due to lower specific weight
Greywacke rock	• Some turbidity issues during initial construction phase • Brittle, means some losses during handling and transport to site
Geotextile container	• High turbidity issues during initial construction phase due to dredging Shorter design life means bag degradation and eventual failure, leading to increased turbidity and environmental concerns due to difficulty of removal. Higher currents over smooth, impermeable structure means increased currents in the lee, posing not only safety concerns but also increased turbidity. • Geotextile material movement and settling posing safety risks • Increased scour potential
Steel/concrete structure	• Some turbidity issues during initial construction phase • Shorter design life due to rust and brittleness • Higher currents over smooth, impermeable structure means increased currents in the lee, posing not only safety concerns but also increased turbidity. • Incremental structure degradation leads to safety concerns • Increased scour potential

Source: RHDHV (2017)

2.3.3 Alternative locations

A number of alternative locations were considered in the feasibility study (RHDHV 2015b), however Middleton Beach was determined to be the prime location for the Albany ASR, due to its wave climate, existing infrastructure and amenities, and proximity to the Port of Albany for construction and vessel harbouring.

Within Middleton Beach, a number of alternative locations were also considered, including moving the ASR a further 200 m south to avoid potential impacts with seagrass, and moving to the northern wave focussing zone at Middleton Beach (Figure 2.2). Neither of these locations were considered suitable by RHDHV for the following reasons:

- The 200 m further south option was not considered feasible since it would require moving the ASR out of the southern wave-focussing zone (Figure 2.2), potentially significantly impacting its performance. The existing proposed location is within an area of low benthic primary producer habitat (BPPH) cover, therefore moving the ASR south was not considered beneficial from the perspective of mitigating BPPH losses.
- The Emu Point location is ~3 km from Ellen Cove. Ellen Cove is the town's main public beach and is where the ASLSC is currently located. One of the main aims for the Albany ASR is driving benefits in tourist, economic development and retention of Albany's younger age demographic (RHDHV 2015b) and the Emu Point location is not suitable in this regard.

3. Regulatory Approvals

3.1 Decision-making authorities

The following key decision-making authorities have been identified for the Project:

- Western Australian Environmental Protection Authority (EPA)
- Australian Government Department of the Environment and Energy (DoEE)
- Western Australian Department of Water and Environment Regulation (DWER)
- Western Australian Department of Biodiversity, Conservation and Attractions (DBCA)
- Western Australian Department of Primary Industries and Regional Development (DPIRD)
- Western Australian Department of Planning, Lands and Heritage (DPLH)
- Western Australian Department of Transport (DoT)
- Southern Ports Port of Albany

3.2 Relevant legislation and guidance material

3.2.1 Environmental Protection Act 1986

The *Environmental Protection (EP) Act 1986* (EP Act) is the principal legislation governing environmental protection and approvals in Western Australia and is applied to State land and waters. Part IV of the EP Act relates to environmental impact assessment of a project, including its referral to, and assessment by, the EPA. Part V of the EP Act relates to the control and licensing of potentially polluting activities and is administered by the DWER.

This document has been prepared to satisfy the requirements of an Environmental Referral to the EPA under the provisions of Part IV of the EP Act and in accordance with the *Administrative Procedures 2016* (EPA 2016a).

The document *Statement of Environmental Principles, Factors and Objectives* denotes key EPA Environmental Factors of: Benthic Communities and Habitat; Coastal Processes; Marine Environmental Quality; Marine Fauna; Flora and Vegetation; Landforms; Subterranean Fauna; Terrestrial Environmental Quality; Terrestrial Fauna; Hydrological Processes; Inland Waters Environmental Quality; Air Quality; Social Surroundings; and Human Health.

Environmental Factor Guidelines (EFGs) pertain to the protection of Environmental Factors; while Environmental Factor Technical Guidance documents provide additional technical detail to the EFGs. For example, guidance relevant to the proposed ASR includes:

- Technical Guidance - Protection of Benthic Communities and Habitats (EPA 2016d)
- Technical Guideline - Coastal Processes (EPA 2016e)
- Technical Guidance - Protecting the Quality of Western Australia's Marine Environment (EPA 2016g)

3.2.2 Environmental Protection and Biodiversity Conservation Act 1999

The *Environmental Protection and Biodiversity Conservation Act 1999* (EPBC Act) is the Australian Government's central piece of environmental legislation, which is administered by the DoEE. The EPBC Act provides a legal framework for the protection and management of nationally and internationally important flora, fauna, ecological communities and heritage places, which are defined in the EPBC Act as matters of national environmental significance (MNES). The Act applies to seven matters of national environmental significance, which are:

- world heritage sites
- national heritage places
- wetlands of international importance
- nationally threatened species and ecological communities
- migratory species
- commonwealth marine areas
- nuclear actions.

A protected matters search tool report was generated for the proposed ASR location (Appendix A). The search tool revealed 10 marine threatened or migratory species within the general area of the proposed ASR. However, it is not anticipated that the proposed ASR will negatively impact on these MNES, and therefore at this stage of Project scoping, referral to the DoEE under the EPBC Act is not considered to be required.

3.2.3 Environmental Protection (Sea Dumping) Act 1981

Through the *Environment Protection (Sea Dumping) Act 1981* (SD Act), the DoEE assesses proposals to load and dump wastes and other materials at sea, permits acceptable activities, and sets conditions of approval to mitigate and manage environmental impacts. The proposed ASR does not require a sea dumping permit as it occurs within waters inside the limits of the State, and not in waters under the jurisdiction of the SD Act (K McLachlan, Assistant Director DoEE Sea Dumping Section. Pers Comm March 2017).

3.2.4 Waterways Conservation Act 1976

DWER administers the *Waterways Conservation Act 1976* (WC Act) and *Waterways Conservation Regulations 1981*(WC Regulations). Middleton Bay is managed by the DoW under the WC Act and WC Regulations. Regulation 9 states that:

a person shall not construct or permit the construction of, any boat ramp, slip, bridge, jetty, boat house, pier, decking, or any other structure, whether floating or otherwise, in, over or contiguous with any waters; or

Therefore the ASR will require a license from the DWER to permit construction.

3.2.5 Biosecurity Act 2015

The *Biosecurity Act 2015* came into effect on 16 June 2016 to provide a regulatory framework for management of biosecurity risks including pests, disease and contaminants (replacing the *Quarantine Act 1908*). This is managed under the Australian Government Department of Agriculture and Water Resources. Decisions made under the Act will depend on the likelihood and consequences of the risk presented resulting in the management of risks more appropriately. The Act includes regulations for ballast water; biofouling and biosecurity risks associated with marine pests, and should be considered for management for these risks (see Section 6.6).

3.2.6 Aboriginal Heritage Act 1972

A search of the Aboriginal Heritage Enquiry System (Appendix B) showed no registered Aboriginal sites within the Project footprint. Therefore, a Notice under Section 18 of the *Aboriginal Heritage Act 1972* will not be required.

The proposed ASR is on land within or adjacent to the Wagyl Kaip Southern Noongar People Indigenous Land Use Agreement (ILUA). On 8 June 2015, six identical Indigenous ILUAs were executed across the South West by the Western Australian Government and, respectively, the Yued, Whadjuk People, Gnaala Karla Booja, Ballardong People, South West Boojarah #2 and

Wagyl Kaip & Southern Noongar groups, and the South West Aboriginal Land and Sea Council (SWALSC).

The ILUAs bind the parties (including 'the State', which encompasses all State Government Departments and certain State Government agencies) to enter into a Noongar Standard Heritage Agreement (NSHA) when conducting Aboriginal Heritage Surveys in the ILUA areas, unless they have an existing heritage agreement.

3.2.7 Other state legislation

The *Navigable Waters Regulations 1958* (referring to the *Shipping and Pilotage Act 1967*, *Jetties Act 1926* and *Western Australian Marine Act 1982*) manage maritime activities in Western Australian navigable waters, including the territorial sea adjacent to the State. As the proposed Project lies within the Port of Albany waters, permission is required from the Southern Ports to install the ASR.

4. Existing Environment

The proposed ASR location sites within King George Sound, a large marine embayment. King George Sound is exposed to the Southern Ocean swells, with the southern and eastern shorelines protected somewhat by islands and headlands (Evangelisti et al 1998). The Sound has two harbours adjoining it, Oyster Harbour in the north and Princess Royal Harbour in the south. The latter is home to the Port of Albany, and both contain extensive, Ecologically important seagrass communities (Wells et al 1990, RHDHV 2015a).

The marine environment of the Albany region is well documented (Bastyan 1986, EPA 1990, Wells et al 1990, Ecologia 2007, Bastyan and Associates 2015). The below sections present a synthesis of available information on the marine environment of Middleton Bay.

4.1 Benthic communities and habitat

Bastyan and Associates (2015) recently mapped the marine BPPH of Middleton Bay (see Section 6.4). Previous mapping has been completed by Ecologia (2007) in support of the Albany Port facilities and channel upgrades and Evangelisti et al (1998). Wells et al (1990) provides an extensive description of the marine flora of the Albany region.

Middleton Bay is dominated by seagrasses and large, bare sandy areas. The dominant seagrass species is *Posidonia coriacea* (Bastyan and Associates 2015) with *P. sinuosa*, *P. australis* and *Amphibolis antarctica* also present within the nearby Princess Royal Harbour (Ecologia 2007). Historically *P. sinuosa* has also been mapped nearby to Emu Point, at the northern end of Middleton Bay (Evangelisti et al 1998). Small areas of pavement reef with algae are found in the shallow extremities of Middleton Bay, along Middleton Beach (Evangelisti et al 1998)

Seagrass meadows in exposed environments like much of King George Sound are highly dynamic, with patches moving over 10 to 50 year time periods (Evangelisti et al 1998). Historically *P. coriacea* beds have been well distributed throughout Middleton Bay; however a large storm in 1984 removed almost all seagrass from the area (Bastyan and Associates 2015). The same storm eroded Lockyer Shoal, Emu Point (Figure 1.1). The erosion of this shoal has resulted in the gradual decrease of seagrass meadows, driven largely though storm events scouring the seabed and dislodging seagrass (URS 2012).

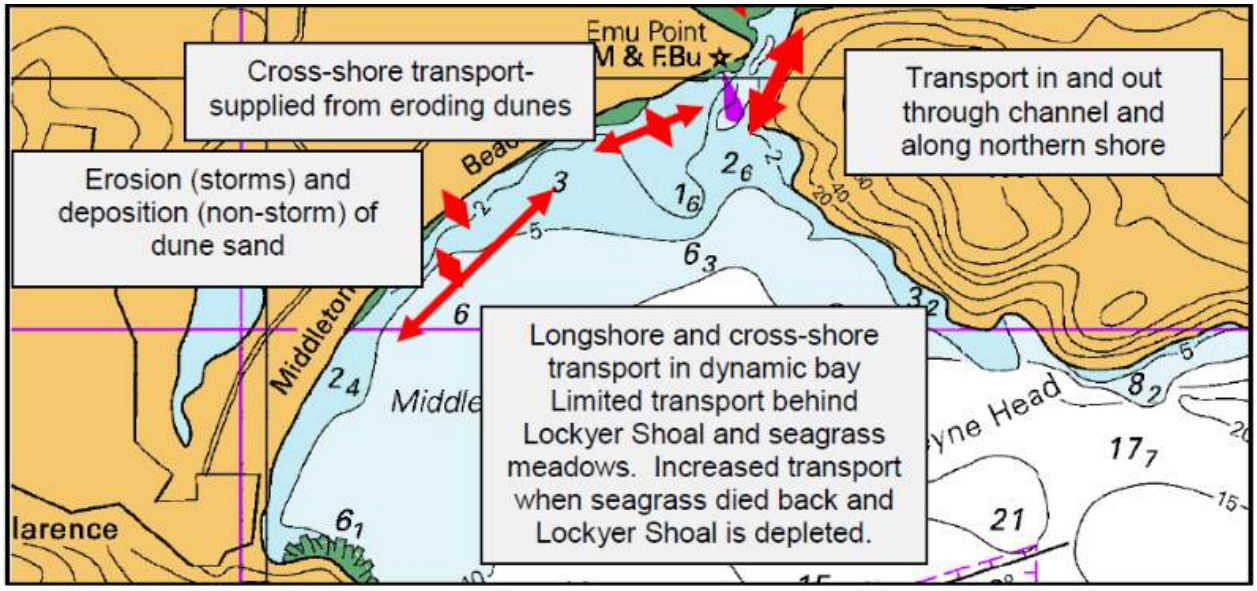
Further offshore, within King George Sound, there are scattered limestone reefs containing numerous species of algae, sponges and black corals (Ecologia 2007). The exposed location limits growth on the seafloor to a minimum on the eastern edges of the Sound (Ecologia 2007). Along the southern end of the Sound there are several patches of *Halophila ovalis*, an ephemeral species of seagrass (Evangelisti et al 1998)

4.2 Coastal processes

Middleton Beach, particularly the northern end near Emu Point, has experienced severe historical erosion (URS 2012). Episodic storm events (high wave energy and high water levels) result in the erosion of the beach and dune system along the majority of Middleton Beach. The protection that was once afforded by the seagrass growing on Lockyer Shoal has been reduced through the erosion of the shoal and the degradation of the seagrass meadows (URS 2012, Bastyan and Associates 2015). There is evidence that the erosion of Lockyer Shoal in the 1984 storm has accelerated the degradation of the meadows.

Analysis of historical vegetation lines shows that the southern end of Middleton Beach is accreting, with a small area of erosion just south of the proposed ASR location (DoT 2012, in

URS 2012, RHDHV 2015a). However, due to the dynamic mix of currents and waves from various sources, Middleton Beach is considered to be in a dynamic equilibrium (Figure 4.1, URS 2012). Other studies have also shown that there is no net loss or gain of fine-medium sand within Middleton Beach (RHDHV 2015a).



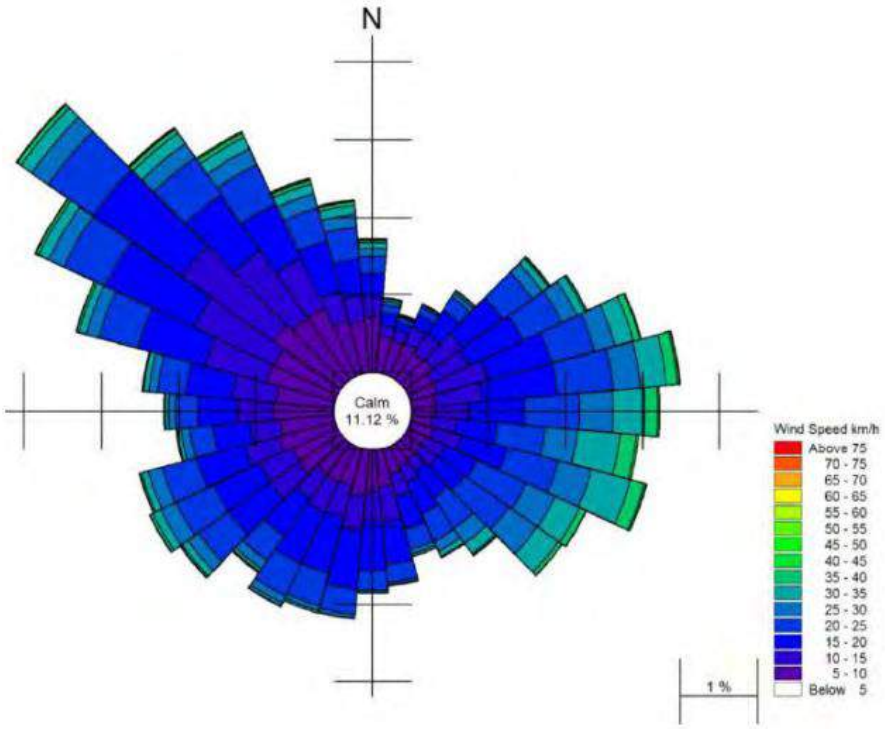
Source: URS (2012)

Figure 4.1 Conceptual model of sediment transport at Middleton Beach

4.3 Wind and wave climate

4.3.1 Meteorology

Two predominant synoptic periods have been identified in King George Sound (RHDHV 2015a). Summer months in Albany are dominated by easterly winds (NE-SE), driven by a ridge of high pressure extending along the SW coastline. The pressure gradient alters more north-easterly, propagating heat troughs and then shifts southwards after the passage of the trough. A rapidly forming high usually then causes strong south-easterly winds (Ecologia 2007). During winter this sub tropical ridge migrates north, causing the Albany region to be affected primarily by strong westerly winds. Strong cold fronts often drive stormy weather during the winter months (Ecologia 2007).



Source: URS (2012)

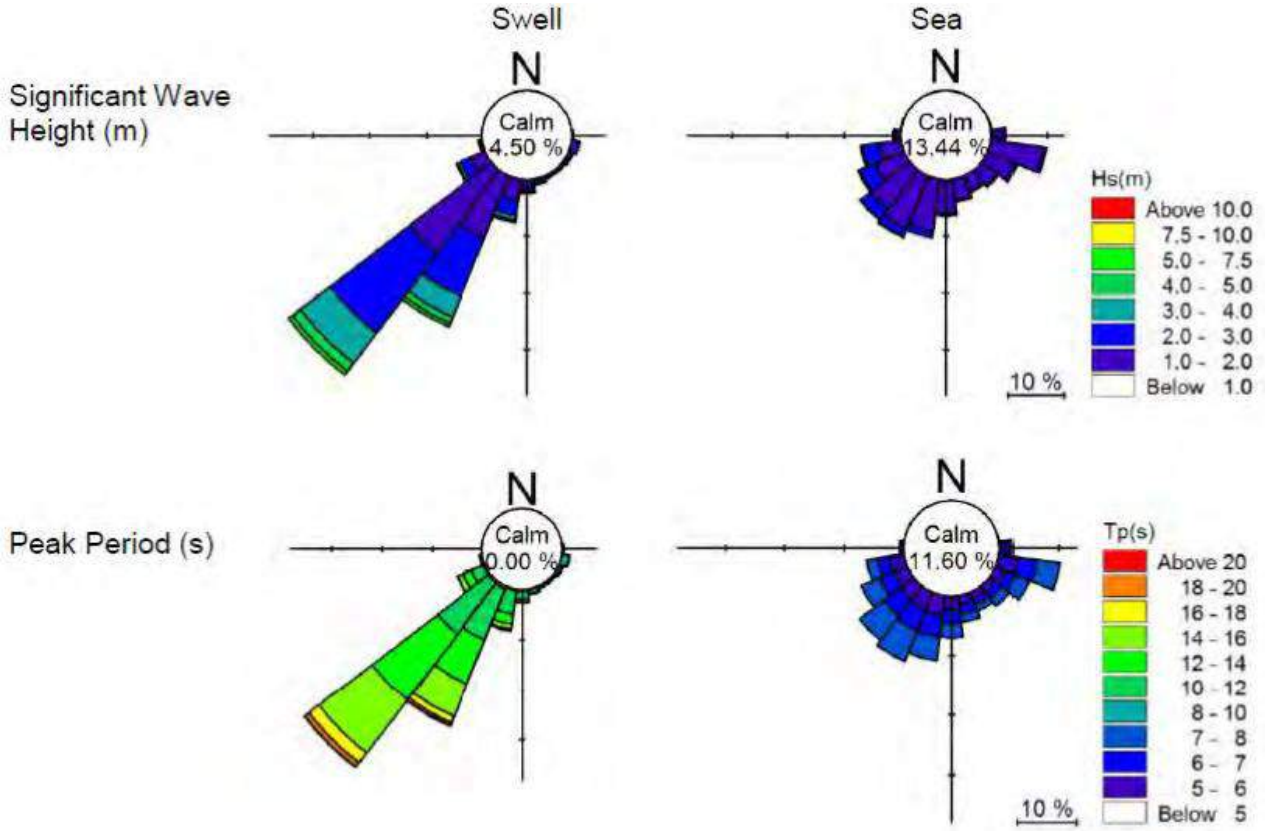
Figure 4.2 Albany airport wind rose for the period 1994-2012

4.3.2 Hydrodynamics

King George Sound is a semi-protected embayment, oriented in a SE-NW alignment. Winter cold fronts and low pressure systems can result in long westerly and south westerly fetches creating local sea-waves. During summer easterly winds result in smaller wind waves, due to the reduced fetch in the Sound. Offshore, the broad, high latitude westerly flow over the Southern and Indian Ocean results in a high-energy environment. Oceanic swell primarily arrives from the southwest direction (Figure 4.3), which propagates around Flinders Peninsula and into King George Sound (Figure 4.4).

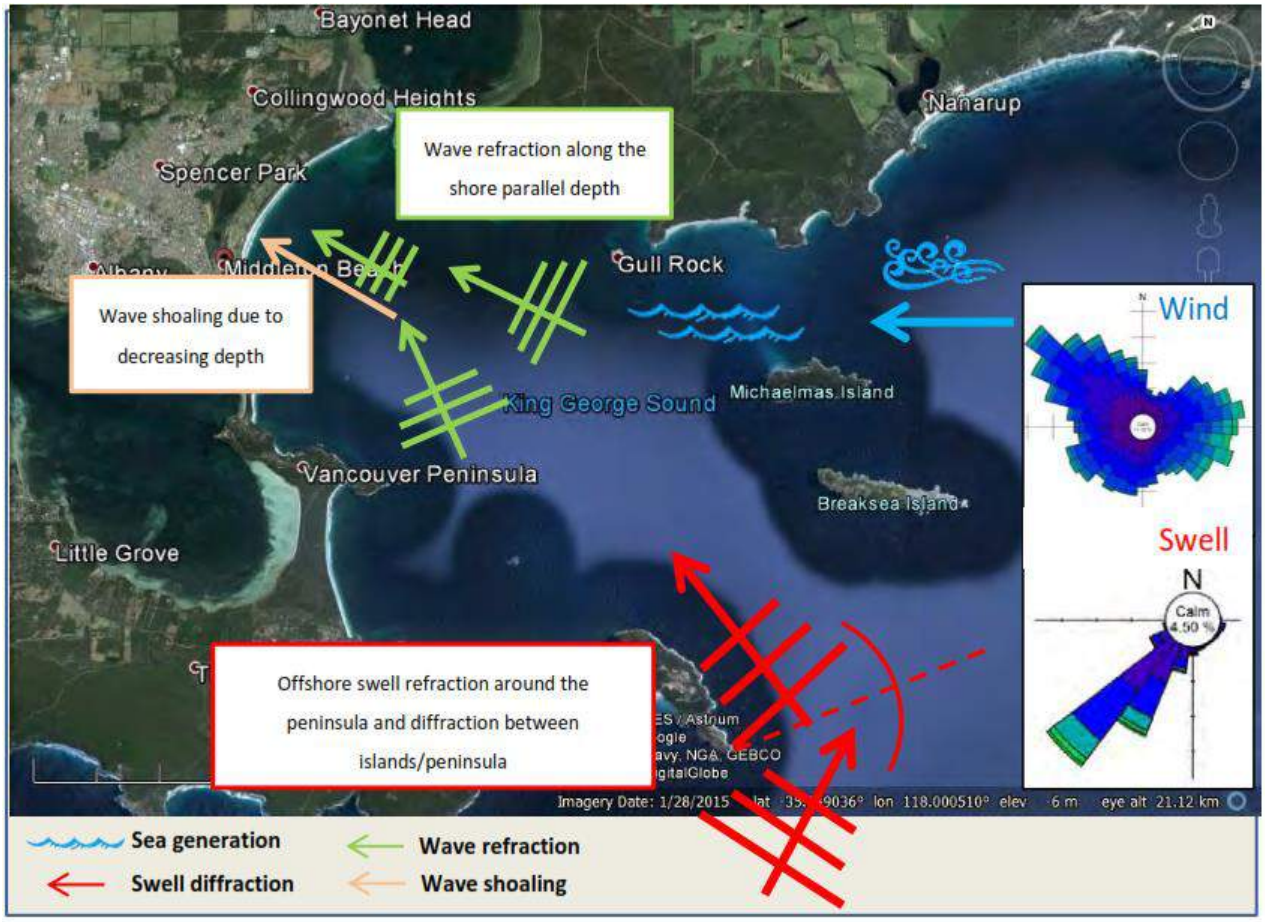
Waves propagating into the Sound from the Southern Ocean are buffered by Breaksea and Michaelmas Islands, to the east, and Flinders Peninsula to the south. Swells and wind driven waves (primarily driven by strong easterly winds in the summer and westerly winds in the winter; Ecologia 2007) tend to focus on two points along Middleton Beach. The southern wave focus point is the proposed location for the ASR, while the northern location is a known area of coastal erosion (Figure 2.2).

Tides in Albany are micro-tidal, with the largest tidal range <1.5 m (URS 2012). Tides are both diurnal and semi-diurnal at various stages of the tidal cycle (RHDHV 2015a). Middleton Beach spring tidal range is 0.9 m, and neap tidal range 0.3 m. Storm surges have been observed to increase water levels by 0.4-1.0 m at Middleton Beach (RHDHV 2015a).



Source: URS (2012)

Figure 4.3 Wave roses for significant wave heights and peak periods derived from the DoT offshore wave rider buoy



Source: RHDHV (2015a)

Figure 4.4 Conceptual model of the wave processes in King George Sound

4.4 Marine environmental quality

Sampling completed by Ecologia (2007) indicated that King George Sound sediments and waters were generally of a high quality. In sediments, tributyltin (TBT) was present within one disposal area, close to Flinders Peninsula, and nearby to the Albany Port Wharf. Hydrocarbons were not present in any samples and metals in sediments were all below the ANZECC/ARMCANZ (2000) ISQG-Low levels (Ecologia 2007).

Water clarity in King George Sound is typically very high with good light penetration (Ecologia 2007, further supported by Bastyan and Associates 2015). However, winter storms and associated high river discharge into the Sound can result in increased turbidity (Ecologia 2007). Reduction in clarity due to waves stirring sediments is generally short lived, due to the low levels of fines within sediments (Ecologia 2007).

While sediment and water sampling has not been completed at the proposed Middleton Beach ASR site, it can be reasonably assumed that the results of Ecologia (2007) would be representative of the marine environmental quality at the site.

4.5 Coastal and marine fauna

An EPBC Act Protected Matters Search Tool report (Appendix A) found 50 threatened species within the proposed Albany ASR location. Of these, 11 are marine species that have the potential to interact with the proposed ASR (Table 4.1).

Table 4.1 Endangered species potentially found at the proposed artificial surf reef site

Species	Common name	Status
Mammals		
Neophoca cinerea	Australian sea lion	Vulnerable
Megaptera novaeangliae	Humpback whale	Vulnerable
Eubalaena australis	Southern right whale	Endangered
Balaenoptera musculus	Blue whale	Endangered
Sharks		
Rhincodon typus	Whale shark	Vulnerable
Carcharodon carcharias	White shark	Vulnerable
Carcharias taurus	Grey nurse shark	Vulnerable
Reptiles		
Dermochelys coriacea	Leatherback turtle	Endangered
Chelonia mydas	Green turtle	Vulnerable
Caretta caretta	Loggerhead turtle	Endangered

Source: DoEE (2016), Appendix A

4.5.1 Mammals

Australian sea lions (*Neophoca cinerea)* are found along the southern coast and islands surrounding Albany (DoEE 2017a). Other Pinniped species include the New Zealand fur seal (*Arctocephalus forsteri*), sub-Antarctic fur seal (*Arctocephalus tropicalis*) and leopard seal (*Hydrurga leptonyx*) are also found in the area (DoEE 2017a). Middleton Beach is not a known sea-lion breeding colony; however sea lions may potentially use the area as a haul-out site (DoEE 2017a).

Whales *(Megaptera novaeangliae, Eubalaena australis,* and *Balaenoptera musculus*) are prevalent throughout Western Australia. The south coast of Western Australia is used as a nursery, migration and feeding area for all three species reported in Table 4.1 (Appendix A). The

proposed ASR location is sited in <5 m of water (RHDHV 2015a). Whale species potentially occurring at the site (*Megaptera novaeangliae, Eubalaena australis,* and *Balaenoptera musculus*) do not typically occur in shallow waters (DoEE 2017b,c,d), and therefore are not anticipated to be impacted by the ASR.

Other marine mammal species not included in the EPBC Act Protected Matters Search Tool report (Appendix A) but that may be present at the proposed ASR location include the bottlenose and common dolphins (*Tursiops truncatus* and *Delphinus delphis*), short finned pilot whale (*Globicephala macrorhynchus*), killer whale (*Orcinus orca*) and false killer whale (*Pseudorca crassidens*) (DoEE 2017e,f, Ecologia 2007). Other than the common and bottlenose dolphin, these species are not frequently sighted in the Albany region (Ecologia 2007). Common and bottlenose dolphins inhabit coastal and offshore regions of Australia, and are not listed as endangered or vulnerable under the EPBC Act (Appendix A). They are likely to be present in the proposed ASR location.

4.5.2 Sharks

Three shark species, the white shark (*Carcharodon carcharias*), grey nurse shark (*Carcharias taurus*) and whale shark (*Rhincodon typus*) are listed as endangered and occurring within the proposed ASR location. Whale sharks typically inhabit warm, oceanic waters around cold water upwellings (DoEE 2017g). Albany is at the very southern limit of the whale shark distribution (DoEE 2017g), therefore it is unlikely that they will be sighted at the proposed ASR location. Grey nurse sharks typically inhabit rocky reefs, in water depths of 15-40 m (DoEE 2017h), and therefore are not anticipated to be found at the proposed ASR location.

White sharks are found in coastal waters of Australia, typically in water depths up to 100 m (DoEE 2017i). White sharks typically congregate around fur seal and sea lion colonies, particularly among the islands along the southern coast of Australia (DoEE 2017i). Individuals are found inshore, around rocky reefs surf beaches and shallow coastal bays, to outer continental shelf and slope areas (DoEE 2017i). Based on past sightings, it may be expected that white sharks will occasionally be in the vicinity of the proposed ASR.

4.5.3 Reptiles

Three species of marine reptile are recorded as potentially occurring within the ASR location (Table 4.1). Leatherback turtle (*Dermochelys coriacea*), green turtle (*Chelonia mydas*) and loggerhead turtle (*Caretta caretta*) are all listed as endangered or vulnerable.

Leather back turtles are typically pelagic, feeding on jellyfish, and venturing close to shore only during nesting season (DoEE 2017j). Green and loggerhead turtles are pelagic for the first 5-15 years of their life, moving inshore to shallow water and intertidal seagrass beds and algae mats for the latter part of their life (DoEE 2017k,l). Green turtles typically occur in tropical areas of northern Australia (DoEE 2017k), and do not typically stray south of the 20ºC isotherm. Loggerhead turtles may occur in southern Western Australia; however more typically occur north of Shark Bay (DoEE 2017l). There are no recorded nesting sites for turtles in southern Western Australia as sand temperatures are inadequate for a successful incubation (DoEE 2017j,k,l). Therefore, marine turtles are not anticipated to occur within the proposed ASR location.

4.5.4 Other marine fauna

King George Sound is home to approximately 203 species of fish (Ecologia 2007), of which the pilchard (*Sardinops sagax*) is the most commercially important. Studies have suggested that the south coast pilchard stock consists of three main aggregations, Albany, Bremer Bay and Esperance (DoF 1999).

Breeding colonies of migratory seabirds, namely the Great Winged Petrel (*Pterodroma macroptera*) and the Flesh-footed Shearwater (*Puffinus carneipes*) breed on Breaksea, Michaelmas, Eclipse and Bald Islands (Ecologia 2007). During breeding season these species rely heavily on pilchards that shoal in the Sound.

The Department of Fisheries has recorded 25 Introduced Marine Species (IMS) in Albany (DoF 2016). Notable IMS species present in Albany Port include (Ecologia 2007, APA 2013, DoF 2016):

- Pacific oyster (*Crassostrea gigas*)
- toxic dinoflagellate (*Gymnodinium catenatum*)
- ascidian tunicate (*Ascidiella asper*sa)
- three species of bryozoans, (*Cryptosula pallasiana, Bugula flabellata*, and *Bugula neritin*a)
- European fan worm (*Sabella spallanzanii*),
- Codium (*Codium fragile* spp*. fragile*).

4.6 Social environment

The City of Albany is home to ~37,000 people, with a median age of 41.3 (ABS 2017). This ageing population is driving the CoA to attract and retain younger people in the city (CoA 2016). Middleton Beach is the closest beach to the Albany Central Business District (CBD), and is heavily used by locals and tourists (RHDHV 2015a). Middleton Beach is also home to the ASLSC, Ellen Cove Swimming Enclosure (installed in March 2016 and is currently being trailed for a period of three years) and a swimming platform that is placed just offshore during the summer months (RHDHV 2015a).

King George Sound is a popular recreational fishing spot. There are three charter operators running whale watching tours from June to October, and one SCUBA dive operator that use the Sound and adjacent waters (Ecologia 2007). The Sound is used for the commercial fishing of Pilchards (*Sardinops sagax*), while both Princess Royal and Oyster Harbours are used as a safe harbour by a small commercial fishing fleet (Ecologia 2007). Twelve aquaculture leases exist for King George Sound, primarily focussed around mussel farming at Mistaken Island (Ecologia 2007).

A search of the DAA Registered Sites database (Appendix B) showed no Aboriginal Heritage sites within the proposed ASR location. Middleton Beach is listed in the Heritage Council State Heritage Office inHerit database, as place number 17520 (Appendix C). Middleton Beach has been given a heritage category of E, meaning it is a historic site with no remaining structures to be managed.

5. Stakeholder Consultation

The CoA has recognised the importance of strong community and stakeholder engagement regarding the ASR Project. The CoA has undertaken extensive community surveys (Appendix D), to determine the position of the Albany community on the ASR Project. The Project has received community support, with 655 out of 728 respondents (90%) supporting of the ASR. Issued raised against the ASR were perceived negative impacts to the environment, and the financial impacts to ratepayers.

In order for community concerns, intentions and aspects of the Project to be met from all stakeholder groups, the city has formed the Albany Artificial Surfing Reef Advisory Steering Group to advise and convey information to appropriate members of the Albany community. The Steering Group have provided input into all stages of the project, including community surveys, beach monitoring surveys, and the completion of a business case for the Project.

An inception workshop was held by the City on 31 March 2015, with members of the Steering Group (SG) and RHDHV. The objectives of the workshop were to introduce the design consultants, allow all stakeholders to voice their background and perspectives of the Project and to provide some background technical discussions. It was not the purpose to evaluate any ASR designs as that work had not yet been undertaken.

The Steering Group, CoA and relevant Project personnel will continue to meet throughout the Project lifetime. Further community consultation will be completed as the Project develops.

6. Environmental Impact Assessment

6.1 Relevant environmental factors and objectives

Based on the review of the existing environment (Section 4), the following environmental factors (EPA 2016b) are believed to be applicable to the construction and operation of the ASR:

- Benthic communities and habitat
- Coastal processes
- Marine environmental quality
- Marine fauna
- Social surroundings

The following factors were considered not relevant to this Project (EPA 2016b):

- Flora and vegetation
- Landforms
- Subterranean fauna
- Terrestrial environmental quality
- Terrestrial fauna
- Hydrological processes
- Inland waters environmental quality
- Air quality
- Human health

Table 6.1 describes the EPA's objectives for each relevant environmental factor. The environmental impacts of the ASR Project should be managed to ensure that the EPA's objectives for each relevant factor are met. The EPA has issued a guideline for each factor, to better outline how the objectives for each factor can be met. The EPA has also issued technical guidance documents for Factors that document expectations for impact assessment (Table 6.1).

Table 6.1 Relevant environmental factors, objectives and guidance documents

Environmental Factor	Factor Objective	EPA Guidance Document	Report section
Benthic communities and habitat	To protect benthic communities and habitats so that biological diversity and ecological integrity are maintained	EPA (2016c) EPA (2016d)	6.4
Coastal processes	To maintain the geophysical processes that shape coastal morphology so that the environmental values of the coast are protected	EPA (2016e)	6.5
Marine environmental quality	To maintain the quality of water, sediment and biota so that environmental values are protected	EPA (2016f) EPA (2016g)	6.6
Marine fauna	To protect marine fauna so that biological diversify and ecological integrity are maintained	EPA (2016h)	6.7
Social surroundings	To protect social surroundings from significant harm	EPA (2016i)	6.8

Source: EPA (2016b)

6.2 Causes of environmental impact

Environmental impacts of the ASR Project may potentially occur during the construction and operation phases.

6.2.1 Construction

During the construction phase, potential environmental impacts may occur as a result of:

- Barge and workboat anchoring
- Placement of ASR material
- Introduction of vessels and/or equipment harbouring introduced marine species
- Hydrocarbon spills and leakage
- Creation of waste material

6.2.2 Operation

During the operational phase, potential environmental impacts may occur as a result of:

- Public interaction
- Movement of the ASR material
- Increased accretion/erosion at Middleton Beach

6.3 Cumulative environmental impact assessment

Although there is a risk an individual project may have potential impacts to the environment, other project impacts can lead to increased deleterious effects on environmental values, if not monitored and/or managed appropriately (EPA 2016b). As such, it is important to consider the cumulative impacts of a project for each environmental factor (Table 6.1), in the context of existing phases of the same development, as well as other developments in the surrounding area.

There are no known projects within the vicinity of the ASR that could contribute to a cumulative impact. The large-scale loss of seagrass offshore from Middleton Beach has been attributed to natural events (URS 2012, Bastyan and Associates 2015). The Ellen Cove jetty, Ellen Cove Swimming Enclosure and the swimming platform, situated to the south of the proposed ASR, does not extend over benthic habitat. Albany Port Dredging operations (and potential expansion) occurs some 2 km away, and do not affect the coastal environment as the proposed ASR (Ecologia 2007). Cumulative benthic community and habitat impacts of the ASR and Albany Port dredging operations are considered in Section 6.4. Erosion issues at Emu Point have not resulted in erosion or accretion at the proposed ASR location (URS 2012, RHDHV 2015a). Therefore it is anticipated that, provided the ASR Project environmental impacts are managed to meet the EPA (2016b) objectives for each factor, there will be no significant cumulative impacts.

6.4 Benthic communities and habitat

Benthic communities and habitat (BCH) play important roles in maintaining the integrity of marine ecosystems and the supply of ecological services (EPA 2016d). The largest impact of the ASR Project is likely to be direct smothering of BCH by the ASR material, and associated scouring, erosion and accretion. The Protection of Benthic Communities and Habitats Technical Guidance Document (EPA 2016d) is specific in how impacts to BCH should be calculated and managed. Assessment of direct and indirect BPPH loss against the EPA (2009) guidelines follows three main steps:

1. Calculation of the area of each habitat type in the Project area
2. Calculation of direct and indirect habitat loss areas from the Project, for each habitat type
3. Calculation of the cumulative loss for each habitat type mapped within the broader habitat management unit prescribed by the EPA (2016d) guidelines

6.4.1 Calculation of the area of each habitat type in the Project area

Seagrass habitats of Middleton Beach were mapped by Bastyan and Associates (2015) (Table 6.2, Figure 6.1). This report provides the most up-to date information on the BCH of the study area. Aerial imagery was captured on 12 April 2014, and classified into BCH based on the coverage of seagrass (predominantly *P. coriacea*) (Bastyan and Associates, 2015). These classifications were ground-truthed using spot dives and towed videos (Bastyan and Associates 2015).

The dominant habitat class was 30-50% mature dense *P. coriacea* (164.53 ha = 42% of the mapped area, Table 6.2). Trace *P. coriacea* (84 ha = 22% of the area), 30% mature *P. coriacea* (43 ha = 11% of the area) and 10-20% small *P. coriacea* plants (44 ha = 11% of the area) were also abundant BCH categories. Bare sand and wrack accounted for 12 ha and 9 ha, of the mapped area, respectively (Table 6.2).

Table 6.2 Mapped benthic community and habitat areas, Middleton Beach

Category	Habitat Description	Mapped area (ha)
1	15-20% small *P. coriacea* plants	12.25
2	Mature *P. coriacea*	15.25
3	Remnant *P. sinuosa*/*P. australis*/*Amphibolis* spp	1.51
4	30-50% mature dense *P. coriacea* seedlings in sand patches	164.53
5	30% mature *P. coriacea* plants	42.84
6	10-20% small *P. coriacea* plants/seedlings	43.88
7	Trace *P. coriacea*	84.62
8	Wrack	9.32
9	Bare Sand	12.37
Total		**386.57**

Source: Bastyan and Associates (2015)



Source: Bastyan and Associates (2015)

Figure 6.1 Benthic communities and habitat of Middleton Beach

6.4.2 Calculation of direct and indirect habitat loss areas from the ASR, for each habitat type

The term 'loss' refers to direct removal or destruction of BCH, commonly associated with activities such as excavation or burial, and most often when this damage to impacted BCH is considered to be irreversible. 'Serious damage' is defined as damage to BCH that is effectively irreversible or where recovery may occur would be unlikely to do so for at least five years following impact (EPA 2016d). It is noted within EPA (2016d) that there is currently no specific EPA guidance for considering short-term reversible impacts upon BCH. To ensure a conservative approach and to demonstrate the minor and temporary impact to BCH as a result of the ASR, the guidance in EPA (2016d) has been adopted.

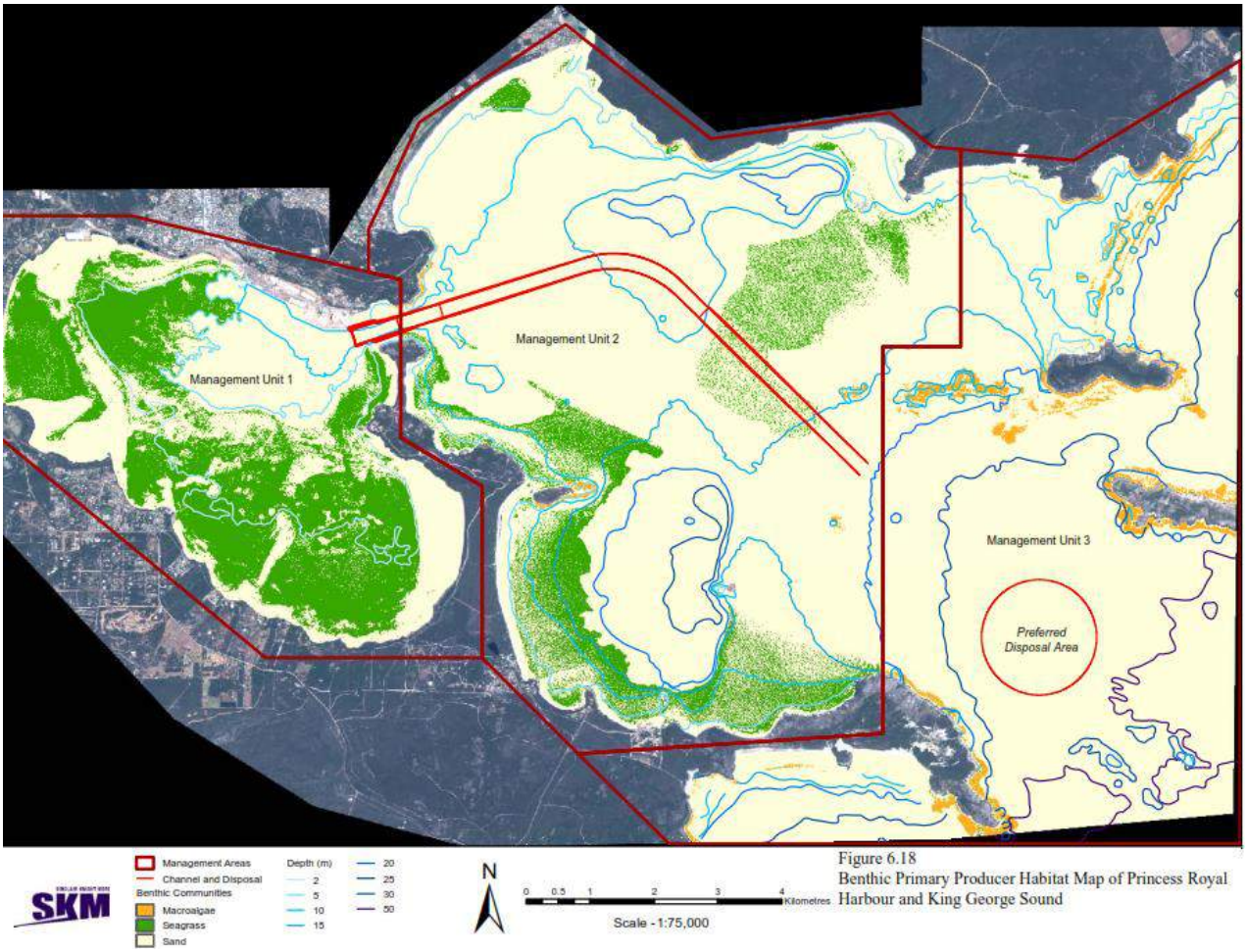
Setting a local assessment unit

Local Assessment Units (LAU) are defined areas within which the impact of a proposal on BCH is spatially assessed (EPA 2016d). LAUs are not standardised, meaning they must be defined individually for each Project, although as a guide a LAU in the order of 50 km^2 (5000 ha) should be used for assessments in Western Australia (EPA 2016d).

For the Albany ASR Project, the mapped area was defined by the area of BCH mapped by Bastyan and Associates (2015; Figure 6.1). A total of 387 ha of seabed was mapped (Table 6.2), significantly smaller than the suggested 5000 ha local assessment unit (EPA 2016d). However, the fine-scale mapping of seagrass at the Project location completed by Bastyan and Associates (2015) means that the impact of the ASR on the local BCH can be assessed.

On a broader scale, Ecologia (2007) defined a management unit (now known as LAU) for inner King George Sound which encompassed the proposed ASR site (Figure 6.2). The management unit/LAU was found to be appropriate by the OEPA in that earlier assessment. The study defined BCH as seagrass, sand or macroalgae, and mapped 817.5 ha of seagrass within inner King George Sound (Table 6.3). Importantly, Ecologia (2007) did not define any seagrass in the area later mapped by Bastyan and Associates (2007), with the exception of a dense patch at the northern end of Middleton Beach (Figure 6.2). This is likely an artefact of the coarse, large scale of the habitats mapped rather than an absence of seagrass at the ASR location.

In order to assess the potential impacts of the ASR on the BCH, in accordance with the EPA (2016d) guidance, loss has been assessed at both the mapped area scale (387 ha) and LAU scale (6540.9 ha; Ecologia 2007).



Source: Ecologia (2007)

Figure 6.2 Benthic primary producer habitats of Princess Royal Harbour and King George Sound

Table 6.3 Benthic primary producer habitat historical and 2007 coverage

Management unit	Bare sand (ha)		Seagrass (ha)		Macroalgae (ha)	
	Historical	2007	Historical	2007	Historical	2007
1 - Princess Royal Harbour	0.0	1453.9	2889.0	1385.0	0.2	0.2
2 - Inner King George Sound	5702.4	5702.4	817.5	817.5	21.0	21.0
3 - Outer King George Sound	5243.5	5243.5	3.0	3.0	232.3	232.3

Source: Ecologia (2007)

Calculating historical loss

A reduction of seagrass has occurred within Middleton Bay as a result of natural events (i.e. storms in 1984 [Bastyan and Associates 2015], natural variation in *P. coriacea* meadows [Evangelisti et al. 1998] and other storm events scouring the seabed [URS 2012]). Anthropogenic loss of BCH in inner King George Sound is negligible (Ecologia 2007; Table 6.3). Human use of the Sound, including the dredging of the existing Albany Port Channel, has resulted in the loss of ~50 m^2 (0.005 ha) of seagrass (Ecologia 2007). Therefore, historical loss for the purposes of a BCH loss assessment against EPA (2016d) has been considered to be nil.

Benthic communities and habitat loss as a result of the Artificial Surf Reef

As the ASR is only in an initial design stage, a worst-case scenario has been adopted for the calculation of BCH loss. This worst case encompasses an area of seabed that would be occupied if all ASR designs currently being considered were built (Figure 6.3), and therefore exaggerates the potential impact of the ASR on BCH (RHDHV 2017). Therefore, this represents

a highly conservative approach, and any as-built impacts to BCH are anticipated to be less than those proposed in this impact assessment. It should be noted that the surest preferred design option (Option B, Figure 6.3) is entirely within an area mapped as wrack.

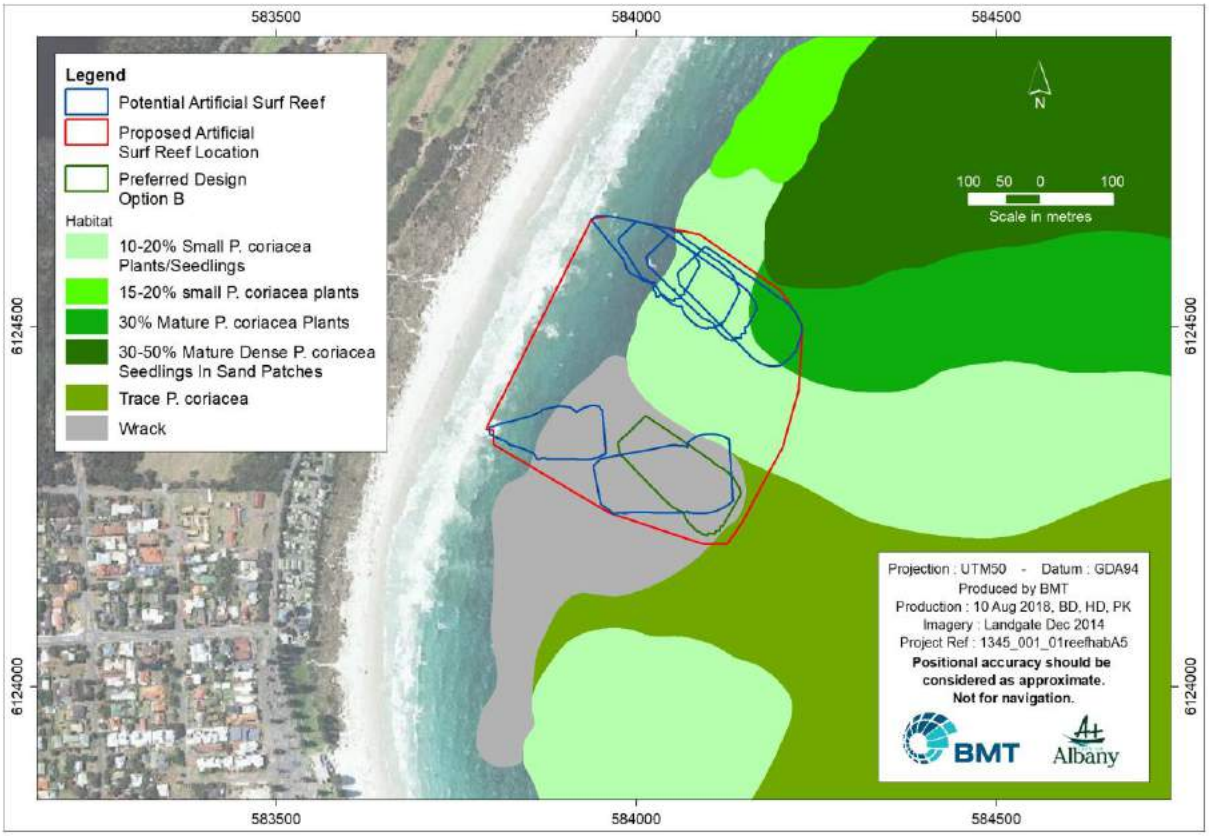


Figure 6.3 Albany Artificial Surf Reef placement options and worst-case scenario used for benthic communities and habitat loss calculations

Direct loss has been considered to be the area of BCH that will be covered by the ASR material, i.e. directly within the footprint. Indirect loss is likely to be considered to be a conservative 50 m wide sand halo around the ASR footprint, within which it is assumed some or all BCH will be lost due to changed currents and water flow. The 50 m halo has been used based on experience with other seawall and breakwater projects, and is considered to be conservative in its estimation of loss. Given the worst case scenario used for these calculations, BCH loss has not been split into direct and indirect.

Other indirect loss pathways, such as reduced water quality from turbidity, vessel strike and misplacement/loss of ASR material are considered to only result in a small, localised and temporary impact, and therefore are not included in loss calculations (EPA 2016d).

The area of each BCH mapped by Bastyan and Associates (2015; Figure 6.1) anticipated to be lost as a result of the ASR is given in Table 6.4. The area of seagrass habitat anticipated to be lost has also been compared to the area of seagrass habitat mapped by Ecologia (2007), to determine the potential impact of the ASR on the BCH of inner King George Sound LAU.

Table 6.4 Indirect and direct benthic community and habitat loss calculations for the Albany Artificial Surf Reef

Habitat Description	Mapped area (ha)	Area of loss
10-20% small *P. coriacea* plants/seedlings[1]	43.88	4.71
30% mature *P. coriacea* plants[1]	42.84	0.66
Trace *P. coriacea*[1]	84.62	0.22
Wrack[1]	9.32	4.33
Seagrass[2]	817.5	5.59

Notes:
1. Mapped by Bastyan and Associates (2015)
2. Mapped by Ecologia (2007)

The cumulative area of BCH anticipated to be permanently lost as a result of the ASR represents 0.26%-46.43% of the BCH mapped by Bastyan and Associates (2015), and 0.68% of the seagrass mapped in the LAU by Ecologia (2007) (Table 6.5). Note that for comparison against the EPA (2016d) guidelines, the Ecologia (2007) LAU has been adopted.

Table 6.5 Cumulative benthic community and habitat loss calculations for Albany Artificial Surf Reef

Habitat Description	Mapped area (ha)	Historical loss[3]	Potential loss from ASR	Cumulative area of loss	% of BCH loss in mapped area	% of BCH loss in LAU
10-20% small *P. coriacea* plants/seedlings[1]	43.88	0	4.71	4.71	10.72	n/a
30% mature *P. coriacea* plants[1]	42.84	0	0.66	0.66	1.54	n/a
Trace *P. coriacea*[1]	84.62	0	0.22	0.22	0.26	n/a
Wrack[1]	9.32	0	4.33	4.33	46.43	n/a
Seagrass[2]	**817.5**	**0**	**5.59**	**5.59**	n/a	**0.68**

Notes:
1. Mapped by Bastyan and Associates (2015)
2. Mapped by Ecologia (2007)
3. From Ecologia (2007)
4. Bold font represents values used for comparison against EPA (2016d) guidelines

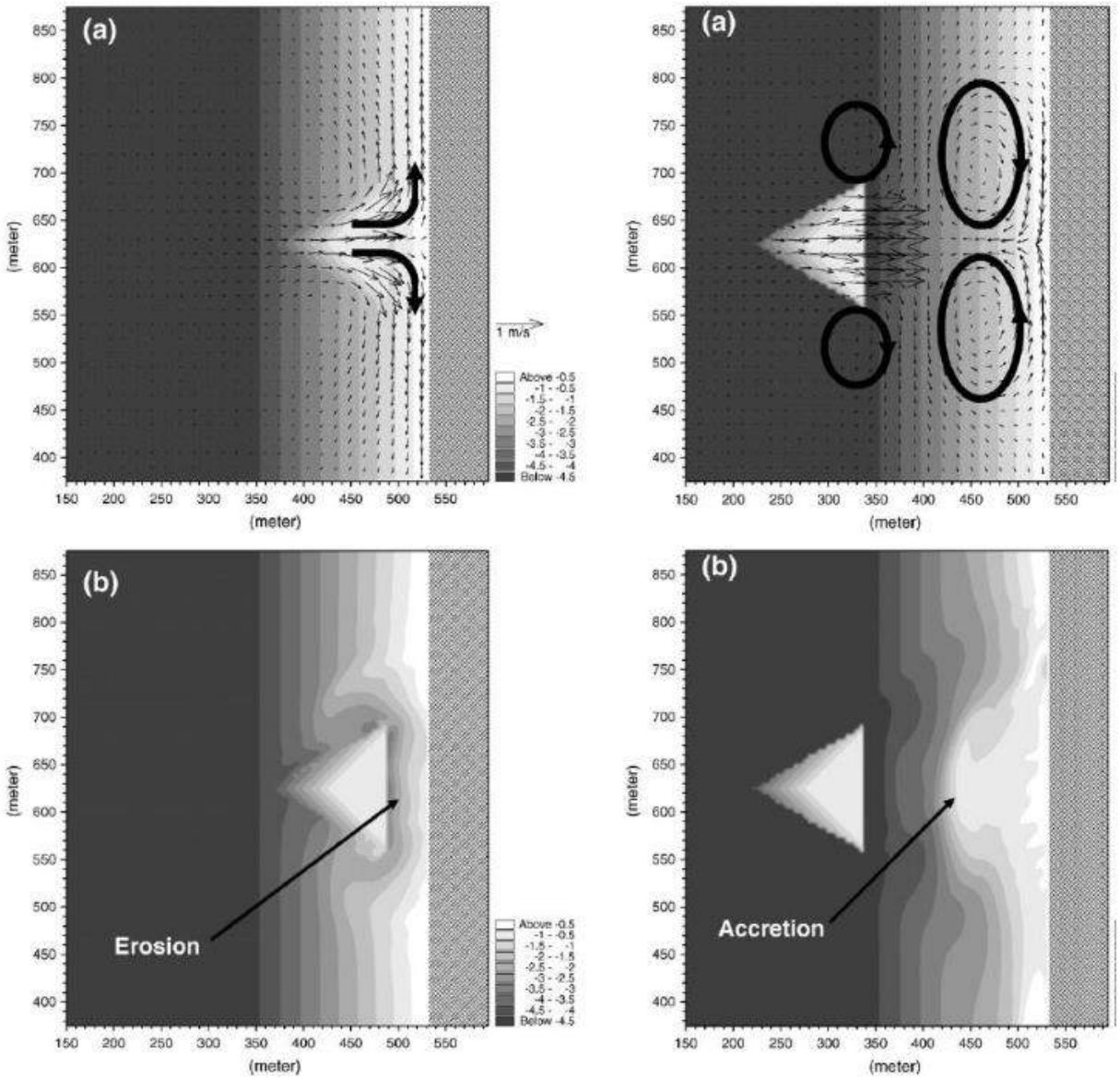
Given the conservative approach used to calculate BCH loss, and the low amount of seagrass cumulative loss anticipated based on the mapped area by Bastyan and Associates (2015), the impact of the ASR on BCH is expected to meet the OEPA's objectives. If considered in the context the LAU (defined by Ecologia 2007) the overall cumulative impact to seagrass is 0.68%. Provided that the ASR final location does not extend outside of the maximum extent used for loss calculations (Figure 6.3), the impact to BCH will meet the EPA (2016d) guidelines. An environmental management framework is provided in Section 7.1.

6.5 Coastal processes

The placement of the ASR offshore of Middleton Beach has the potential to alter hydrodynamic and sediment transport processes, changing the equilibrium of Middleton Beach (RHDV 2015a). Disruptions to longshore currents as well as cross-shore sediment transport processes by the introduction of structures within the active littoral zone have the potential to alter the long-term bathymetric features of a coastline.

Ranasinghe and Turner (2005) reviewed the near shore processes of a number of submerged parallel structures during shore normal wave attack and concluded that a strong onshore flow is

generated over submerged structures. The study identified two different modes of near shore currents that developed in the lee of the structure due to water level set up. These two different circulation patterns (two cell and four cell, Figure 6.4) can potentially cause erosion or accretion of sediments in the lee of the ASR, depending on the distance offshore (RHDV 2015a).



Source: RHDV (2015a)
Note:

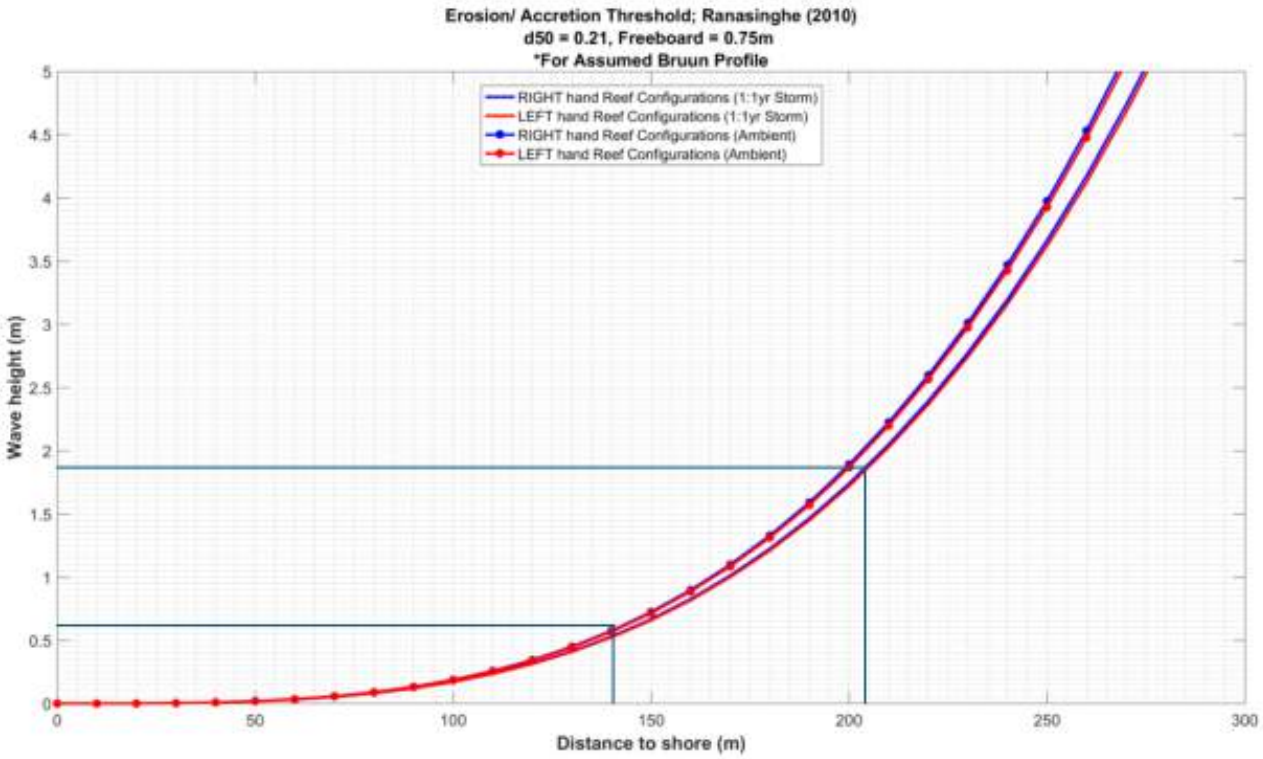
1. The resulting near shore currents (a) and beach response (b), the shoreline response can be seen as erosive (left) and accreting (right).

Figure 6.4 Interaction of a single submerged offshore shore parallel structure with an incoming wave field placed at two distances from the shore, close–proximity (left) and at a distance from the shore (right).

However, there is an equilibrium point, at which the placement of an offshore structure (i.e. ASR) will not cause erosion or accretion to the shoreline structure (RHDV 2015a). Ranasinghe et al. (2010) investigated this phenomenon using numerical modelling and formed an empirical relationship between the shoreline response and distance to shore of a submerged structure.

RHDHV (2018) developed this empirical relationship specific to the ASR and Middelton Beach (a copy of the report is provided in Appendix E). The study demonstrated that the minimum offshore placement distance of the ASR for a null shoreline response is approximately 205 m for a 1 year

average re-occurrence interval wave event from the landward extent of the structure or 140 m under ambient wave conditions (Figure 6.5).

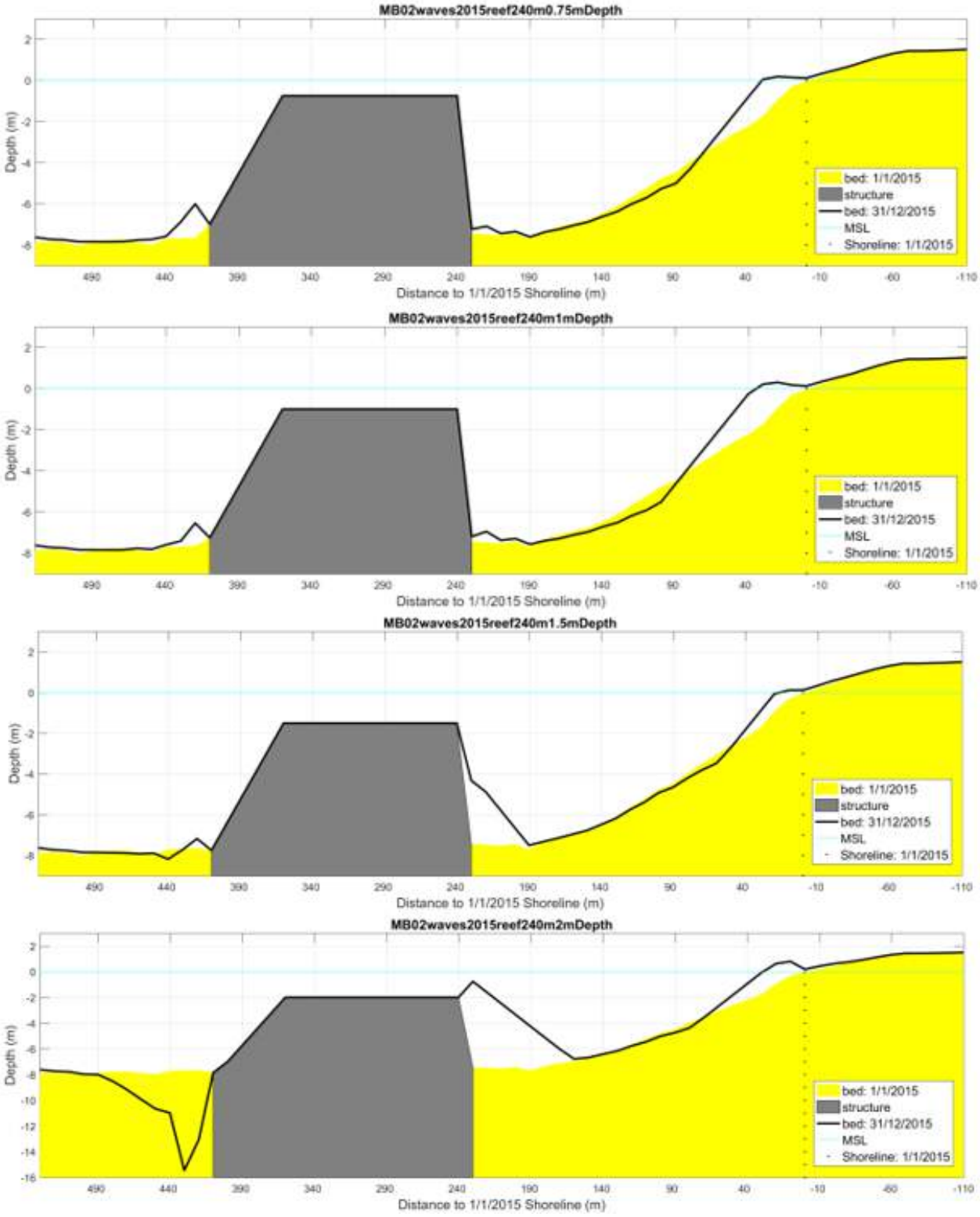


Source: RHDHV (2018)
Note:

1. Curves represent updated calculations based on Ranasinghe (2010)

Figure 6.5 Null point distance offshore for the Albany Artificial Surf Reef for a 1 year average reoccurrence interval, and under ambient conditions,

Wave and current modelling was undertaken for a range of reef configurations and offshore positions under a variety of event-based wave conditions (RHDHV 2018). The modelling showed, similarly to the empirical calculations, that the ASR layouts either produced a 2 or 4-cell current circulation pattern in their lee, and offshore placement (Figure 6.6) would be expected to produce no significant impact on the shoreline compared to inshore placement (Figure 6.7).

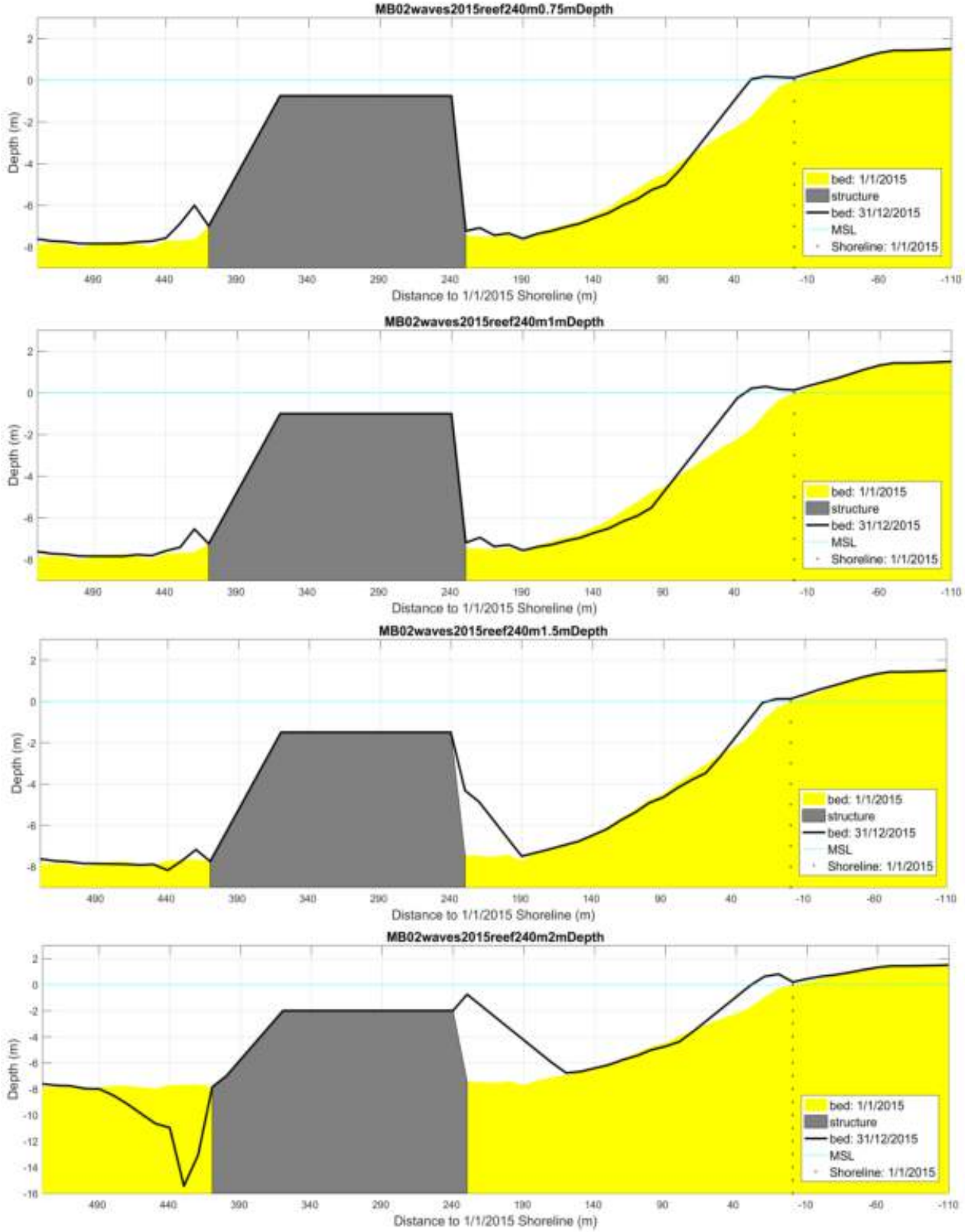


Source: RHDHV (2018)

Note:

1. Results demonstrate shoreline response between 1 January 2015 (yellow) and 31 December 2015 (black line), at ASR crest depth range from 0.75 m (top) to 2 m (bottom).

Figure 6.6 Simulated shoreline response to the Albany Artificial Surf Reef placed 240 m offshore



Source: RHDHV (2018)

Note:

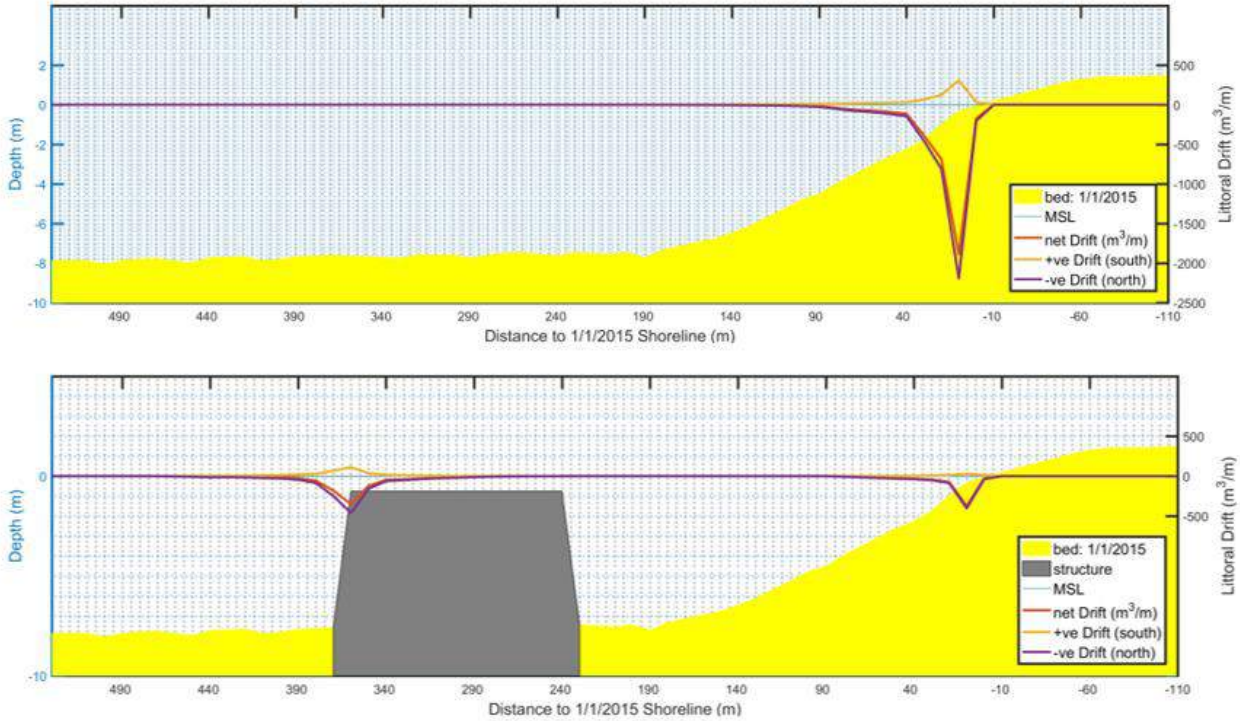
1. Results demonstrate shoreline response between 1 January 2015 (yellow) and 31 December 2015 (black line), at ASR crest depth range from 0.75 m (top) to 2 m (bottom).

Figure 6.7 Simulated shoreline response to the Albany Artificial Surf Reef placed 90 m offshore

One dimensional longshore and cross-shore sediment transport modelling was undertaken of an idealised beach profile at the Surfers location at Middleton Beach, both with and without the ASR in place at the offshore location for a representative year (RHDHV 2018).

The introduction of the reef was seen to dissipate the current along the shoreline (and sediment transport) in this location with the ASR installed 240 m offshore (Figure 6.8). Some increase in longshore sediment drift was observed on the seaward edge of the ASR (Figure 6.8), however this is not significant, and appears to cause minor accretion at the cross shore toe of the ASR (observed in the 240 m offshore, 0.75 m crest depth simulation, Figure 6.6).

Results of modelling demonstrated that the ASR can be constructed within the proposed footprint, while not having a significant impact on coastal processes at Middleton Beach (RHDHV 2018). Design Option B (Figure 6.9) has the least impact to coastal processes. Further detailed design and modelling will determine final depth, location, shape, orientation and structure volumes prior to construction.



Source: RHDHV (2018)
Note:

1. Longshore littoral drift (m^3/m) from the LITDRIFT simulations for (top) base case (no reef) and (bottom) inclusion of reef structure at 240 m offshore.

Figure 6.8 Simulated longshore sediment drift for Middelton Beach, with and without the Albany Artificial Surf Reef installed

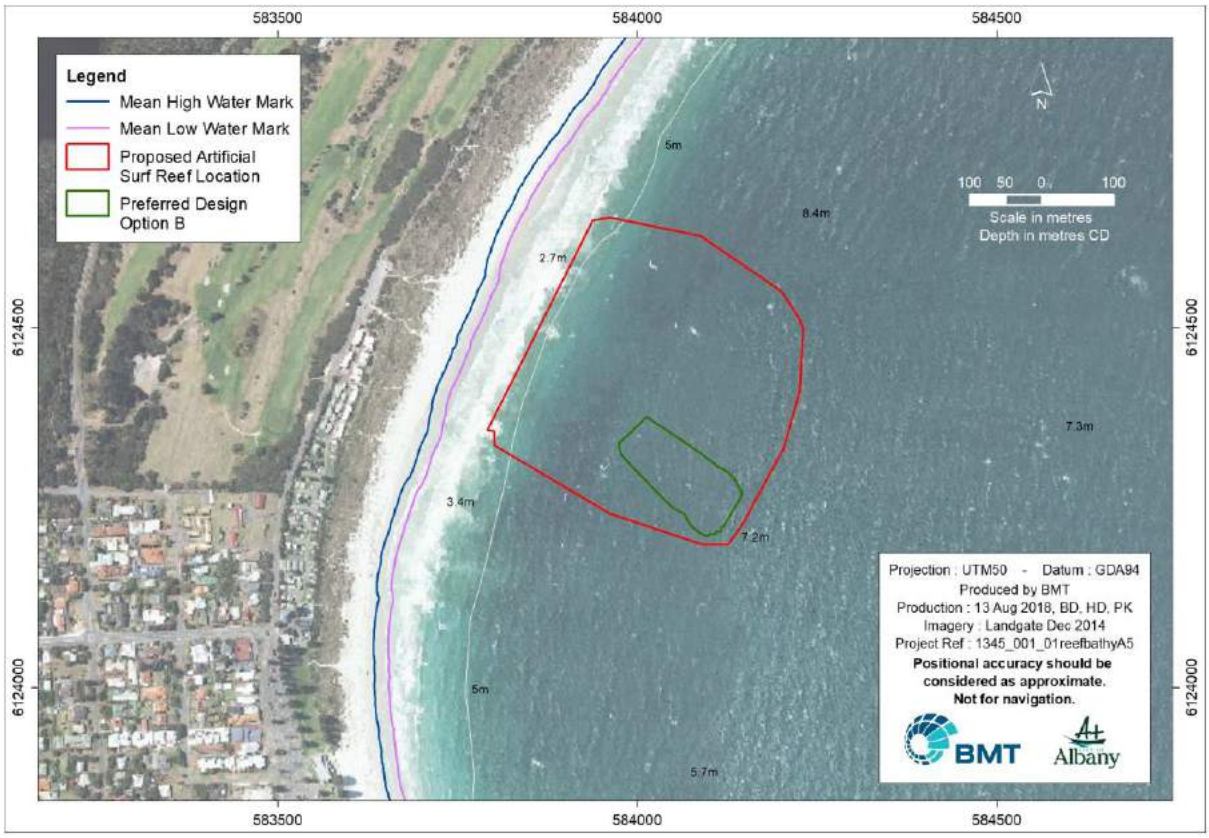


Figure 6.9 Albany Artificial Surf Reef design Option B and development footprint with depth contours

As requested by DWER, RHDHV (2018) was reviewed by Department of Transport (DoT) Coastal Management Department. DoT provided the following comments and recommendations (pers. com. F. Li June 2018):

- Based on the literature review and modelling outcome documented in the report, DoT is convinced that the impacts of the proposed ASR structure on adjacent Middleton Beach coastline will be moderate and manageable should appropriate modelling and design work are carried out carefully during the detailed design phases of the project.
- As the model predicted impacts are indicative only, an adequate coastal impact monitoring program will be critical to resolve the uncertainties of modelling results and work out a set of realistic and adequate impact management options for the short, medium, and long-term future.
- Given the inherent inaccuracies typically associated with the calculation of inflection point and sediment movement patterns and volumes, a significant contingency should be included when considering the potential impacts on adjoining foreshore areas. There is a risk that the impacts of the ASR may be greater than estimated. The contingencies for additional impacts and management requirements should be adequately planned for.

DoT's recommendations will be adopted by the CoA during the development of an environmental management plan for the ASR Project. An environmental management framework is provided in Section 7.1

6.6 Marine environmental quality

6.6.1 Introduced marine species

The risk of introducing IMS to the Project area via vessel movements and usage is low. The CoA propose to use local vessels if possible, however it is likely that vessels from elsewhere in the state, or potentially overseas, will be utilised if local vessels are not available.

CoA will verify each vessel's operational history, fouling control coating and ballast water details are accurate and reliable before contracting the vessel. All vessels to be used on the Project shall complete the DPIRD vessel check biofouling risk assessment tool (available at https://vesselcheck.fish.wa.gov.au/).

Information to be provided in the vessel check includes:

- Evidence that sediment and ballast water has, or will be, managed to prevent IMSs entering and moving within Western Australia. Alternatively, a maintained ballast water management plan and record book should be provided
- Vessel's log entries showing operational history since last antifouling coating application or IMS inspection, or a maintained biofouling management plan and record book
- The most recent in-water cleaning or dry dock/slip report, and IMS inspection report
- Evidence of either an active marine growth prevention system or a suitable manual treatment regime for internal seawater pipe works
- The most recent antifouling coating application certificate or original receipts or invoices stating the coating type, volume purchased, vessel name (if possible) and date of application
- Type of vessel.

CoA will ensure that vessels have a risk assessment rating of "low" prior to mobilising for the Project. Further IMS management will be undertaken as required by the outcomes of the tool and in consultation with DPIRD. An environmental management framework is provided in Section 7.1.

6.6.2 Sediment quality

It is unlikely that the proposed ASR will have any detrimental effect on the sediment quality of Middleton Beach. Material for the construction of the ASR will be inert, with no potential for release of contaminants into the environment. Contamination of sediment may occur through the discharge of wastes or hydrocarbons (i.e. oil spills) during construction. Therefore, waste and hydrocarbon spill management will be required for the Project. An environmental management framework is provided in Section 7.1.

6.6.3 Water quality

Potential impacts to marine water quality associated with ASR Project construction are primarily associated with turbidity generation from installation, spills and waste.

There may be the potential for turbid sediment plumes to be created when the ASR material is placed on the seabed. The ASR location is a high wave energy environment (RHDHV 2015a), and as such it is anticipated that turbidity generation as a result of installation activities will be rapidly dissipated with minimal impact to the water quality expected.

The potential for deteriorated water quality as a result of spills and waste (e.g. generated from vessel movement) is predicted to be low, however waste and hydrocarbon spill management will be required for the Project. An environmental management framework is provided in Section 7.1.

6.7 Marine fauna

Section 4.5 lists the marine fauna that may be present at the Proposed ASR location. The small footprint of the Project infrastructure means that mobile fauna (such as fish, sharks, mammals, reptiles and birds) will likely not be impacted during any phase of the Project. It is likely that the presence of ASR will lead to increases in marine flora and fauna at the site (RHDHV 2015a). Artificial reefs have been demonstrated to successfully increase fish abundances above that of surrounding bare sand habitat (Nielsen & Wells 1997).

6.7.1 Vessel strikes

The construction process is expected to involve slow moving vessels and as such poses negligible risk, however a detailed EMP will be developed which includes management of vessel movement in relation to marine mammals. An environmental management framework is provided in Section 7.1.

6.7.2 Underwater noise

Underwater noise is likely to be generated during the construction phase of the ASR Project primarily from vessel movements. Underwater noise has the potential to impact upon sensitive marine fauna (mammals, fish and turtles) by interfering with communication, causing changes in behaviour or in extreme cases causing physiological damage to auditory organs (Southall et al. 2007). The potential for impacts from noise-generating activities is dependent on a range of factors, including the intensity and frequency of the noise, prevailing ambient noise levels and proximity of noise to sensitive species (Richardson et al. 1995, NRC 2005, Southall et al. 2007, Popper & Hastings 2009).

Dolphin whistles and fish chorusing were recorded while pile driving occurred in the Fremantle inner harbour in Western Australia, which regularly experiences a high level of anthropogenic sound from vessel traffic, dredging operations, and trains passing over Bridges (Salgado Kent et al. 2012). Given the ASR Project proximity to the Albany Port (and associated vessel movements), the low level of underwater noise expected to be generated (i.e. no piling, only vessel movements), underwater noise is not likely to cause physiological damage to sensitive marine fauna and has been assessed as low residual risk.

6.8 Social surroundings

The impacts to social surroundings of the Project have been split up into heritage (European and Aboriginal), safety, and public amenity.

6.8.1 Heritage impacts

There are no Aboriginal heritage sites in the vicinity of the proposed ASR (Section 4.6). Therefore, it is anticipated that no impacts to Aboriginal heritage will occur as a result of the ASR Project. Middleton Beach is a registered European Heritage place (4.6). It is not anticipated that the proposed ASR Project will have any impact on the European heritage values of Middleton Beach, however advice should be sought from the State Heritage Office as to any permitting that may be required.

6.8.2 Maritime safety

There is a potential risk relating to maritime safety associated with the proposed ASR Project, including the loss of materials from the construction vessels and barges, and disturbance to the navigation of other vessels in the area.

Maritime safety during the operational phase of the Project has been identified as having a medium residual risk rating, due to the presence of the ASR units close to the water surface. The

ASR crest will remain in place at ~0.75 m below the water surface. The presence of both the ASR and construction vessels may cause safety issues for unauthorised boats entering the development area (either accidentally or deliberately).

Prior to construction commencing, the CoA will apply to the DoT for a notice to mariners (NTM), outlining the position of the ASR and start date for construction activities. This NTM will be promulgated by the DoT to relevant maritime personnel, including the commercial and recreational boating community. An environmental management framework is provided in Section 7.1.

6.8.3 Public amenity

It is highly unlikely that recreational or commercial fisheries will be negatively impacted upon as a result of the Project. Construction works may require that part of Middleton Beach is closed for a period of time. The impact of this closure to public amenity is anticipated to be low, due to the small area that would require closure (if any), distance from the main swimming part of Middleton Beach. Works will be timed to occur predominantly during winter, when beach use is at its lowest. An environmental management framework is provided in Section 7.1.

7. Environmental Management

A construction EMP will be written for the Project, to ensure that the EPA's objectives for each environmental factor impacted by the Project (see Section 6.1) are met. In addition, the construction contractor will be required to abide with the *City Guidelines – Responsibility of Contractors.* This document sets out the CoA's guidelines for work conducted by contractors on behalf of the CoA and is complementary to other documents relating to the tendering, acceptance and review of contracts and quotations. The CoA will engage a superintendent to oversee the Project and implementation of the EMP.

Operational impacts of the ASR will be managed in accordance with the management framework provided in Table 7.1. Details of methods and reporting for management and monitoring will be included in an operational EMP.

7.1 Environmental management framework

The following framework will be used to determine if the ASR construction and operation is having a significant impact on the environment. Table 7.1 outlines management targets for each environmental factor (as defined by EPA 2016b), as well as associated monitoring, triggers for management and management actions should a trigger be exceeded.

Table 7.1 Summary of environmental management commitments for the Albany Artificial Surf Reef Project

Factor	Management Objective	Potential Impact	Monitoring				Management Trigger	Management			
			Action	Responsibility	Timing/Frequency	Evidence		Action	Responsibility	Frequency/Timing	Evidence
Benthic Communities and Habitat (BCH)	No loss (direct or indirect) of BCH outside of development footprint during construction	Direct loss or burial of subtidal BCH outside of defined development footprint (Figure 6.3).	Review of ASR position via GPS logger data	Environmental representative	Weekly during construction	Weekly progress report	GPS logger data shows construction has occurred outside of defined development footprint	Review ASR design and potential impacts to BCH. Move ASR back inside development footprint where possible.	Proponent	As required	Weekly progress report
		Indirect loss of BCH	See marine environmental quality, below								
Coastal processes	No significant impacts to coastal processes of Middelton Beach	Significant erosion or accretion of beach adjacent to ASR	Detailed design process and model verification to determine impacts to coastal processes from final design are not-significant	Project design engineers	Prior to construction	Detailed design report	Erosion or accretion beyond natural variability is predicted to occur as a result of the ASR construction	Inform DoT and DWER Change design or move ASR to reduce impacts to coastal processes.	Proponent	Prior to construction	Final details design and correspondence with DWER/DoT
			Ongoing monitoring of beach profiles through the City of Albany Middleton Beach profile study, as per the ASR EMP.	Proponent	3 monthly	Annual report	Erosion or accretion above that modelled in detailed design phase and directly attributable to ASR	Inform DoT and DWER, to determined best methods for remediation. This may include changing design or move ASR to reduce impacts to coastal processes.	Proponent	Within 1 month of monitoring report being issued.	Correspondence with regulators
Marine Environmental Quality	To maintain the quality of water, sediment and biota	Increase in water column turbidity	Daily plume sketches and site photos	Contractor	Daily during construction	Imagery and plume sketches provided to Proponent	Turbid plume beyond the defined development footprint	Modify construction method and design turbidity control measures to reduce turbidity	Contractor	Within one week of turbidity trigger being exceeded	Summarised in weekly progress report
		Hydrocarbon spills and waste into the environment	Inspect and maintain equipment	Contractor	Daily during construction	Weekly progress report summarising any spills, malfunctioning equipment	Evidence of hydrocarbon leaks on construction equipment, or within the receiving marine environment	Manage the spill/waste and review equipment/work method to ensure no further spills Determine if additional environmental sampling or notification to DWER/DBCA/DoT is required	Contractor	As required	Summarised in weekly progress report
			Visual inspection of work area for spills and rubbish	Contractor	Daily during construction		Evidence of any water/rubbish not contained in an appropriate manner		Proponent	Within 48 hrs of a spill being reported	Incident report to Proponent and regulators as required
		Introduced marine species	All construction equipment and related vessels to be cleaned prior to mobilisation.	Contractor	Prior to mobilisation	Photos supplied to Proponent	Evidence that vessel and equipment not cleaned	Instruct Contractor to re-clean equipment	Proponent	As required	Photos of re-cleaned equipment
			All vessels entering State Waters should abide by Port of Albany and/or DPIRD IMS procedures and requirements	Contractor	Prior to mobilisation	Correspondence with Port of Albany/DPIRD	As per Port of Albany/DPIRD procedures and requirements. Including assessment by https://vesselcheck.fish.wa.gov.au/				
Marine Fauna	No significant impacts to fauna from construction	Vessel strikes Noise from construction equipment	Dedicated MFO on board construction vessels and or equipment	Contractor	While underway or construction is occurring	Weekly fauna observation log	Fauna (marine mammal or pinniped) approaches <300 m from vessel or equipment while operating	If possible, move away from fauna or slow down to avoid impact. Shut down equipment until fauna	Contractor	As required	Reported in weekly fauna observation log and dredge log

Factor	Management Objective	Potential Impact	Monitoring				Management Trigger	Management			
			Action	Responsibility	Timing/Frequency	Evidence		Action	Responsibility	Frequency/Timing	Evidence
								moves >100 m from equipment.			
Social Surroundings	Ensure public and navigational safety at all times	Public safety (including navigational safety)	Public to be informed of navigational risk and construction	Proponent	Prior to construction commencing	NTM or similar issued by DoT or Port of Albany Public signage and dissemination of information	Public or navigational safety incident	Cease works if required to prevent further incidents Inform Proponent immediately Complete incident report and investigation as required	Contractor	Immediately following incident	Incident report to Proponent and other relevant regulators as required
			Equipment/vessel master to maintain visual contact with approaching vessels Correct lighting displayed on construction equipment and vessels at all times Public access restricted to any construction areas	Contractor	Throughout construction	Incident report to Proponent					

Note:

1. DPIRD - Department of Primary Industry and Resource Development, DoT - Department of Transport, DBCA - Department of Biodiversity, Conservation and Attractions, DWER - Department of Water and Environmental Regulation, NTM - Notice to Mariners, ASR - Artificial Surf Reef, BCH - Benthic Communities and Habitat, IMS - Introduced Marine Species, EMP - Environmental Management Plan

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Appendix A

DoEE Protected Matters Search Tool Report

EPBC Act Protected Matters Report

This report provides general guidance on matters of national environmental significance and other matters protected by the EPBC Act in the area you have selected.

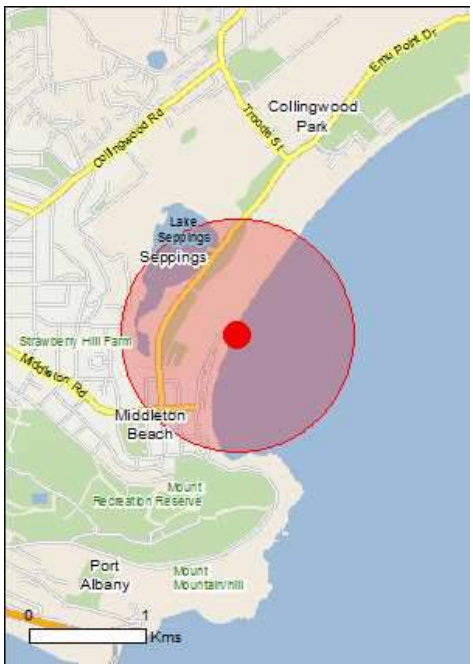
Information on the coverage of this report and qualifications on data supporting this report are contained in the caveat at the end of the report.

Information is available about Environment Assessments and the EPBC Act including significance guidelines, forms and application process details.

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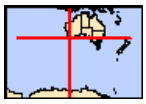
Summary
Details
- Matters of NES
- Other Matters Protected by the EPBC Act
- Extra Information

Caveat
Acknowledgements



This map may contain data which are ©Commonwealth of Australia (Geoscience Australia), ©PSMA 2010

Coordinates
Buffer: 1.0Km



Summary

Matters of National Environmental Significance

This part of the report summarises the matters of national environmental significance that may occur in, or may relate to, the area you nominated. Further information is available in the detail part of the report, which can be accessed by scrolling or following the links below. If you are proposing to undertake an activity that may have a significant impact on one or more matters of national environmental significance then you should consider the Administrative Guidelines on Significance.

World Heritage Properties:	None
National Heritage Places:	None
Wetlands of International Importance:	None
Great Barrier Reef Marine Park:	None
Commonwealth Marine Area:	None
Listed Threatened Ecological Communities:	None
Listed Threatened Species:	50
Listed Migratory Species:	53

Other Matters Protected by the EPBC Act

This part of the report summarises other matters protected under the Act that may relate to the area you nominated. Approval may be required for a proposed activity that significantly affects the environment on Commonwealth land, when the action is outside the Commonwealth land, or the environment anywhere when the action is taken on Commonwealth land. Approval may also be required for the Commonwealth or Commonwealth agencies proposing to take an action that is likely to have a significant impact on the environment anywhere.

The EPBC Act protects the environment on Commonwealth land, the environment from the actions taken on Commonwealth land, and the environment from actions taken by Commonwealth agencies. As heritage values of a place are part of the 'environment', these aspects of the EPBC Act protect the Commonwealth Heritage values of a Commonwealth Heritage place. Information on the new heritage laws can be found at http://www.environment.gov.au/heritage

A permit may be required for activities in or on a Commonwealth area that may affect a member of a listed threatened species or ecological community, a member of a listed migratory species, whales and other cetaceans, or a member of a listed marine species.

Commonwealth Land:	None
Commonwealth Heritage Places:	None
Listed Marine Species:	75
Whales and Other Cetaceans:	12
Critical Habitats:	None
Commonwealth Reserves Terrestrial:	None
Commonwealth Reserves Marine:	None

Extra Information

This part of the report provides information that may also be relevant to the area you have nominated.

State and Territory Reserves:	None
Regional Forest Agreements:	None
Invasive Species:	21
Nationally Important Wetlands:	None
Key Ecological Features (Marine)	None

Details

Matters of National Environmental Significance

Listed Threatened Species		[Resource Information]
Name	Status	Type of Presence
Birds		
Botaurus poiciloptilus Australasian Bittern [1001]	Endangered	Species or species habitat known to occur within area
Calidris canutus Red Knot, Knot [855]	Endangered	Species or species habitat known to occur within area
Calidris ferruginea Curlew Sandpiper [856]	Critically Endangered	Species or species habitat likely to occur within area
Calidris tenuirostris Great Knot [862]	Critically Endangered	Species or species habitat known to occur within area
Calyptorhynchus banksii naso Forest Red-tailed Black-Cockatoo, Karrak [67034]	Vulnerable	Species or species habitat likely to occur within area
Calyptorhynchus baudinii Baudin's Cockatoo, Baudin's Black-Cockatoo, Long-billed Black-Cockatoo [769]	Vulnerable	Breeding known to occur within area
Calyptorhynchus latirostris Carnaby's Cockatoo, Carnaby's Black-Cockatoo, Short-billed Black-Cockatoo [59523]	Endangered	Species or species habitat known to occur within area
Cereopsis novaehollandiae grisea Cape Barren Goose (south-western), Recherche Cape Barren Goose [25978]	Vulnerable	Species or species habitat may occur within area
Charadrius leschenaultii Greater Sand Plover, Large Sand Plover [877]	Vulnerable	Species or species habitat known to occur within area
Charadrius mongolus Lesser Sand Plover, Mongolian Plover [879]	Endangered	Species or species habitat known to occur within area
Dasyornis longirostris Western Bristlebird [515]	Vulnerable	Species or species habitat likely to occur within area
Diomedea antipodensis Antipodean Albatross [64458]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea dabbenena Tristan Albatross [66471]	Endangered	Species or species habitat may occur within

Name	Status	Type of Presence
		area
Diomedea epomophora (sensu stricto) Southern Royal Albatross [1072]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea exulans (sensu lato) Wandering Albatross [1073]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea sanfordi Northern Royal Albatross [64456]	Endangered	Foraging, feeding or related behaviour likely to occur within area
Limosa lapponica baueri Bar-tailed Godwit (baueri), Western Alaskan Bar-tailed Godwit [86380]	Vulnerable	Species or species habitat may occur within area
Limosa lapponica menzbieri Northern Siberian Bar-tailed Godwit, Bar-tailed Godwit (menzbieri) [86432]	Critically Endangered	Species or species habitat may occur within area
Macronectes giganteus Southern Giant-Petrel, Southern Giant Petrel [1060]	Endangered	Species or species habitat may occur within area
Macronectes halli Northern Giant Petrel [1061]	Vulnerable	Species or species habitat may occur within area
Numenius madagascariensis Eastern Curlew, Far Eastern Curlew [847]	Critically Endangered	Species or species habitat likely to occur within area
Pachyptila turtur subantarctica Fairy Prion (southern) [64445]	Vulnerable	Species or species habitat likely to occur within area
Sternula nereis nereis Australian Fairy Tern [82950]	Vulnerable	Breeding likely to occur within area
Thalassarche cauta cauta Shy Albatross, Tasmanian Shy Albatross [82345]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Thalassarche cauta steadi White-capped Albatross [82344]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Thalassarche impavida Campbell Albatross, Campbell Black-browed Albatross [64459]	Vulnerable	Species or species habitat may occur within area
Thalassarche melanophris Black-browed Albatross [66472]	Vulnerable	Species or species habitat may occur within area
Mammals		
Balaenoptera musculus Blue Whale [36]	Endangered	Species or species habitat likely to occur within area
Dasyurus geoffroii Chuditch, Western Quoll [330]	Vulnerable	Species or species habitat known to occur within area
Eubalaena australis Southern Right Whale [40]	Endangered	Breeding known to occur within area
Megaptera novaeangliae Humpback Whale [38]	Vulnerable	Species or species habitat likely to occur within area

Name	Status	Type of Presence
Neophoca cinerea Australian Sea-lion, Australian Sea Lion [22]	Vulnerable	Species or species habitat may occur within area
Parantechinus apicalis Dibbler [313]	Endangered	Species or species habitat likely to occur within area
Pseudocheirus occidentalis Western Ringtail Possum, Ngwayir, Womp, Woder, Ngoor, Ngoolangit [25911]	Vulnerable	Species or species habitat likely to occur within area
Plants		
Banksia brownii Brown's Banksia, Feather-leaved Banksia [8277]	Endangered	Species or species habitat may occur within area
Banksia verticillata Granite Banksia, Albany Banksia, River Banksia [8333]	Vulnerable	Species or species habitat likely to occur within area
Caladenia harringtoniae Harrington's Spider-orchid, Pink Spider-orchid [56786]	Vulnerable	Species or species habitat likely to occur within area
Caladenia winfieldii Majestic Spider-orchid [64504]	Endangered	Species or species habitat may occur within area
Chordifex abortivus Manypeaks Rush [64868]	Endangered	Species or species habitat likely to occur within area
Diuris drummondii Tall Donkey Orchid [4365]	Vulnerable	Species or species habitat likely to occur within area
Drakaea micrantha Dwarf Hammer-orchid [56755]	Vulnerable	Species or species habitat likely to occur within area
Isopogon uncinatus Hook-leaf Isopogon [20871]	Endangered	Species or species habitat likely to occur within area
Kennedia glabrata Northcliffe Kennedia [16452]	Vulnerable	Species or species habitat likely to occur within area
Sphenotoma drummondii Mountain Paper-heath [21160]	Endangered	Species or species habitat may occur within area
Reptiles		
Caretta caretta Loggerhead Turtle [1763]	Endangered	Breeding likely to occur within area
Chelonia mydas Green Turtle [1765]	Vulnerable	Breeding likely to occur within area
Dermochelys coriacea Leatherback Turtle, Leathery Turtle, Luth [1768]	Endangered	Breeding likely to occur within area
Sharks		
Carcharias taurus (west coast population) Grey Nurse Shark (west coast population) [68752]	Vulnerable	Species or species habitat likely to occur within area
Carcharodon carcharias White Shark, Great White Shark [64470]	Vulnerable	Foraging, feeding or related behaviour known to occur within area

Name	Status	Type of Presence
Rhincodon typus Whale Shark [66680]	Vulnerable	Species or species habitat may occur within area

Listed Migratory Species [Resource Information]

* Species is listed under a different scientific name on the EPBC Act - Threatened Species list.

Name	Threatened	Type of Presence
Migratory Marine Birds		
Apus pacificus Fork-tailed Swift [678]		Species or species habitat likely to occur within area
Diomedea antipodensis Antipodean Albatross [64458]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea dabbenena Tristan Albatross [66471]	Endangered	Species or species habitat may occur within area
Diomedea epomophora (sensu stricto) Southern Royal Albatross [1072]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea exulans (sensu lato) Wandering Albatross [1073]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea sanfordi Northern Royal Albatross [64456]	Endangered	Foraging, feeding or related behaviour likely to occur within area
Macronectes giganteus Southern Giant-Petrel, Southern Giant Petrel [1060]	Endangered	Species or species habitat may occur within area
Macronectes halli Northern Giant Petrel [1061]	Vulnerable	Species or species habitat may occur within area
Puffinus carneipes Flesh-footed Shearwater, Fleshy-footed Shearwater [1043]		Foraging, feeding or related behaviour likely to occur within area
Sterna caspia Caspian Tern [59467]		Foraging, feeding or related behaviour known to occur within area
Thalassarche cauta (sensu stricto) Shy Albatross, Tasmanian Shy Albatross [64697]	Vulnerable*	Foraging, feeding or related behaviour likely to occur within area
Thalassarche impavida Campbell Albatross, Campbell Black-browed Albatross [64459]	Vulnerable	Species or species habitat may occur within area
Thalassarche melanophris Black-browed Albatross [66472]	Vulnerable	Species or species habitat may occur within area
Thalassarche steadi White-capped Albatross [64462]	Vulnerable*	Foraging, feeding or related behaviour likely to occur within area
Migratory Marine Species		
Balaenoptera edeni Bryde's Whale [35]		Species or species habitat may occur within area
Balaenoptera musculus Blue Whale [36]	Endangered	Species or species habitat likely to occur

Name	Threatened	Type of Presence within area
Caperea marginata Pygmy Right Whale [39]		Species or species habitat may occur within area
Carcharodon carcharias White Shark, Great White Shark [64470]	Vulnerable	Foraging, feeding or related behaviour known to occur within area
Caretta caretta Loggerhead Turtle [1763]	Endangered	Breeding likely to occur within area
Chelonia mydas Green Turtle [1765]	Vulnerable	Breeding likely to occur within area
Dermochelys coriacea Leatherback Turtle, Leathery Turtle, Luth [1768]	Endangered	Breeding likely to occur within area
Eubalaena australis Southern Right Whale [40]	Endangered	Breeding known to occur within area
Lagenorhynchus obscurus Dusky Dolphin [43]		Species or species habitat may occur within area
Lamna nasus Porbeagle, Mackerel Shark [83288]		Species or species habitat may occur within area
Manta alfredi Reef Manta Ray, Coastal Manta Ray, Inshore Manta Ray, Prince Alfred's Ray, Resident Manta Ray [84994]		Species or species habitat known to occur within area
Manta birostris Giant Manta Ray, Chevron Manta Ray, Pacific Manta Ray, Pelagic Manta Ray, Oceanic Manta Ray [84995]		Species or species habitat known to occur within area
Megaptera novaeangliae Humpback Whale [38]	Vulnerable	Species or species habitat likely to occur within area
Orcinus orca Killer Whale, Orca [46]		Species or species habitat may occur within area
Rhincodon typus Whale Shark [66680]	Vulnerable	Species or species habitat may occur within area
Migratory Terrestrial Species		
Motacilla cinerea Grey Wagtail [642]		Species or species habitat may occur within area
Migratory Wetlands Species		
Actitis hypoleucos Common Sandpiper [59309]		Species or species habitat known to occur within area
Arenaria interpres Ruddy Turnstone [872]		Species or species habitat known to occur within area
Calidris acuminata Sharp-tailed Sandpiper [874]		Species or species habitat known to occur within area
Calidris alba Sanderling [875]		Species or species habitat known to occur within area

Name	Threatened	Type of Presence
Calidris canutus Red Knot, Knot [855]	Endangered	Species or species habitat known to occur within area
Calidris ferruginea Curlew Sandpiper [856]	Critically Endangered	Species or species habitat likely to occur within area
Calidris ruficollis Red-necked Stint [860]		Species or species habitat known to occur within area
Calidris tenuirostris Great Knot [862]	Critically Endangered	Species or species habitat known to occur within area
Charadrius bicinctus Double-banded Plover [895]		Species or species habitat known to occur within area
Charadrius leschenaultii Greater Sand Plover, Large Sand Plover [877]	Vulnerable	Species or species habitat known to occur within area
Charadrius mongolus Lesser Sand Plover, Mongolian Plover [879]	Endangered	Species or species habitat known to occur within area
Heteroscelus brevipes Grey-tailed Tattler [59311]		Species or species habitat known to occur within area
Limnodromus semipalmatus Asian Dowitcher [843]		Species or species habitat known to occur within area
Limosa lapponica Bar-tailed Godwit [844]		Species or species habitat known to occur within area
Limosa limosa Black-tailed Godwit [845]		Species or species habitat known to occur within area
Numenius madagascariensis Eastern Curlew, Far Eastern Curlew [847]	Critically Endangered	Species or species habitat likely to occur within area
Numenius phaeopus Whimbrel [849]		Species or species habitat known to occur within area
Pandion haliaetus Osprey [952]		Species or species habitat known to occur within area
Pluvialis fulva Pacific Golden Plover [25545]		Species or species habitat known to occur within area
Pluvialis squatarola Grey Plover [865]		Species or species habitat known to occur within area
Tringa nebularia Common Greenshank, Greenshank [832]		Species or species habitat known to occur within area
Tringa stagnatilis Marsh Sandpiper, Little Greenshank [833]		Species or species habitat known to occur within area

Name	Threatened	Type of Presence
Xenus cinereus Terek Sandpiper [59300]		Species or species habitat known to occur within area

Other Matters Protected by the EPBC Act

Listed Marine Species		[Resource Information]
* Species is listed under a different scientific name on the EPBC Act - Threatened Species list.		
Name	Threatened	Type of Presence
Birds		
Actitis hypoleucos Common Sandpiper [59309]		Species or species habitat known to occur within area
Apus pacificus Fork-tailed Swift [678]		Species or species habitat likely to occur within area
Ardea alba Great Egret, White Egret [59541]		Species or species habitat known to occur within area
Ardea ibis Cattle Egret [59542]		Species or species habitat may occur within area
Arenaria interpres Ruddy Turnstone [872]		Species or species habitat known to occur within area
Calidris acuminata Sharp-tailed Sandpiper [874]		Species or species habitat known to occur within area
Calidris alba Sanderling [875]		Species or species habitat known to occur within area
Calidris canutus Red Knot, Knot [855]	Endangered	Species or species habitat known to occur within area
Calidris ferruginea Curlew Sandpiper [856]	Critically Endangered	Species or species habitat likely to occur within area
Calidris ruficollis Red-necked Stint [860]		Species or species habitat known to occur within area

Name	Threatened	Type of Presence
Calidris tenuirostris Great Knot [862]	Critically Endangered	Species or species habitat known to occur within area
Cereopsis novaehollandiae grisea Cape Barren Goose (south-western), Recherche Cape Barren Goose [25978]	Vulnerable	Species or species habitat may occur within area
Charadrius bicinctus Double-banded Plover [895]		Species or species habitat known to occur within area
Charadrius leschenaultii Greater Sand Plover, Large Sand Plover [877]	Vulnerable	Species or species habitat known to occur within area
Charadrius mongolus Lesser Sand Plover, Mongolian Plover [879]	Endangered	Species or species habitat known to occur within area
Charadrius ruficapillus Red-capped Plover [881]		Species or species habitat known to occur within area
Diomedea antipodensis Antipodean Albatross [64458]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea dabbenena Tristan Albatross [66471]	Endangered	Species or species habitat may occur within area
Diomedea epomophora (sensu stricto) Southern Royal Albatross [1072]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea exulans (sensu lato) Wandering Albatross [1073]	Vulnerable	Foraging, feeding or related behaviour likely to occur within area
Diomedea sanfordi Northern Royal Albatross [64456]	Endangered	Foraging, feeding or related behaviour likely to occur within area
Haliaeetus leucogaster White-bellied Sea-Eagle [943]		Species or species habitat known to occur within area
Heteroscelus brevipes Grey-tailed Tattler [59311]		Species or species habitat known to occur within area
Himantopus himantopus Black-winged Stilt [870]		Species or species habitat known to occur within area
Limnodromus semipalmatus Asian Dowitcher [843]		Species or species habitat known to occur within area
Limosa lapponica Bar-tailed Godwit [844]		Species or species habitat known to occur within area
Limosa limosa Black-tailed Godwit [845]		Species or species habitat known to occur within area
Macronectes giganteus Southern Giant-Petrel, Southern Giant Petrel [1060]	Endangered	Species or species habitat may occur within area

Name	Threatened	Type of Presence
Macronectes halli Northern Giant Petrel [1061]	Vulnerable	Species or species habitat may occur within area
Merops ornatus Rainbow Bee-eater [670]		Species or species habitat may occur within area
Motacilla cinerea Grey Wagtail [642]		Species or species habitat may occur within area
Numenius madagascariensis Eastern Curlew, Far Eastern Curlew [847]	Critically Endangered	Species or species habitat likely to occur within area
Numenius phaeopus Whimbrel [849]		Species or species habitat known to occur within area
Pachyptila turtur Fairy Prion [1066]		Species or species habitat likely to occur within area
Pandion haliaetus Osprey [952]		Species or species habitat known to occur within area
Pluvialis fulva Pacific Golden Plover [25545]		Species or species habitat known to occur within area
Pluvialis squatarola Grey Plover [865]		Species or species habitat known to occur within area
Puffinus carneipes Flesh-footed Shearwater, Fleshy-footed Shearwater [1043]		Foraging, feeding or related behaviour likely to occur within area
Recurvirostra novaehollandiae Red-necked Avocet [871]		Species or species habitat known to occur within area
Sterna caspia Caspian Tern [59467]		Foraging, feeding or related behaviour known to occur within area
Thalassarche cauta (sensu stricto) Shy Albatross, Tasmanian Shy Albatross [64697]	Vulnerable*	Foraging, feeding or related behaviour likely to occur within area
Thalassarche impavida Campbell Albatross, Campbell Black-browed Albatross [64459]	Vulnerable	Species or species habitat may occur within area
Thalassarche melanophris Black-browed Albatross [66472]	Vulnerable	Species or species habitat may occur within area
Thalassarche steadi White-capped Albatross [64462]	Vulnerable*	Foraging, feeding or related behaviour likely to occur within area
Thinornis rubricollis Hooded Plover [59510]		Species or species habitat likely to occur within area
Tringa nebularia Common Greenshank, Greenshank [832]		Species or species habitat known to occur within area

Name	Threatened	Type of Presence
Tringa stagnatilis Marsh Sandpiper, Little Greenshank [833]		Species or species habitat known to occur within area
Xenus cinereus Terek Sandpiper [59300]		Species or species habitat known to occur within area
Fish		
Acentronura australe Southern Pygmy Pipehorse [66185]		Species or species habitat may occur within area
Campichthys galei Gale's Pipefish [66191]		Species or species habitat may occur within area
Heraldia nocturna Upside-down Pipefish, Eastern Upside-down Pipefish, Eastern Upside-down Pipefish [66227]		Species or species habitat may occur within area
Hippocampus breviceps Short-head Seahorse, Short-snouted Seahorse [66235]		Species or species habitat may occur within area
Histiogamphelus cristatus Rhino Pipefish, Macleay's Crested Pipefish, Ring-back Pipefish [66243]		Species or species habitat may occur within area
Leptoichthys fistularius Brushtail Pipefish [66248]		Species or species habitat may occur within area
Lissocampus caudalis Australian Smooth Pipefish, Smooth Pipefish [66249]		Species or species habitat may occur within area
Lissocampus runa Javelin Pipefish [66251]		Species or species habitat may occur within area
Maroubra perserrata Sawtooth Pipefish [66252]		Species or species habitat may occur within area
Nannocampus subosseus Bonyhead Pipefish, Bony-headed Pipefish [66264]		Species or species habitat may occur within area
Notiocampus ruber Red Pipefish [66265]		Species or species habitat may occur within area
Phycodurus eques Leafy Seadragon [66267]		Species or species habitat may occur within area
Phyllopteryx taeniolatus Common Seadragon, Weedy Seadragon [66268]		Species or species habitat may occur within area
Pugnaso curtirostris Pugnose Pipefish, Pug-nosed Pipefish [66269]		Species or species habitat may occur within area
Solegnathus lettiensis Gunther's Pipehorse, Indonesian Pipefish [66273]		Species or species habitat may occur within area
Stigmatopora argus Spotted Pipefish, Gulf Pipefish, Peacock Pipefish [66276]		Species or species habitat may occur within area

Name	Threatened	Type of Presence
Stigmatopora nigra Widebody Pipefish, Wide-bodied Pipefish, Black Pipefish [66277]		Species or species habitat may occur within area
Stigmatopora olivacea a pipefish [74966]		Species or species habitat may occur within area
Urocampus carinirostris Hairy Pipefish [66282]		Species or species habitat may occur within area
Vanacampus margaritifer Mother-of-pearl Pipefish [66283]		Species or species habitat may occur within area
Vanacampus phillipi Port Phillip Pipefish [66284]		Species or species habitat may occur within area
Vanacampus poecilolaemus Longsnout Pipefish, Australian Long-snout Pipefish, Long-snouted Pipefish [66285]		Species or species habitat may occur within area
Mammals		
Arctocephalus forsteri Long-nosed Fur-seal, New Zealand Fur-seal [20]		Species or species habitat likely to occur within area
Neophoca cinerea Australian Sea-lion, Australian Sea Lion [22]	Vulnerable	Species or species habitat may occur within area
Reptiles		
Caretta caretta Loggerhead Turtle [1763]	Endangered	Breeding likely to occur within area
Chelonia mydas Green Turtle [1765]	Vulnerable	Breeding likely to occur within area
Dermochelys coriacea Leatherback Turtle, Leathery Turtle, Luth [1768]	Endangered	Breeding likely to occur within area

Whales and other Cetaceans [Resource Information]

Name	Status	Type of Presence
Mammals		
Balaenoptera acutorostrata Minke Whale [33]		Species or species habitat may occur within area
Balaenoptera edeni Bryde's Whale [35]		Species or species habitat may occur within area
Balaenoptera musculus Blue Whale [36]	Endangered	Species or species habitat likely to occur within area
Caperea marginata Pygmy Right Whale [39]		Species or species habitat may occur within area
Delphinus delphis Common Dophin, Short-beaked Common Dolphin [60]		Species or species habitat may occur within area
Eubalaena australis Southern Right Whale [40]	Endangered	Breeding known to occur within area
Grampus griseus Risso's Dolphin, Grampus [64]		Species or species

Name	Status	Type of Presence
		habitat may occur within area
Lagenorhynchus obscurus Dusky Dolphin [43]		Species or species habitat may occur within area
Megaptera novaeangliae Humpback Whale [38]	Vulnerable	Species or species habitat likely to occur within area
Orcinus orca Killer Whale, Orca [46]		Species or species habitat may occur within area
Tursiops aduncus Indian Ocean Bottlenose Dolphin, Spotted Bottlenose Dolphin [68418]		Species or species habitat likely to occur within area
Tursiops truncatus s. str. Bottlenose Dolphin [68417]		Species or species habitat may occur within area

Extra Information

Invasive Species [Resource Information]

Weeds reported here are the 20 species of national significance (WoNS), along with other introduced plants that are considered by the States and Territories to pose a particularly significant threat to biodiversity. The following feral animals are reported: Goat, Red Fox, Cat, Rabbit, Pig, Water Buffalo and Cane Toad. Maps from Landscape Health Project, National Land and Water Resouces Audit, 2001.

Name	Status	Type of Presence
Birds		
Anas platyrhynchos Mallard [974]		Species or species habitat likely to occur within area
Columba livia Rock Pigeon, Rock Dove, Domestic Pigeon [803]		Species or species habitat likely to occur within area
Streptopelia senegalensis Laughing Turtle-dove, Laughing Dove [781]		Species or species habitat likely to occur within area
Sturnus vulgaris Common Starling [389]		Species or species habitat likely to occur within area
Mammals		
Felis catus Cat, House Cat, Domestic Cat [19]		Species or species habitat likely to occur within area
Mus musculus House Mouse [120]		Species or species habitat likely to occur within area
Oryctolagus cuniculus Rabbit, European Rabbit [128]		Species or species habitat likely to occur within area

Name	Status	Type of Presence
Rattus rattus Black Rat, Ship Rat [84]		Species or species habitat likely to occur within area
Sus scrofa Pig [6]		Species or species habitat likely to occur within area
Vulpes vulpes Red Fox, Fox [18]		Species or species habitat likely to occur within area
Plants		
Asparagus aethiopicus Asparagus Fern, Ground Asparagus, Basket Fern, Sprengi's Fern, Bushy Asparagus, Emerald Asparagus [62425]		Species or species habitat likely to occur within area
Asparagus asparagoides Bridal Creeper, Bridal Veil Creeper, Smilax, Florist's Smilax, Smilax Asparagus [22473]		Species or species habitat likely to occur within area
Asparagus declinatus Bridal Veil, Bridal Veil Creeper, Pale Berry Asparagus Fern, Asparagus Fern, South African Creeper [66908]		Species or species habitat likely to occur within area
Asparagus scandens Asparagus Fern, Climbing Asparagus Fern [23255]		Species or species habitat likely to occur within area
Genista monspessulana Montpellier Broom, Cape Broom, Canary Broom, Common Broom, French Broom, Soft Broom [20126]		Species or species habitat likely to occur within area
Genista sp. X Genista monspessulana Broom [67538]		Species or species habitat may occur within area
Lantana camara Lantana, Common Lantana, Kamara Lantana, Large-leaf Lantana, Pink Flowered Lantana, Red Flowered Lantana, Red-Flowered Sage, White Sage, Wild Sage [10892]		Species or species habitat likely to occur within area
Pinus radiata Radiata Pine Monterey Pine, Insignis Pine, Wilding Pine [20780]		Species or species habitat may occur within area
Protasparagus densiflorus Asparagus Fern, Plume Asparagus [5015]		Species or species habitat likely to occur within area
Rubus fruticosus aggregate Blackberry, European Blackberry [68406]		Species or species habitat likely to occur within area
Ulex europaeus Gorse, Furze [7693]		Species or species habitat likely to occur within area

Caveat

The information presented in this report has been provided by a range of data sources as acknowledged at the end of the report.

This report is designed to assist in identifying the locations of places which may be relevant in determining obligations under the Environment Protection and Biodiversity Conservation Act 1999. It holds mapped locations of World and National Heritage properties, Wetlands of International and National Importance, Commonwealth and State/Territory reserves, listed threatened, migratory and marine species and listed threatened ecological communities. Mapping of Commonwealth land is not complete at this stage. Maps have been collated from a range of sources at various resolutions.

Not all species listed under the EPBC Act have been mapped (see below) and therefore a report is a general guide only. Where available data supports mapping, the type of presence that can be determined from the data is indicated in general terms. People using this information in making a referral may need to consider the qualifications below and may need to seek and consider other information sources.

For threatened ecological communities where the distribution is well known, maps are derived from recovery plans, State vegetation maps, remote sensing imagery and other sources. Where threatened ecological community distributions are less well known, existing vegetation maps and point location data are used to produce indicative distribution maps.

Threatened, migratory and marine species distributions have been derived through a variety of methods. Where distributions are well known and if time permits, maps are derived using either thematic spatial data (i.e. vegetation, soils, geology, elevation, aspect, terrain, etc) together with point locations and described habitat; or environmental modelling (MAXENT or BIOCLIM habitat modelling) using point locations and environmental data layers.

Where very little information is available for species or large number of maps are required in a short time-frame, maps are derived either from 0.04 or 0.02 decimal degree cells; by an automated process using polygon capture techniques (static two kilometre grid cells, alpha-hull and convex hull); or captured manually or by using topographic features (national park boundaries, islands, etc). In the early stages of the distribution mapping process (1999-early 2000s) distributions were defined by degree blocks, 100K or 250K map sheets to rapidly create distribution maps. More reliable distribution mapping methods are used to update these distributions as time permits.

Only selected species covered by the following provisions of the EPBC Act have been mapped:

- migratory and
- marine

The following species and ecological communities have not been mapped and do not appear in reports produced from this database:

- threatened species listed as extinct or considered as vagrants
- some species and ecological communities that have only recently been listed
- some terrestrial species that overfly the Commonwealth marine area
- migratory species that are very widespread, vagrant, or only occur in small numbers

The following groups have been mapped, but may not cover the complete distribution of the species:

- non-threatened seabirds which have only been mapped for recorded breeding sites
- seals which have only been mapped for breeding sites near the Australian continent

Such breeding sites may be important for the protection of the Commonwealth Marine environment.

Coordinates

-35.01797 117.91827

Acknowledgements

This database has been compiled from a range of data sources. The department acknowledges the following custodians who have contributed valuable data and advice:

-Office of Environment and Heritage, New South Wales
-Department of Environment and Primary Industries, Victoria
-Department of Primary Industries, Parks, Water and Environment, Tasmania
-Department of Environment, Water and Natural Resources, South Australia
-Department of Land and Resource Management, Northern Territory
-Department of Environmental and Heritage Protection, Queensland
-Department of Parks and Wildlife, Western Australia
-Environment and Planning Directorate, ACT
-Birdlife Australia
-Australian Bird and Bat Banding Scheme
-Australian National Wildlife Collection
-Natural history museums of Australia
-Museum Victoria
-Australian Museum
-South Australian Museum
-Queensland Museum
-Online Zoological Collections of Australian Museums
-Queensland Herbarium
-National Herbarium of NSW
-Royal Botanic Gardens and National Herbarium of Victoria
-Tasmanian Herbarium
-State Herbarium of South Australia
-Northern Territory Herbarium
-Western Australian Herbarium
-Australian National Herbarium, Canberra
-University of New England
-Ocean Biogeographic Information System
-Australian Government, Department of Defence
Forestry Corporation, NSW
-Geoscience Australia
-CSIRO
-Australian Tropical Herbarium, Cairns
-eBird Australia
-Australian Government – Australian Antarctic Data Centre
-Museum and Art Gallery of the Northern Territory
-Australian Government National Environmental Science Program
-Australian Institute of Marine Science
-Reef Life Survey Australia
-American Museum of Natural History
-Queen Victoria Museum and Art Gallery, Inveresk, Tasmania
-Tasmanian Museum and Art Gallery, Hobart, Tasmania
-Other groups and individuals

The Department is extremely grateful to the many organisations and individuals who provided expert advice and information on numerous draft distributions.

Please feel free to provide feedback via the Contact Us page.

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Appendix B

Department of Aboriginal Affairs Registered Sites Search

Search Criteria

0 Registered Aboriginal Sites in Custom search area; 583361.21mE, 6123252.44mN z50 (MGA94) : 585354.61mE, 6125388.84mN z50 (MGA94)

Disclaimer

The *Aboriginal Heritage Act 1972* preserves all Aboriginal sites in Western Australia whether or not they are registered. Aboriginal sites exist that are not recorded on the Register of Aboriginal Sites, and some registered sites may no longer exist.

The information provided is made available in good faith and is predominately based on the information provided to the Department of Aboriginal Affairs by third parties. The information is provided solely on the basis that readers will be responsible for making their own assessment as to the accuracy of the information. If you find any errors or omissions in our records, including our maps, it would be appreciated if you email the details to the Department at heritageenquiries@daa.wa.gov.au and we will make every effort to rectify it as soon as possible.

South West Settlement ILUA Disclaimer

Your heritage enquiry is on land **within or adjacent to** the following Indigenous Land Use Agreement(s): Wagyl Kaip Southern Noongar People ILUA

On 8 June 2015, six identical Indigenous Land Use Agreements (ILUAs) were executed across the South West by the Western Australian Government and, respectively, the Yued, Whadjuk People, Gnaala Karla Booja, Ballardong People, South West Boojarah #2 and Wagyl Kaip & Southern Noongar groups, and the South West Aboriginal Land and Sea Council (SWALSC).

The ILUAs bind the parties (including 'the State', which encompasses all State Government Departments and certain State Government agencies) to enter into a Noongar Standard Heritage Agreement (NSHA) when conducting Aboriginal Heritage Surveys in the ILUA areas, unless they have an existing heritage agreement. It is also intended that other State agencies and instrumentalities enter into the NSHA when conducting Aboriginal Heritage Surveys in the ILUA areas. It is recommended a NSHA is entered into, and an 'Activity Notice' issued under the NSHA, if there is a risk that an activity will 'impact' (i.e. by excavating, damaging, destroying or altering in any way) an Aboriginal heritage site. The Aboriginal Heritage Due Diligence Guidelines, which are referenced by the NSHA, provide guidance on how to assess the potential risk to Aboriginal heritage.

Likewise, from 8 June 2015 the Department of Mines and Petroleum (DMP) in granting Mineral, Petroleum and related Access Authority tenures within the South West Settlement ILUA areas, will place a condition on these tenures requiring a heritage agreement or a NSHA before any rights can be exercised.

If you are a State Government Department, Agency or Instrumentality, or have a heritage condition placed on your mineral or petroleum title by DMP, you should seek advice as to the requirement to use the NSHA for your proposed activity. The full ILUA documents, maps of the ILUA areas and the NSHA template can be found at https://www.dpc.wa.gov.au/lantu/Claims/Pages/SouthWestSettlement.aspx.

Further advice can also be sought from the Department of Aboriginal Affairs (DAA) at heritageenquiries@daa.wa.gov.au.

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Coordinate Accuracy
Accuracy is shown as a code in brackets following the coordinates. Map coordinates (Latitude/Longitude and Easting/Northing) are based on the GDA 94 Datum. The Easting/Northing map grid can be across one or more zones. The zone is indicated for each Easting on the map, i.e. '500000mE:Z50' means Easting=500000, Zone=50.

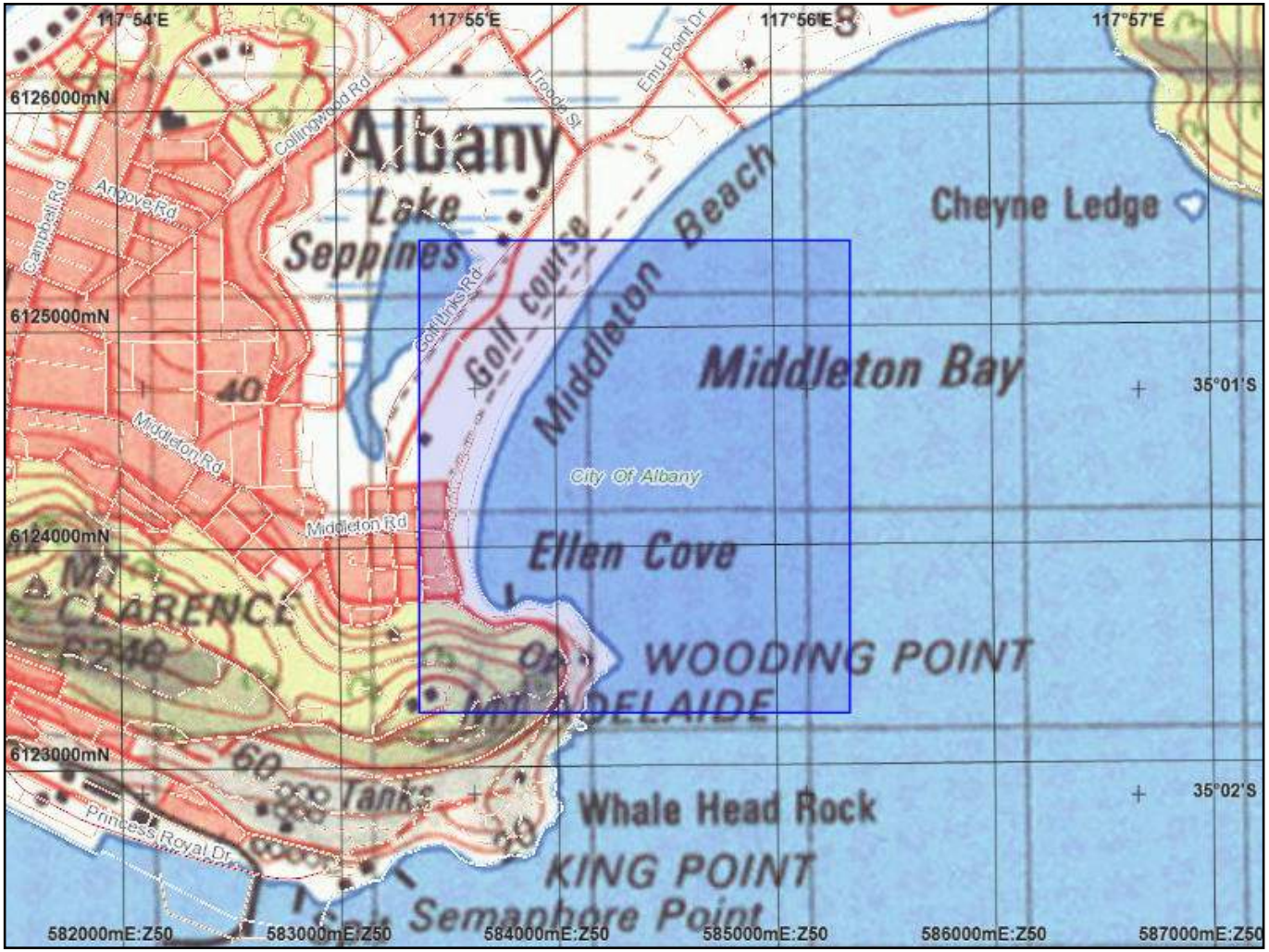
Terminology (NB that some terminology has varied over the life of the legislation)
Place ID/Site ID: This a unique ID assigned by the Department of Aboriginal Affairs to the place
Status:
- **Registered Site**: The place has been assessed as meeting Section 5 of the *Aboriginal Heritage Act 1972*
- **Other Heritage Place which includes:**
  - **Stored Data / Not a Site:** The place has been assessed as not meeting Section 5 of the *Aboriginal Heritage Act 1972*
  - **Lodged:** Information has been received in relation to the place, but an assessment has not been completed at this stage to determine if it meets Section 5 of the *Aboriginal Heritage Act 1972*

Status Reason: e.g. Exclusion - Relates to a portion of an Aboriginal site or heritage place as assessed by the Aboriginal Cultural Material Committee (ACMC). e.g. such as the land subject to a section 18 notice.
Origin Place ID: Used in conjuction with Status Reason to indicate which Registered Site this Place originates from.
Access and Restrictions:
- **File Restricted = No:** Availability of information (other than boundary) that the Department of Aboriginal Affairs holds in relation to the place is not restricted in any way.
- **File Restricted = Yes:** Some of the information that the Department of Aboriginal Affairs holds in relation to the place is restricted if it is considered culturally sensitive. This information will only be made available if the Department of Aboriginal Affairs receives written approval from the informants who provided the information. Download the Request to Access Restricted Information letter and form.
- **Boundary Restricted = No:** place location is shown as accurately as the information lodged with the Registrar allows.
- **Boundary Restricted = Yes:** To preserve confidentiality the exact location and extent of the place is not displayed on the map. However, the shaded region (generally with an area of at least 4km^2) provides a general indication of where the place is located. If you are a landowner and wish to find out more about the exact location of the place, please contact DAA.
- **Restrictions:**
  - **No Restrictions:** Anyone can view the information.
  - **Male Access Only:** Only males can view restricted information.
  - **Female Access Only:** Only females can view restricted information

Legacy ID: This is the former unique number that the former Department of Aboriginal Sites assigned to the place. This has been replaced by the Place ID / Site ID.

List of Registered Aboriginal Sites with Map

No Results



Legend

Selected Heritage Sites

- Registered Sites
- Aboriginal Community Occupied
- Aboriginal Community Unoccupied
- Town
- Search Area

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Geothermal Application, Geothermal Title, Mining Tenement, Petroleum Application, Petroleum Title boundary data copyright © the State of Western Australia (DMP) (2017.1)

For further important information on using this information please see the Department of Aboriginal Affairs' Terms of Use statement at http://www.daa.wa.gov.au/Terms-Of-Use/

Appendix C

Heritage Council inHerit Report for Middleton Beach

Middleton Beach

AUTHOR Heritage Council **PLACE NUMBER** 17520

LOCATION

Albany

LOCATION DETAILS

Arising from nomination of P17771 Norfolk Pine Trees

LOCAL GOVERNMENT Albany **REGION** Great Southern

CONSTRUCTION DATE

Constructed from 1940

DEMOLITION YEAR N/A

Statutory Heritage Listings

TYPE	STATUS	DATE	DOCUMENTS
Heritage List	YES	30 Dec 1983	

Other Heritage Listings and Surveys

TYPE	STATUS	DATE	GRADING/MANAGEMENT CATEGORY
Municipal Inventory	Adopted	23 Sep 1999	Category E
RHP - To be assessed	Current	23 Mar 2007	

Child Places

17771 Norfolk Pines

Creation Date 02 Jun 2006 **Last Update** 01 Jan 2017 **Publish place record online (inHerit):** Approved

Disclaimer

This information is provided voluntarily as a public service. The information provided is made available in good faith and is derived from sources believed to be reliable and accurate. However, the information is provided solely on the basis that readers will be responsible for making their own assessment of the matters discussed herein and are advised to verify all relevant representations, statements and information.

Appendix D

City of Albany Artificial Surf Reef Feasibility Study Community Feedback Survey

City of Albany Artificial Surf Reef Feasibility Survey

Friday, February 12, 2016



728

Total Responses

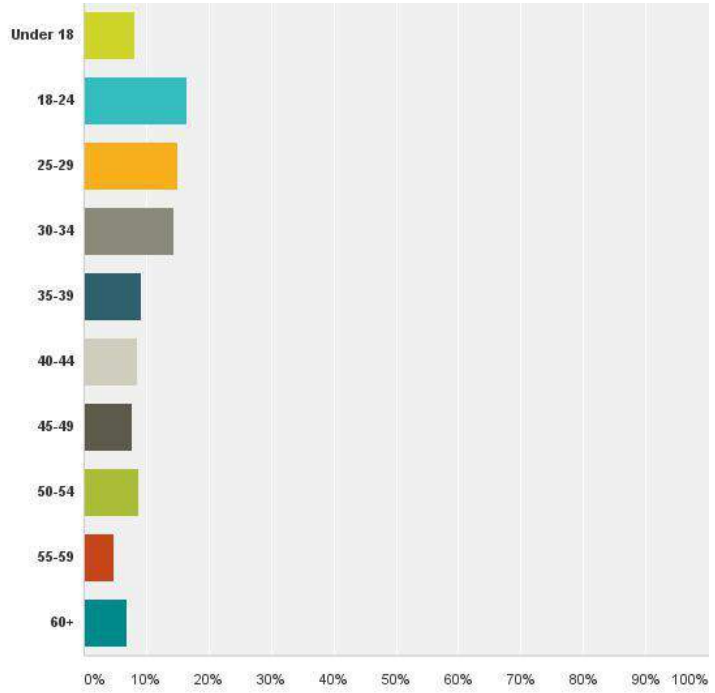
Date Created: Monday, June 15, 2015

Complete Responses: 695

Questions 1 -6 Summary

Qualitative Demographics:

- Most of the respondents were aged between 18-24 years (17%).
- The second largest group were aged 25-29 years (15%).
- The third largest group of respondents were aged between 30-34 years (14%).



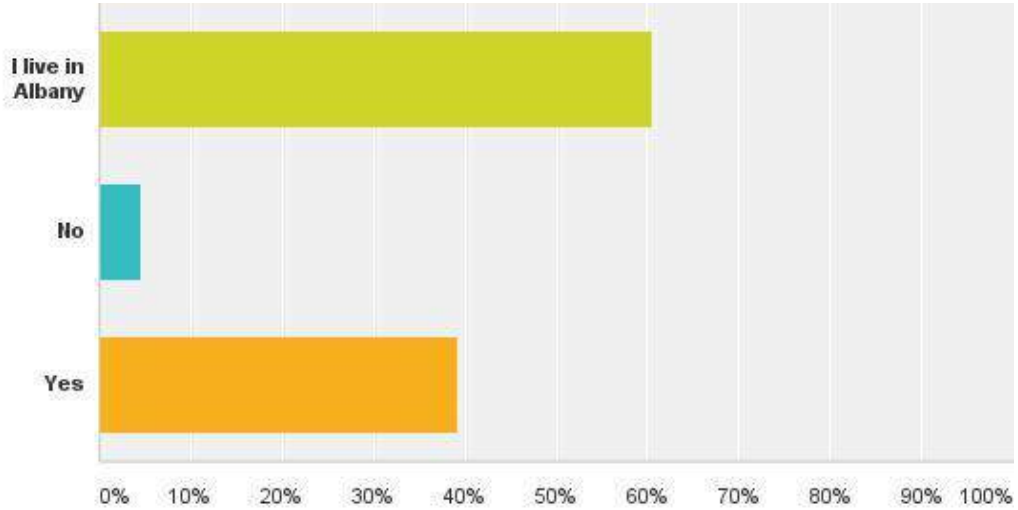
This highest response rate from those aged 18- 24 years represents 7.5% of the Albany population (source: Census data 2011). This is the third smallest group within the population. However, there has been a growth trend within this group since 2001, and traditionally this group has been the age when young people leave Albany for education and employment in Perth. The service age group is described as tertiary education and independents and identified as making up 2,507 of 33,648 in population (2011).

- The largest group of respondents lived in Albany (62%).
- The second largest group were from Perth (27%), mostly from the Northern Coastal suburbs.
- The third largest group were from Denmark WA.

This represents support from areas outside of Albany and provides opportunity for the proposal to increase visitor numbers mainly from the Northern Coastal parts of Perth. There were 19 respondents from outside WA.

Q7: If you live outside of Albany, would you visit Albany more often if the surf/wave riding conditions at Middleton Beach were improved?

Answered: 713 Skipped: 15



- The majority indicated that they would visit Albany more often (39%) if the surf/wave conditions were improved. 5% indicated that they would not visit more often.

Q8: Tell us in five words that come to mind when you think about your Middleton Beach experience today?

Answered: 710 Skipped: 18

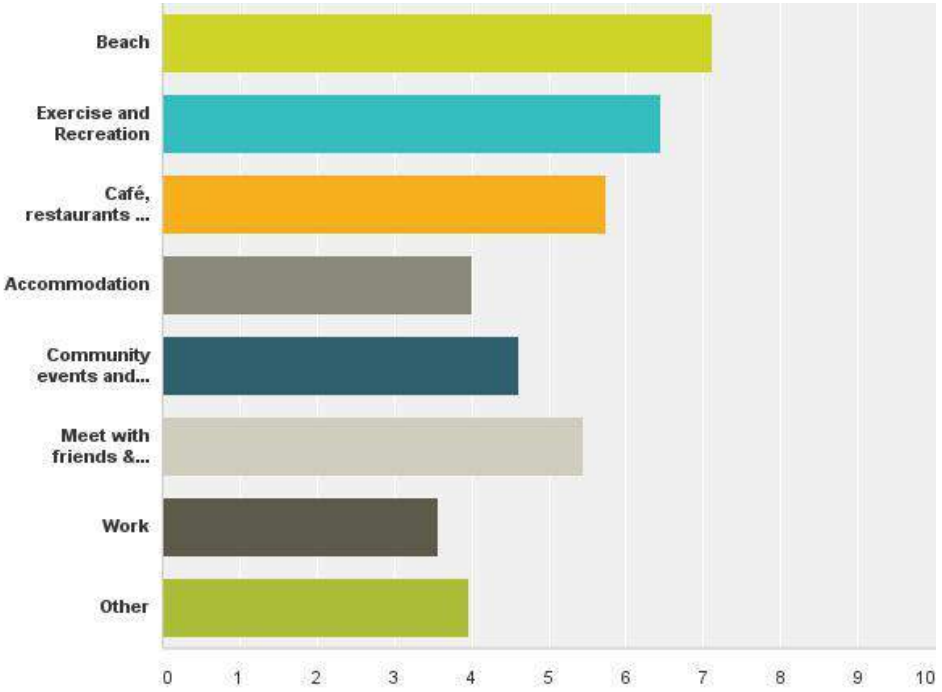
- Surfing (259)
- Beautiful (226)
- Scenery (117)

And…

- Walking, time with family, and a peaceful and relaxing experience.

Q9: What are the main things that bring you to Middleton Beach?(rank your top 5)

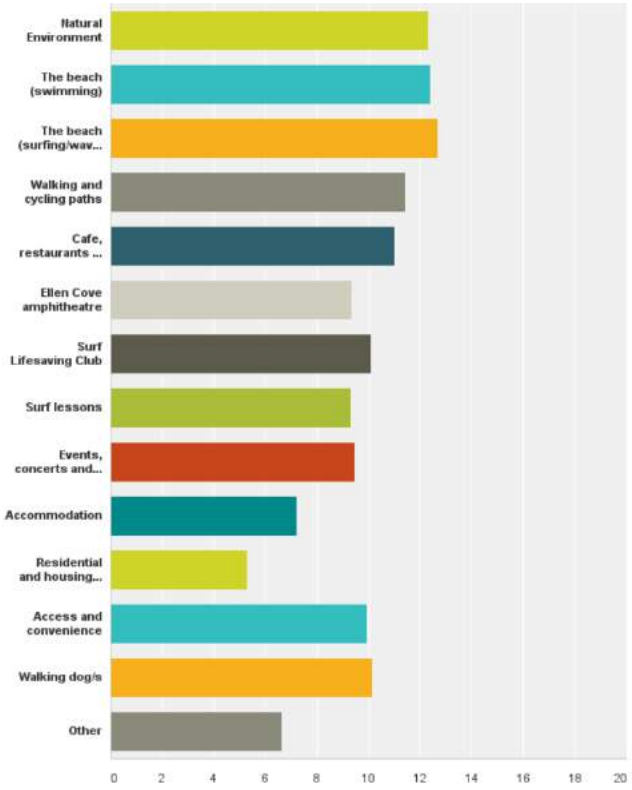
Answered: 710 Skipped: 18



- The beach (63%)
- For exercise and recreation (19%)
- And visits to cafés, restaurants and bars (8%), to meet with friends and family (6%)

Q10: We'd like to know what you think are the best things about Middleton Beach. (rank 1 to 5, 1 being most important & 5 being least important)

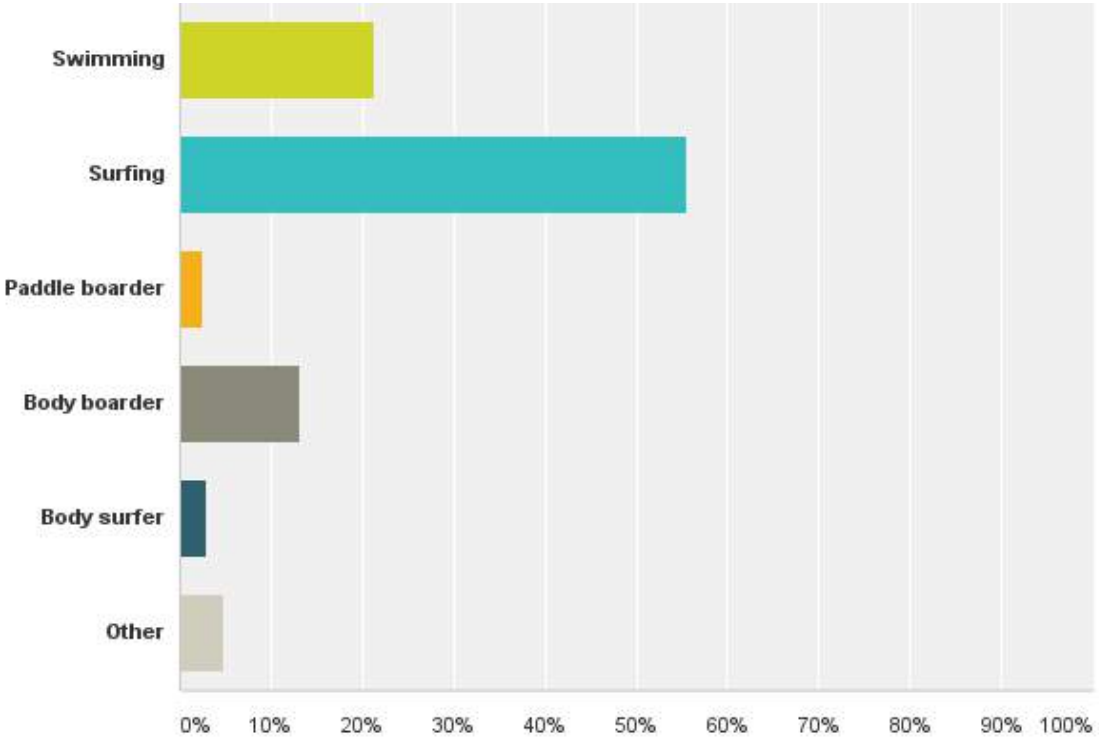
Answered: 704 Skipped: 24



- Surfing & wave riding (49%)
- Natural environment (36%)
- Swimming (36%)

Q11: What best describes your ocean use?

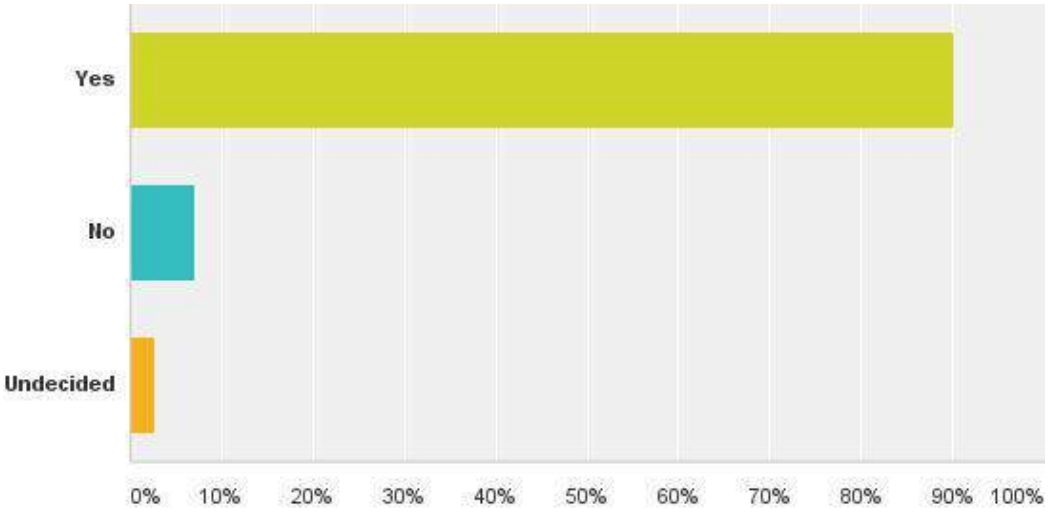
Answered: 723 Skipped: 5



- Surfing (55%)
- Swimming (21%)
- Body boarding (13%)

Q12: Do you support, in principle, the proposal to create an Artificial Surf Reef at Middleton Beach?

Answered: 725 Skipped: 3



- Majority support (90%)
- 300 qualitative comments

Q12: Qualitative Comments



IN SUPPORT:

- Potential to increase tourism (26%)
- Better surfable wave (13%)
- Less travel/more accessible (10%)
- Decrease antisocial activity
- Increase health & wellbeing
- Activate Middleton Beach

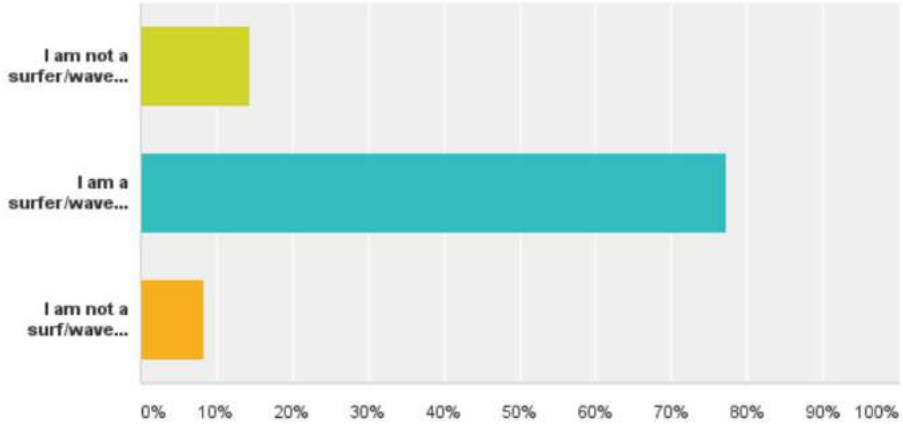
NOT IN SUPPORT:

- Negative impact to environment (13%)
- Financial impact to ratepayer (4%)



Q13: If you are a surf/wave rider at Middleton Beach, it's important we know what level of wave rider you are...

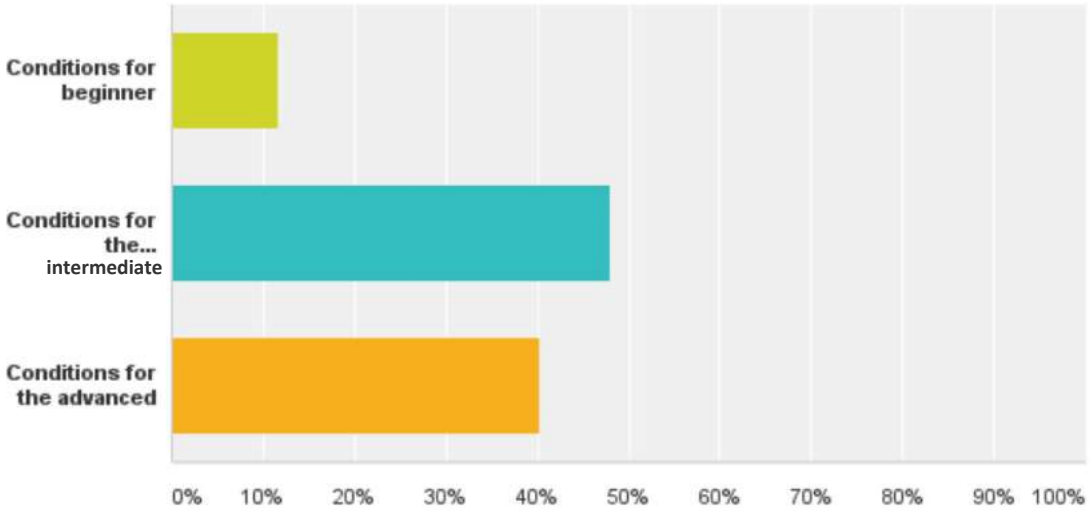
Answered: 713 Skipped: 15



77% surfer/wave riders

Q14: What level of surfing would suit your interests?

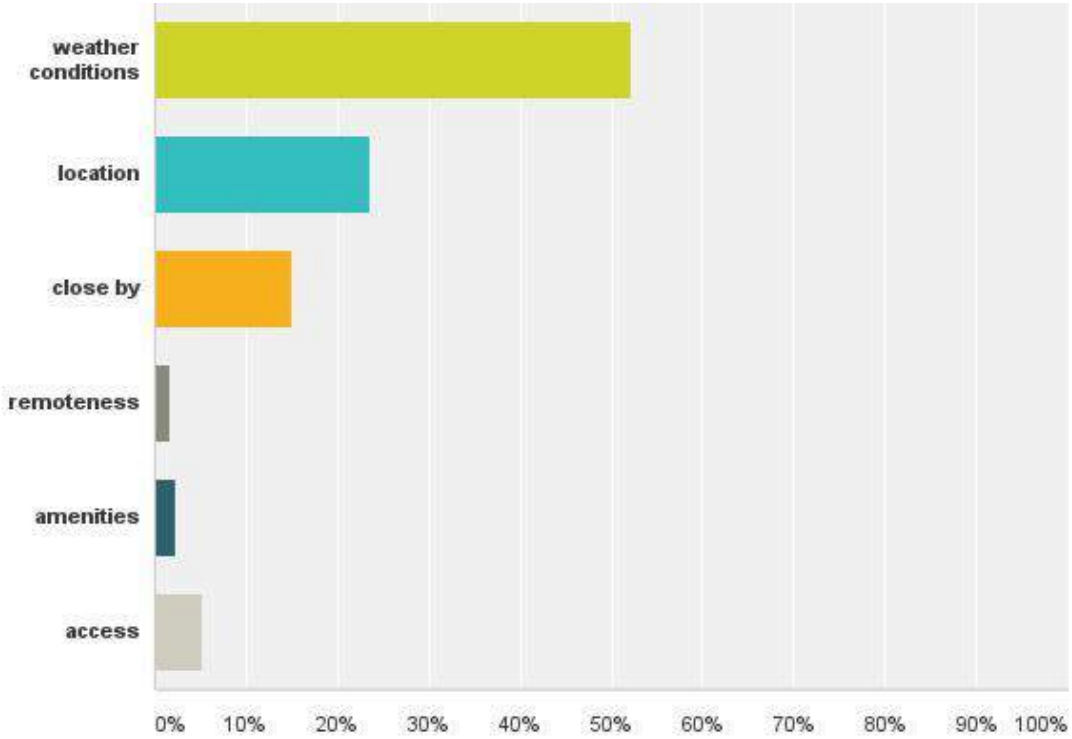
Answered: 646 Skipped: 82



47% intermediate

Q15: What conditions would influence your decision on where to go for surf/wave riding?

Answered: 674 Skipped: 54



- Weather (52%)
- Location/place (24%)
- Proximity (15%)

Answered: 674 Skipped: 54

Q16: If the proposed Middleton Beach Artificial Surf Reef was to become feasible, do you have ideas on what other improvements would be needed to support the project at Middleton Beach?

Main themes for amenities:

- Additional parking (170)
- Additional toilets (112)
- Lookouts (84)
- Showers (58)
- Access (55)







Q16: Should the proposed Middleton Beach Artificial Surf Reef become feasible, do you have any ideas on how the community could raise the funds to make it happen?

Answered: 674 Skipped: 54

GOVERNMENT FUNDING	
Royalties for Regions	Lotterywest
National Stronger Regions Fund	Landcorp
Department of Sport & Recreation	Southern Seas Port Authority, Albany
Regional Events Scheme	Healthway
Women Leaders in Sport	Coastcare - amenities
Department of Fisheries	Our Neighbourhood Grants - youth activities
Regional Arts Fund - festivals	



Powered by

CORPORATE (naming rights, corporate responsibility, tourism campaign rights)	COMMUNITY (foundation)
Middleton Beach developers	Grass roots funding through local clubs/schools hosting events and fundraising campaigns
Large scale businesses – surf industry operators, Red Bull, Richard Branson, mining companies, McDonalds	Seek charity support through organisations such as Philanthropy Australia, Albany Community Foundation, and individuals
Local business - tourism operators, retailers, restaurants, café owners	Use online platforms to attract donations – Crowdfunding.com, Kickstarter, Ozcrowd, Pozible, Indiegogo, Crowdsource, Gofund4me




CREATIVE COMMUNITIES	
International pro surf competition/event Beach Festival - DJ Beach Party, market day, pro surf workshops/lessons Community Concert - local and national musicians Swimming Competition - Iron man event Weekend beach activities- car park carwash, bake sale, sausage sizzle and markets Board riders competition Raffles Quiz nights	Community Beach Ball/Cocktail event Beach Olympics/carnival Beach Film Nights Auction FunRun Music festival Volunteer program Tourism campaign to support fundraising – “A drop in the Ocean” Girls go Surfing Day Small levy Night Surf with pontoon lights

Q18: Share with us your one, big idea that will make the proposed Middleton Beach Artificial Surf Reef succeed and become one of the South Coast's best surfing destinations.

WHAT DOES SUCCESS LOOK LIKE?



WAVE
- ✓ Quality
- ✓ Consistency

PLANNING
- ✓ Strategic thinkers
- ✓ Marketing
- ✓ Sustainability/Environment
- ✓ Education
- ✓ Business case models
- ✓ Fundraising
- ✓ Advocacy

PROMOTION
- ✓ High profile events
- ✓ Famous surfer dudes
- ✓ Partner with arts/culture – festivals
- ✓ Technology
- ✓ Social Media
- ✓ Exposure/web

Submissions:

- 4 submissions - 3 from individuals and 1 from a community group.

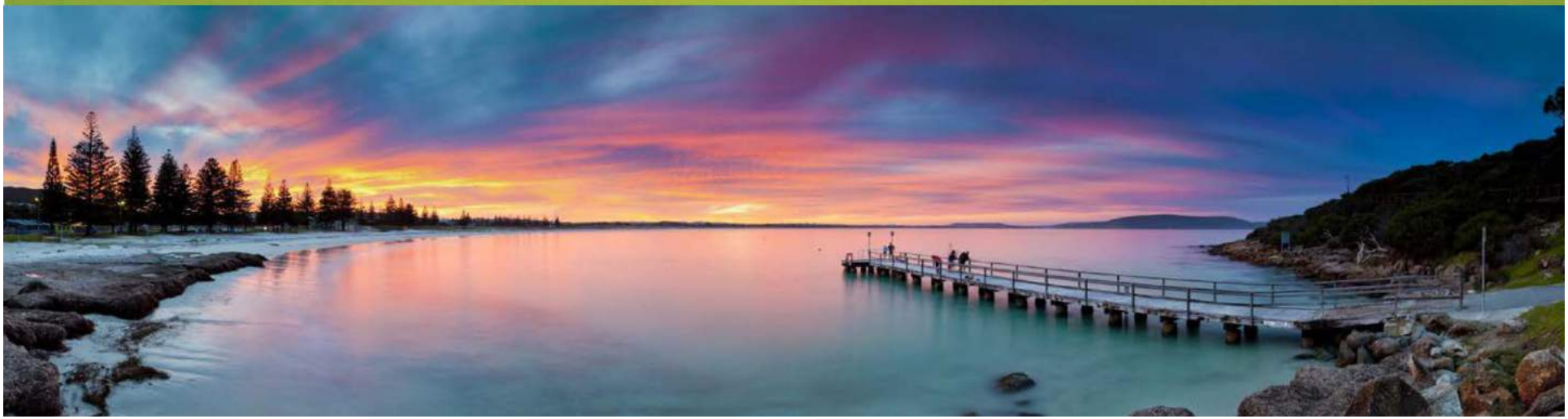
  Of the four submissions – one was supportive and three were not supportive.

  The main concerns in the objections was the perceived negative environmental impact, concern for a negative impact on marine life and the financial cost.

Summary:

- Has support from surfing community
- Challenges – environmental & financial
- Meeting community expectations (yes or no. It has been talked about for over a decade)
- Other comments included expanding the working group to plan and investigate a business case – including advocating for funding, demonstrating economic and social development benefits to the region.

DISCUSSIONS



Appendix E

Albany Artificial Surfing Reef Preliminary Shoreline Modelling Report

Royal HaskoningDHV
Enhancing Society Together

CITY OF Albany

REPORT

Albany Artificial Surfing Reef

Preliminary Shoreline Modelling

Client: City of Albany

Reference: M&APA1805R001F0.0
Revision: 0.0/Final
Date: 27 June 2018



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Document title:	Albany Artificial Surfing Reef
Document short title:	AASR - Shoreline Modelling
Reference:	M&APA1805R001F0.0
Revision:	0.0/Final
Date:	27 June 2018
Project name:	Albany Shoreline Modelling
Project number:	PA1805
Author(s):	James Lewis, Evan Watterson
Drafted by:	James Lewis
Checked by:	Evan Watterson
Date / initials:	**27/06/2018 EW**
Approved by:	Evan Watterson
Date / initials:	**27/06/2018 EW**

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Appendices

Appendix A – Joint frequency Analysis of RHDHV AWAC Site from 38 year hindcast

Appendix B – benthic fauna habitats (BMT, 2017)

1 Introduction

The Environmental Protection Agency (EPA) on advice from the Western Australian Department of Transport (DoT) needs to assess short and long-term coastal response to a proposed Artificial Surfing Reef at Middleton Beach, Albany. The structure that has been proposed is detailed in the Albany Artificial Surfing Reef (AASR) Feasibility Report (RHDHV, 2015). Royal HaskoningDHV (RHDHV) has been engaged by the City of Albany (the City) to undertake initial numerical modelling to assess possible impacts to coastal processes at Middleton Beach.

1.1 Project Background

The City commenced the AASR Project with the aim to provide enhanced recreational surfing amenity at Middleton Beach. This enhanced amenity was to be achieved through the construction of an Artificial Surfing Reef (ASR). The project seeks to create a surfable wave which maximises available swell conditions and is central to Albany, driving benefits in tourism, economic development and retention of Albany's younger age demographic. The structure was to complement the existing level of coastal protection afforded by Middleton Beach.

The subsequent Feasibility Report (RHDHV, 2015) produced a recommended Concept Design (Option B) which was accepted by the City. The following recommendations were made by the Advisory Steering Group (ASG) to be investigated further into future design stages:

- the structure is to be scalable depending on project funding availability;
- a left breaking wave was preferred for visual amenity and to meet safety concerns;
- moving the structure closer to the shore for surfer accessibility; and
- Placing the structure approximately 200m to the south to avoid any potential seagrass impact areas.

The City has commenced the environmental approvals process with investigations being undertaken by BMT Western Australia (BMT, 2017). The document detailed a 'footprint' of possible ASR configurations and layouts offshore of Middleton Beach based on the recommendations of the Feasibility Report and the considerations put forward by the ASG. The footprint and reef layouts can be seen in Figure 1, benthic fauna habitat map can be seen in Appendix B for reference.

Concurrently, a metocean data collection campaign has been undertaken since 2013 and an extensive stakeholder consultation process followed, which showed a 90% positive response to the project from residents complementing the business case for the AASR project. In addition, the continuation of an examination of an appropriate funding mechanism and approvals process has taken place prior to the anticipated Detailed Design and Construction Phases.

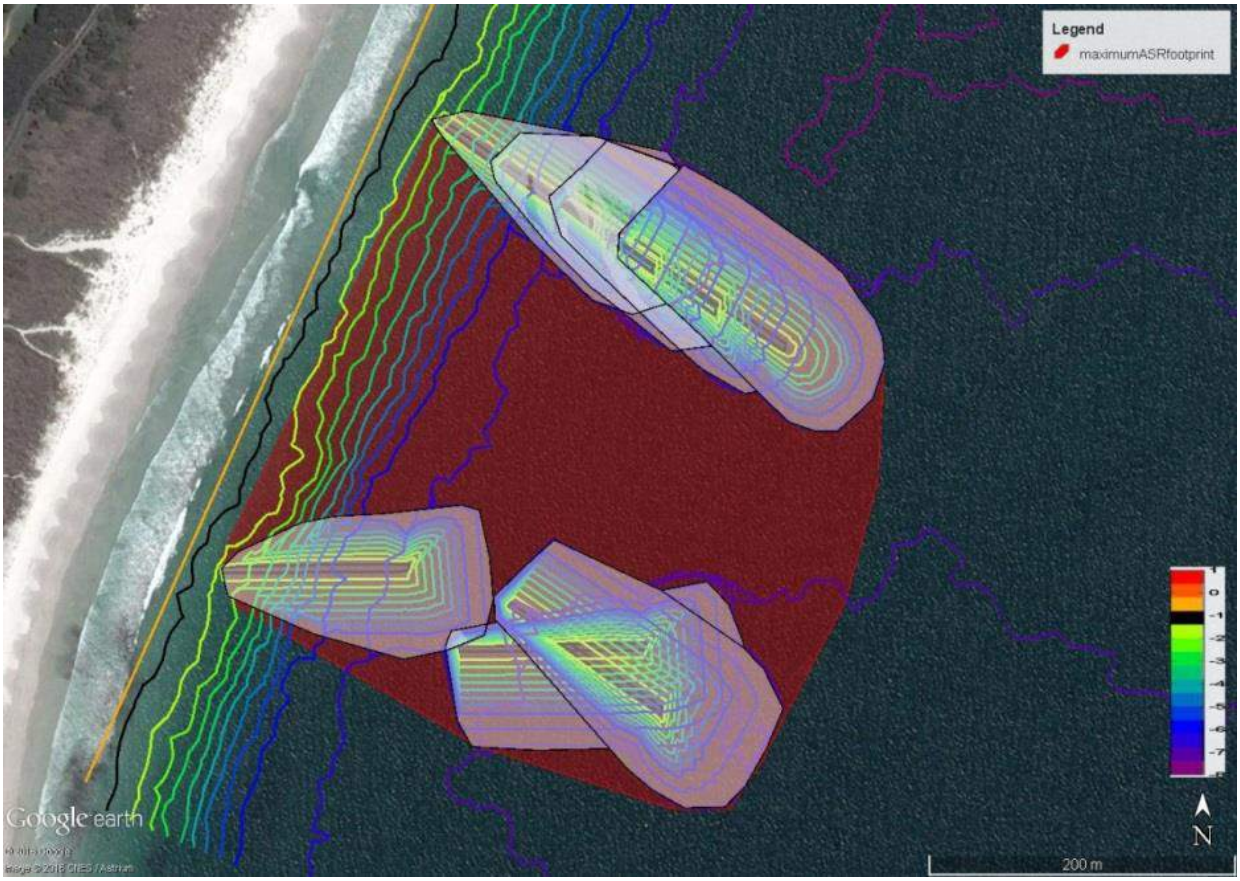


Figure 1 Maximum spatial footprint encompassing all preliminary ASR design iterations Source: RHDHV (left)

1.2 Project Objectives

The objective of this report is to present initial findings, based on numerical modelling, on the possible impacts on the overall coastal processes of Middleton Beach due to the introduction of the proposed AASR structure. This scope of works is intended to support the approvals process and can be seen as a preliminary step to better inform the forward modelling works (both physical and numerical) involved with the Detailed Design Phase.

1.3 Project Scope

The scope has been broken down into the following four (4) tasks:

1. Focussed literature review of similar ASR projects and empirical formulae to ascertain possible coastal response.
2. Numerical wave and current pattern modelling of the proposed ASR.
3. Based on empirical formulations determine the envelope within which the actual shoreline response of the AASR will lie.
4. Numerical sediment transport modelling of beach profiles in the vicinity of the proposed submerged structure.

The methodology and result for each of the above tasks is presented in the following section of the report.

2 Methodology and Results

2.1 Site Conditions

This section summarises the metocean setting of the study site utilising recently acquired wave data as well as hindcast modelling results. The recommended AASR layout (RHDHV, 2015) and material specification is then presented for reference.

The recently completed Emu Point to Middleton Beach Coastal Adaptation Study (RHDHV, 2017) incorporated a regional spectral wave and hydrodynamic model driven at its offshore boundary by NOAA WWIII hindcast wave data from 1979 - 2016. The model was also forced by hindcast coastal winds as well as predicted tide and was calibrated to long-term wave and current data recorded at Emu Point (maintained by DoT: 2013 - 2017) and offshore of Middleton Beach (maintained by RHDHV: 2015 – 2017, see Figure 2). Details of the calibration are provided in RHDHV, 2017.The annual as well as seasonal wave climate statistics for the Middleton Beach location taken from the hindcast model can be seen in Table 1.



Figure 2 Approximate RHDHV AWAC location (image sourced from seagrass map: G. Bastyan).

Table 1 Average as well as seasonal wave statistics calculated for the 38 year hindcast model extracted at the RHDHV AWAC location

Parameter	Statistic	Annual	Winter	Autumn	Summer	Spring
Hs(m)	Average	0.56	0.52	0.57	0.61	0.54
	20%ile	0.33	0.33	0.34	0.32	0.32
	90%ile	0.96	0.83	0.96	1.08	0.93
	Max	3.07	3.00	3.03	3.07	2.64
Tp (s)	Average	12.7	13.9	12.9	11.1	12.7
	20%ile	11.3	12.5	11.9	7.8	11.4
	90%ile	15.5	16.3	15.5	14.7	15.5
	% of Time Sea (Tp<8s)	9%	2%	8%	20%	9%
	% of Time Swell (Tp>8s)	91%	98%	92%	80%	91%
Peak Wave Direction (˚N)	Weighted Average	119.2	120.4	119.2	118.1	119.2
	Average	121.0	121.2	120.6	120.7	121.3
	STD	4.7	4.1	4.1	5.0	4.0

**Further information on the deployment and metocean conditions of Middleton Beach can be seen in (RHDHV, 2017).*

It can be seen that the ASR site is dominated by low energy, long period swell waves with a seasonal increase in the percentage of sea state in the summer to the winter months (from 2%–20% Tp >8sec). The waves are unidirectional (~121°) with a very small standard deviation in the peak wave direction of between 4° in the winter and 5° in the summer. The slight increase in the variability in wave direction is due to the locally generated sea state in the summer (south south-east (SSE)) coming from a different direction to the dominant swell direction (south-west (SW)).

The weighted average wave directions were calculated based upon the wave energy of each reading, as follows:

$$Dp_{weighted} = (H_s^2 \times T_p \times D_p) / \text{sum}\ (H_s^2 \times T_p)$$

As the wave climate is unidirectional this is considered a suitable method to describe the long-term wave direction.

2.2 AASR Design

The recommended AASR design brought forward from the Feasibility Report was Concept Design Option B; a granite rock structure offshore of 'Surfers' at Middleton Beach, orientated 125°N, with a 120m long crest length with its deepest point (offshore toe) in approximately -7m depth, the total structure volume is approximately 52,000 m^3. A contour map of the structure and initial placement location can be seen in Figure 2.

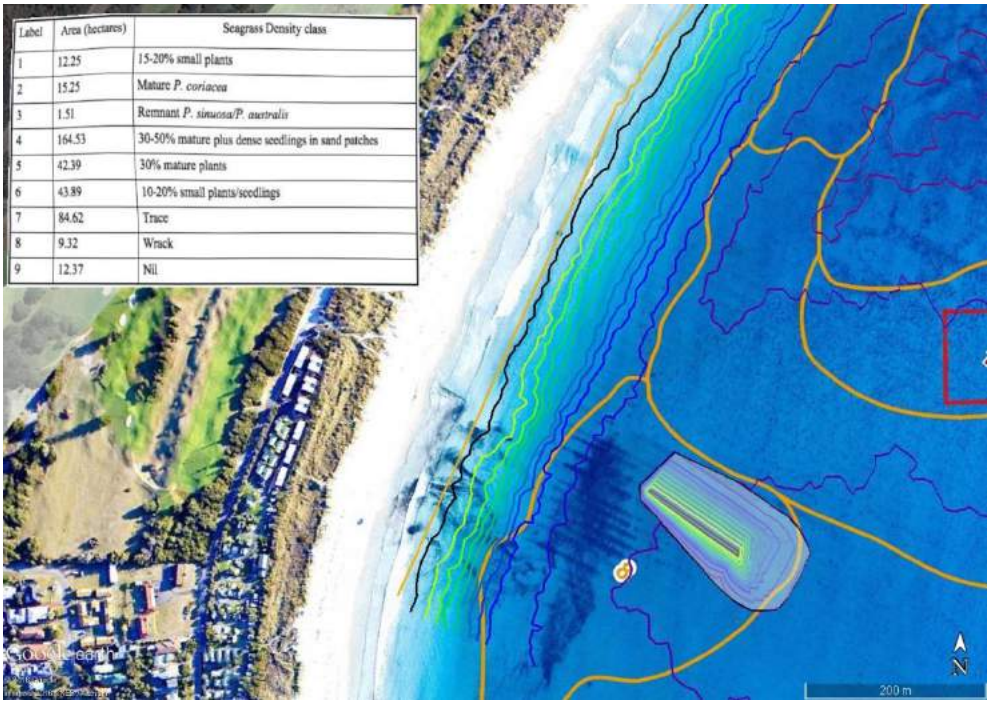


Figure 3 AASR Concept Design option B approximate contour map atop seagrass distribution map (source seagrass map: G. Bastyan).

2.3 Literature Review

The following subsection begins with a directed investigation into artificial reef projects and their documented effect on local coastal processes. It then follows with a summary of empirical formulation derived from numerical and physical modelling used to quantify coastal and shoreline response to nearshore artificial reefs.

Past Projects

In order to compare similar structures, it is important to highlight the difference between submerged constructed reefs (SCR); artificial reefs, multi-purpose reefs and submerged breakwaters. Blacka et al. (2013) defines each of these structures as follows:

Artificial Reef - An artificial reef structure, typically submerged during most tides, with a single intended purpose. This may include enhancing surfing amenity, enhancing marine habitat and associated amenity or for coastal protection. Other Common Names; Artificial Surfing Reef (ASR), Artificial Fishing Reef, Offshore Artificial Reef (OAR).

Multi-Purpose Reef (MPR) - An artificial reef structure, typically submerged during most tides, where the structure is intended to achieve multiple objectives. These may include erosion protection, marine habitat, recreational amenity including surfing, diving, fishing, etc. Other Common Names include: Multi-Purpose Artificial Reef and Multi-Function Artificial Reef.

Submerged Breakwater -A submerged artificial structure detached from the shoreline and intended primarily to reduce wave energy and promote the accretion of sand in its lee. A reef breakwater is a specific subcategory of submerged breakwater where the armouring is a homogeneous rubble grading which is designed to reshape under wave attack. Other common names; low-crested breakwater, low-crested structure, submerged rubble mound, submerged detached breakwater.

Based upon these definitions, the AASR can be defined as an Artificial Surfing Reef (ASR) as the structure is being designed for the primary function of surf amenity. Coastal protection has not been listed as a project objective, although it is stated that the reef will have no net impact on shoreline response at the site over the long-term.

2.3.1 Similar ASR Projects

The following is a targeted review of similar ASR projects that have been constructed, their linkages to the AASR, subsequent shoreline response and effect on local coastal processes. These include:

- Cables Station Reef
- Borth Multi-Purpose-Reef

The material specification detailed in the AASR Feasibility Report (RHDHV, 2015) recommended that the most suitable construction material be that of local rock (granite) boulders. Two cases of submerged rock structures with an objective to either provide shoreline protection or to enhance surfing amenity (or both) have been built to date the Cables Station Reef (Perth, WA) and Borth Multi-Purpose-Reef (Borth, UK).

Cables Station Reef

This is geographically the closest constructed ASR to the proposed site. Cables Station ASR was constructed utilising granite rocks and was completed around mid-1999. Cable Station ASR is discussed in detail in Bancroft (1999), Pattiaratchi (1999), and Pattiaratchi (2003). The information presented in these references has been used to compile this summary.

The site is situated on the northern end of Leighton Beach between the suburbs of Mosman Park and North Fremantle. It is approximately 275m offshore and can be seen in Figure 3. The final reef design was V-shaped in planform, with a longshore length of approximately 140m, cross shore width of 70m, and crest submergence of 1 - 2.5m below MSL (see the insert of Figure 3).



Figure 4 Cable Station Reef site, approximately 3km south of Cottesloe Beach, Perth (left). Final reef design (inset).

Note: The predominantly stable, rocky foreshore can be seen in this figure.

Metocean conditions vary somewhat from that of Middleton Beach, with a very distinct seasonal shift as seen in the analysis of the nearshore Cottesloe Waverider Buoy (managed by DoT), Table 2. The Cottesloe Waverider Buoy is located approximately 6km to the North West of Cables Station Reef. It can be seen there is a clear seasonal difference in wave heights, with a greater percentage of short-period wind waves experienced in the summer months whereas during winter the waves are comparable to that of the AASR site (although larger) with average wave height and period of 1.11m and 12.9sec respectively.

Table 2 Average as well as seasonal wave statistics calculated for the 9years of non-directional (1999-2008) and 8 years of directional (2009-2017) wave data recorded at the DoT waverider buoy offshore of Cottesloe Beach

Parameter	Statistic	Average	Winter	Autumn	Summer	Spring
Hs(m)	Average	0.95	1.11	0.84	0.82	1.04
	20%ile	0.62	0.71	0.57	0.58	0.68
	90%ile	1.51	1.83	1.28	1.19	1.64
	Max	3.56	3.35	3.13	2.54	3.56
Tp (s)	Average	12.2	12.9	12.6	10.9	12.4
	20%ile	10.5	11.1	11.1	7.1	11.1
	90%ile	15.4	16.7	15.4	14.3	15.4
	% of Time Sea (Tp<8s)	13%	12%	9%	22%	11%
	% of Time Swell (Tp>8s)	87%	88%	91%	78%	89%
Peak Wave Direction (°N)	Weighted Average	257.7	260.2	256.1	251.7	257.5
	Average	254.6	257.16	253.76	251.8	255.3
	STD	18.6	19.1	19.1	18.4	17.2

In contrast to the geological setting of Middleton Beach, the local geology of Cables consists mainly of Holocene sands overlying Pleistocene Tamala Limestone, which rests on older sandstone, siltstone, claystone and shales. The Tamala Limestone is calcarenite and forms small rocky headlands and nearshore reef platforms (Searle and Semeniuk 1985; Sanderson and Eliot 1999, and BMT WA 2015).

Aerial photography analysis was undertaken by Tingay (1999) prior to the construction of the Cable Station Reef. The analysis identified a general trend of accretion from 1965 to 1998 for the majority of Port and Leighton beaches (except the southern area of Port Beach). Subsequent studies have shown that the northern part of Leighton Beach (Cables Station location) is subject to seasonal longshore sediment transport rates in the same order as that found for the AASR site (i.e. 'Surfers' location, Middleton Beach) in RHDHV (2017) of +/-10,000m^3/yr. This seasonal variability has an effect on the shoreline width in the lee of Cables Station Reef due to the geomorphology of the beach, the sediment transport regime as detailed in DPU (2004) and seen in Figure 5.

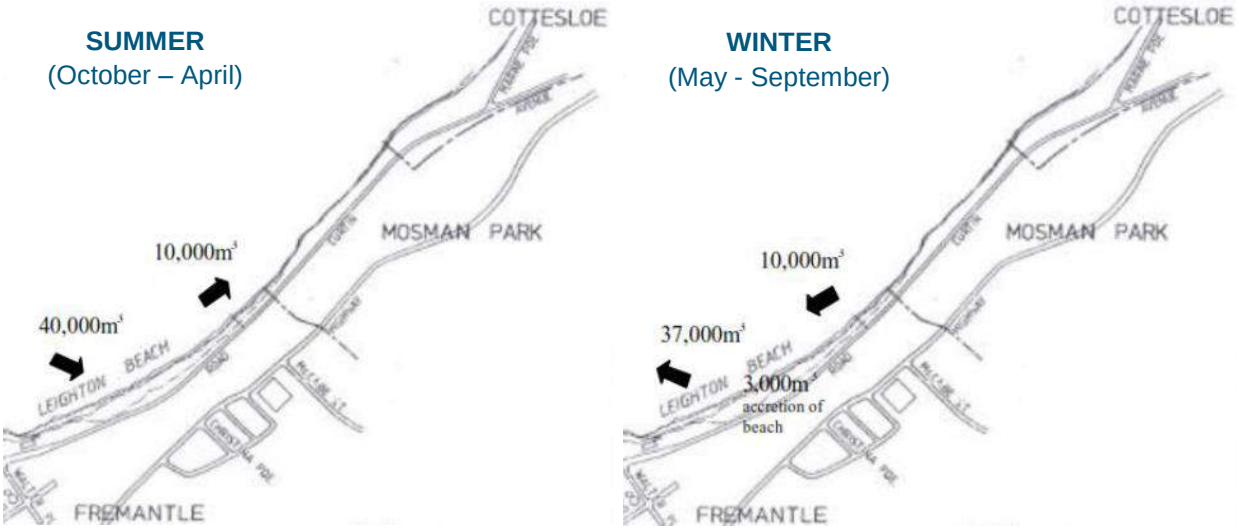


Figure 5 Conceptual contemporary model of sediment transport at Leighton Beach (DPI 2004)

The Mosman Beach Management Plan, prepared by the Town of Mosman Park (ToMP) states *"The soils are typically fine to coarse leached sands."* Beach sand *"overlays the submarine Tamala limestone shelf known as the Cottesloe Fringing bank to a depth of 1.5m, however in winter much of the shelf is exposed."* (ToMP 2003). Further analysis of hydrographic survey and historical photogrammetry data was undertaken predominantly at Port Beach to the south following the extension of Rous Head.
The investigations were undertaken by M.P. Rogers & Associates initially from 2004 to 2009 (MPRA, 2010) and again from 2009 to 2013 (MPRA, 2014). Analysis of the data further reiterated the seasonal variability of the sediment cell with northward movement of sediments during summer and southward movement of sediments during winter storms.

The key findings of the study relating to Port and Leighton Beaches were:

- The movement of sediment into offshore sand banks during winter storms.
- Subsequently returns to the beach face during summer swell.
- Sediment moves north along the cell during summer under the prevailing sea breeze events. At the end of summer there is a relatively wide beach at the northern end of the study area (Leighton Beach, in vicinity of Cables).
- At the end of winter this beach is very narrow with rocks exposed along the shoreline.
- Movement of sediment south during north-westerly winter storms. The width of the beach at the southern end of Port Beach is greater following the winter months when sand is transported from the north to the south during storm events.

The reports provide no estimate of storm induced or long-term erosion or accretion of the shoreline; the investigation is primarily concerned with bathymetric changes offshore. The report also makes no mention of response of the shoreline in the direct vicinity of the reef or any noticeable change (in comparison to adjacent areas) due to the presence of the reef.

Bancroft (1999) undertook simplistic beach width analysis directly following the construction of the Cables Reef in order to investigate possible deposition and erosion of sediment on the beach in its lee. Two sites were selected for measurement in the lee of the reef; north and south as well as a benchmark site 3km further north at Cottesloe Beach. The beach widths were taken from a reference point landward of the mean high water (MHW) level. Beach measurement was undertaken at approximately midday on each of the monitoring dates approximately 2 weeks apart. The width was recorded as the distance from a

landward reference point to where heavier material was being deposited by waves on the shoreline using a tape measure.

Although there are clear concerns about the validity of the methodology used for the beach width monitoring, when the results (Figure 4) are interpreted qualitatively, it can be seen that there was little change in beach width measurement over the monitoring period.

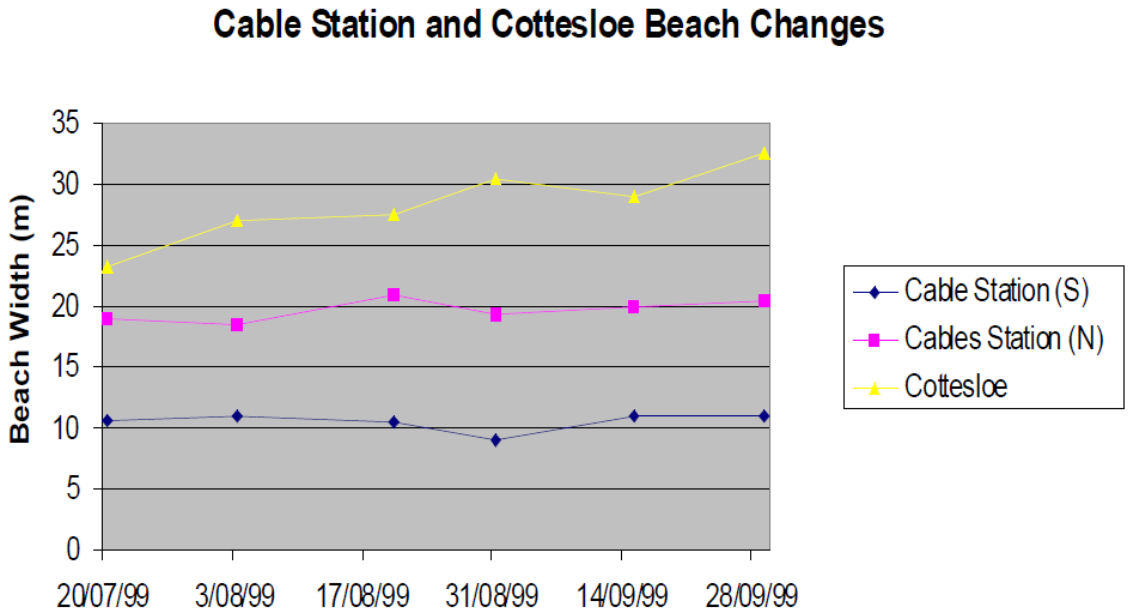


Figure 6 Beach width measurements for the three monitoring sites taken from Bancroft (1999).

The WA Department of Planning and Infrastructure (DPI) undertook detailed, successive hydrographic surveys in the areas in the lee of the Cable Station Reef for a 3 year period commencing just prior to its construction in March 1999. A total of thirteen transects were analysed from 5 surveys between 1999 and 2002, covering the seasonal erosive/accretive trends described above, the transect locations can be seen in Figure 7.

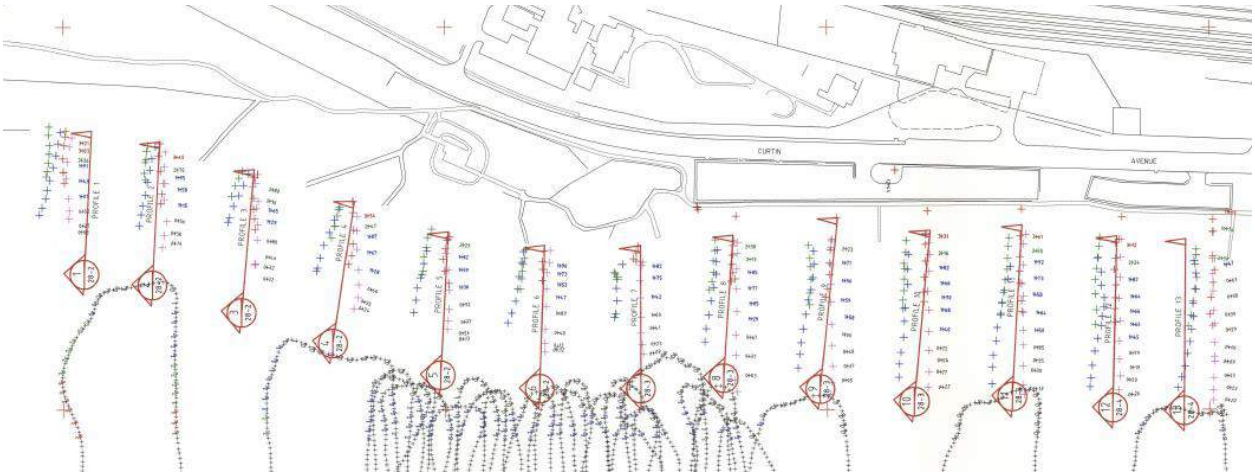


Figure 7 Layout of beach transects 1-13 undertaken by the DPI at South Cottesloe – Cables Artificial Surf Reef based on January 2002 Survey (source: DoT).

Analysis of the transects to the north, south and in the direct lee (Profiles 1, 6 and 12) can be seen in Figure 8. The analyses shows that in the 3 year post-construction of the Cables Station Reef, the shoreline and upper beach profiles to the north and south of the reef (Profiles 1 and 12) show very little change in profile volume (i.e. the observed change was within the bounds of expected seasonal variability). However, in the profile in the direct lee of the reef (Profile 6), it can be seen that there appears to be a reduction in upper beach volume (from +1 to +3m) for all four post construction surveys. From further investigation, it also appears that all the profiles in the direct lee of the Cable Station Reef (Profiles 3 - 7, Figure 9) experienced significant upper beach volume loss over the survey period when compared to profiles to the north (Profiles 1 - 2) and south (Profiles 8 - 13). The upper beach loss is at its largest in Profile 6, situated shore-normal to the centre of the offshore structure.

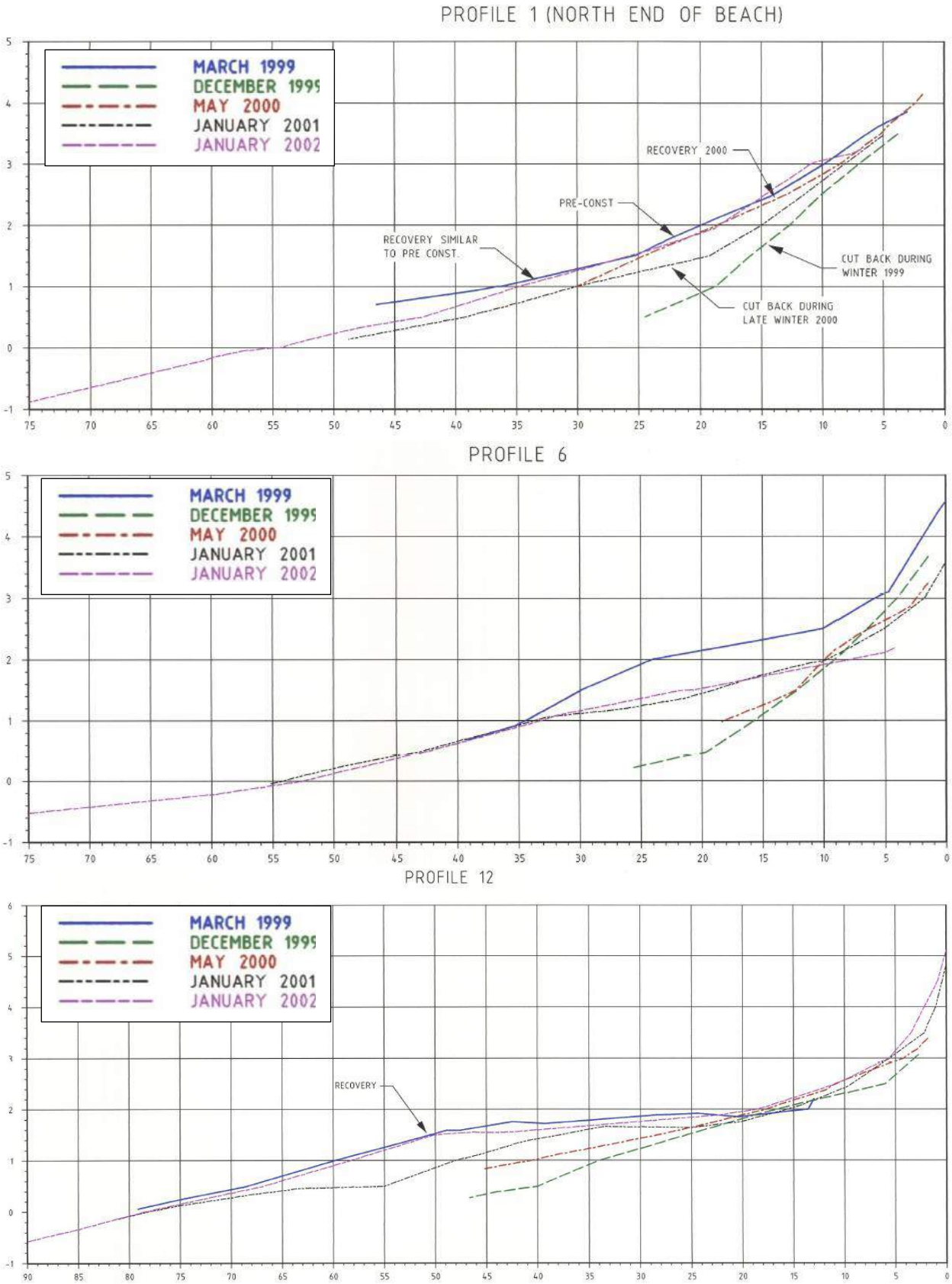


Figure 8 Analysis of Profiles 1, 6 and 12 taken from DPI beach surveys of South Cottesloe Beach from 1999-2002. (source: DoT)

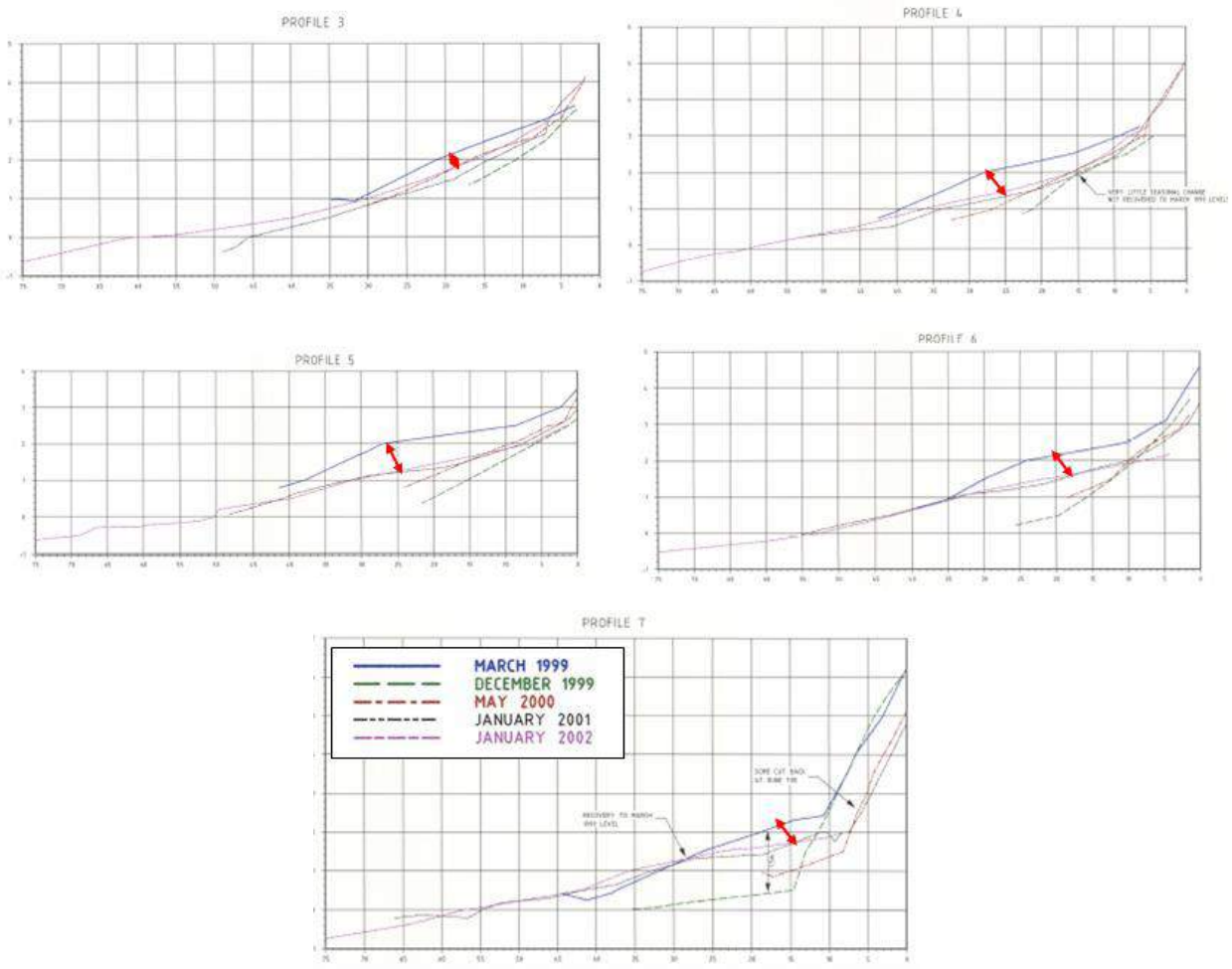


Figure 9 Analyses of pre- and post-construction beach profiles in the lee of Cables Station Reef for Profiles 3-7 taken from 1999-2002. The red arrow shows the change in upper beach profile height from pre-construction (March 1999) to 3 years following (January 2002). (Source: DoT).

Due to the relatively short timescale of the survey (3 years) it is difficult to attribute this volume reduction to the introduction of the reef, it can also be seen that several large energy wave events occurred in the period from September 1999 September 2002, with Hs exceeding the 99.5th percentile of 2.75m (see Figure 10). These events may have deposited sediments into offshore bars as described in (MPRA, 2010 and MPRXA 2013) and were not able to be redistributed onshore as seen in the recovery rates of the Profiles to the north and south of the reef where almost all profiles returned to that of the pre-construction following the 3 year monitoring period.

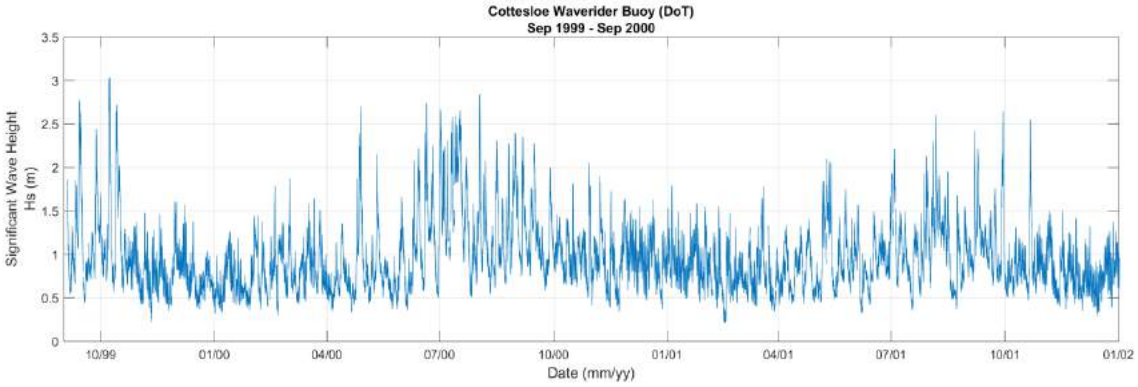


Figure 10 Significant Wave Height, Hs (m) recorded at the Cottesloe Waverider Buoy from September 1999 to September 2002.

As such, it is necessary to investigate a longer beach profile dataset to sufficiently determine the effect the Cable Station Reef has had on the natural littoral processes of the study site. The City of Mosman has only recently re-commenced beach monitoring survey in the vicinity of the reef. This survey data covers the last 2 years and has only recently become available for incorporation into this study. However key data for the years from 2002 to 2016 are not available.

Borth Multi-Purpose-Reef

The Borth Reef is a MPR built primarily for shoreline protection as part of the Borth Coastal Defence Scheme, a scheme designed to provide an upgrade to the local areas degrading coastal defences. Due to the growing tourism industry in the area, it was recommended that the submerged control structures proposed for coastal protection serve an additional function of providing recreational surfing amenity.

The Borth MPR was constructed primarily of rock and is located in the small Welsh coastal town of Borth on the Irish Sea. The site has a low energy wave climate; predominantly sea and/or short period swell (8-9 sec). It also has a large tidal range, meaning that the structure is emergent for a portion of the tidal cycle, as seen in Figure 11. The reef portion of the structure was able to be built using land-based construction due to the large tidal range, however its distance from shore is approximately 140m from MHW, the structure is fully submerged at MSL.



Figure 11 The Borth MPR emergent on a lower tide (left) and a wave breaking on high water (right), (source: eCoast).

Based upon the extensive Ceredigion Beach Profile Monitoring (RHDHV, 2014), the beaches of Borth have been seen to meet coastal protection objectives following Phase 1 and 2 of the Borth Coastal Defence Scheme as seen in Figure 12. The immediate salient response following construction of Phase 1 in 2012 is clearly evident in Figure 12 (2011 – 2012 LiDAR plate). Following a large storm event in 2014, a small pocket of erosion was seen to occur in the lee of the Borth MPR showing that nearshore scour was dominant during this extreme event. Sediment was redistributed along the shoreline during this period. Under the average wave conditions that followed, the salient formation continued to grow, proving the effectiveness of the reef to promote shoreline accretion during normal (or ambient) conditions.

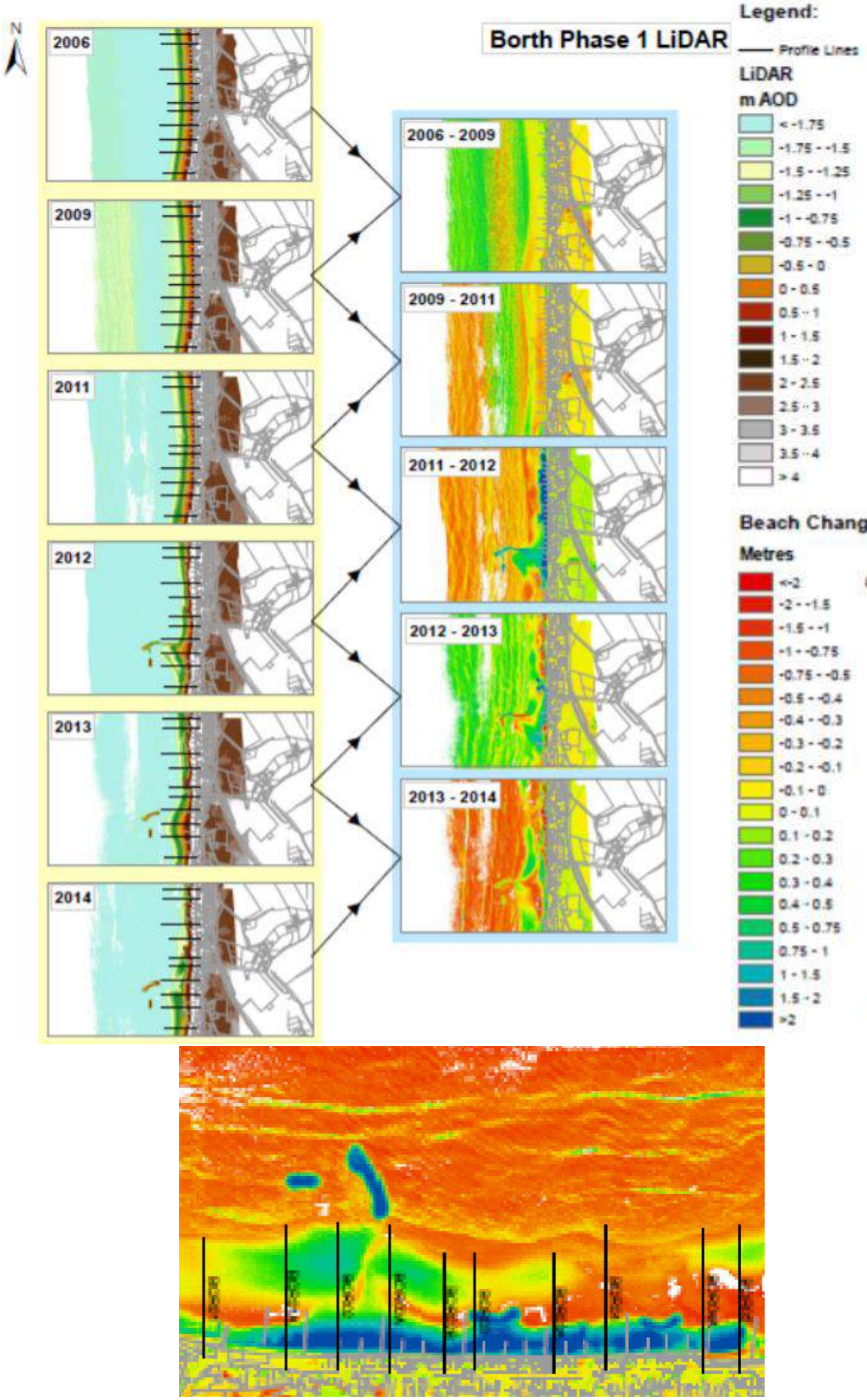


Figure 12 Results of historical survey data at Borth 2006-2014 showing the evolution the foreshore following the completion of Phase 1 and 2 of the Borth Coastal Defence Scheme. The lower figure is 2014-2006.

Translating the results of each of these case studies to the present study is very difficult due to the disparity between each in; structure size, configuration and layout as well as the large differences in site conditions, such as water level variation and incoming wave conditions. The other key consideration is that each structure design, placement location and layout is based on a specified design wave condition. In reality and most evident in the case studies mentioned here, the smaller the variation in metocean variability at each site; water level, wave height period and direction (and seabed morphology), the easier it is to anticipate possible coastal response to the introduction of a submerged constructed reef (SCR). This response has been studied extensively in both physical and numerical models where the effect of altering one parameter (metocean or structural) can be examined to determine its effect on local hydrodynamic and coastal response.

A further comparison of past artificial reef projects was undertaken in the Feasibility Report (RHDHV, 2015) and draws on a lot of the work undertaken in the review by (Blacka et al. 2013).

2.4 Empirical formulation

The following section outlines a range of empirical formulations determined through physical and numerical modelling to determine the effects of SCRs on coastal process and shoreline response.

2.4.1 Wave Transmission

As mentioned in the previous section, one of the key recommendations of the AASR Feasibility Report was that the structure be constructed from rock. Due to its long history of application, a great deal is known about the construction, resilience and design of submerged rock structures, groynes, breakwaters and jetties. There exist well-founded guidelines and texts relating to the design and construction of rock structures within the marine environment and hence there is a greater confidence with the use of rock for a submerged structure than other materials. Rock structures are permeable i.e. have voids in the structure through which water can flow. The key influence that a permeable (e.g. rock) submerged structure will have on local hydrodynamic processes will be in terms of wave, transmission wave breaking and local wave-driven currents.

When compared to impermeable structures, submerged rock structures influence the following:

- Permeable structures reduce the wave driven currents due to the increased roughness of the structure.
- Rock-built structures reduce wave transmission due to increased friction and porosity of the structure.
- Rock-built structures result in better performance in terms of wave breaking and surf quality due to increased roughness. This is discussed further below.
- The surfability of breaking waves over rock-built structures diminishes at a faster rate than for impermeable structures in less energetic wave conditions due to higher energy absorption. Also discussed below.

In regard to the two surfing amenity points raised above, HR Wallingford (2010) and ASR (2010) undertook detailed three-dimensional (3D) physical modelling of the Borth multi-purpose-reef and assessed the impacts of impermeable and permeable construction materials on wave breaking, with a focus on surfability. They found that impermeable structures (e.g. large geotextile containers or solid concrete steel structures) had a negative impact on the wave breaking quality in terms of recreational surfing amenity due to the reduced roughness. Impermeable structures only produced high-quality surfing waves at larger wave heights and higher water levels in comparison to the more permeable rock-built structures.

Furthermore, they found that the smooth, impermeable structures produced faster and more energetic breaking waves compared to the breaking observed over the rock structures. These types of breaking conditions were suited to very advanced level stand-up surfers or experienced body-boarders, limiting the number of boardriders to find amenity on the structure (see example of Boscombe ASR in Figure 13).



Figure 13 Example of intense breaking wave at Boscombe ASR one year after construction (Source: BBC).

Smooth, impermeable structures do not allow for the vertical displacement of water through the structure, resulting in enhanced set-down during wave breaking and exposing the structure (ASR (2010) and HR Wallingford (2010)). The enhanced set-down observed during wave breaking in the Borth physical modelling tests forced the water over the structure to drain off the crest resulting in currents interfering with the incoming breaking waves, causing waves to break prior to that anticipated during design (in a two-stage process or step), reducing surfing quality. This has also been observed in the prototype situation, see example of Narrowneck MPR in Figure 14.



Figure 14 Exposure of GSC crest due to wave set-down at the Narrowneck MPR (Source: Jackson et al., 2007).

WRL (2013) states that wave heights observed in the lee of a structure is a function of both two-dimensional transmission over and through the structure, and wave propagation around the structure by

refraction and diffraction processes. The gradient in wave heights in the direct vicinity of the structure drives localised currents and is the key mechanism for sediment transport adjacent to and inshore of structures. This sediment transport can result in scour at the toe of the structure (settlement) and also in varying modes of beach response in its lee.

There are several empirical methods for describing wave transmission over an SCR as discussed in the WRL report. These are: Ahrens, 1987, Van der Meer and Daemen1994, d'Angremond et al., 1996, Seabrook and Hall, 1998 and Bleck, 2006. The most comprehensive study to date, was undertaken by (Burcharth et al., 2007) following a large number of physical model tests, the following relationships were derived:

$K_t = -0.4\frac{R_c}{H_i} + \left(\frac{B}{H_i}\right)^{-0.31}\left(1 - e^{0.5\xi}\right)0.64$	Narrow Crested Structures with $B/H_{si} < 10$	K_t =transmission co-efficient H_i = incident wave height B = structure crest width ξ= Iribarren number = tanα/(H/L)$^{0.5}$ R_c = crest height above SWL
$K_t = -0.35\frac{R_c}{H_i} + 0.51\left(\frac{B}{H_i}\right)^{-0.65}\left(1 - e^{-0.41\xi}\right)$	Wide Crested Structures with $B/H_{si} > 10$	

It should be noted however that this formulation does not account for detailed two-dimensional (2D) processes such as wave refraction and diffraction around submerged structures which play a significant role for complex 3D structures and those with narrow crests.

Ideally, what is required is the establishment of a relationship between structure dimension, location, metocean conditions and expected mode of shoreline response; accretion or erosion.

2.4.2 Hydrodynamic Response

The importance of the offshore distance to the placement of an artificial reef was introduced in the Feasibility Report (RHDHV, 2015). The initial offshore placement location was determined using the formulation of Ranasinghe et al. (2006). Ranasinghe undertook numerical hydrodynamic modelling, using MIKE 21, of a range of simplified nearshore triangular structures to identify current patterns in the lee. The numerical models were then replicated with large scale laboratory mobile bed physical models to validate the results. Ranasinghe hypothesized that depending on the distance to the surf zone of the structure (which varies with wave height) either a two-cell or four-cell current circulation pattern develops in its lee.

The resulting physical modelling showed the following:

- structure located nearer to the surf zone (shore): two-cell circulation, divergent current, erosive shoreline response; and
- Structure located further from the surf zone (shore): four-cell circulation, convergent current, accretionary shoreline response.

Similar investigations undertaken by Yoshioka et al. (1993) and later Burcharth et al. (2007) identified the two-cell/four-cell systems for submerged structures. Yoshioka et al. (1993) work covered a wider range of cases (submerged long crested breakwaters and structures with very small crest length) and identified a larger range in the cell circulation patterns. Burcharth et al. (2007) found that the circulation patterns were also evident for emergent detached structures. The circulation patterns determined by each of the authors can be seen in Figure 14, Figure 15 and Figure 16 5, respectively.

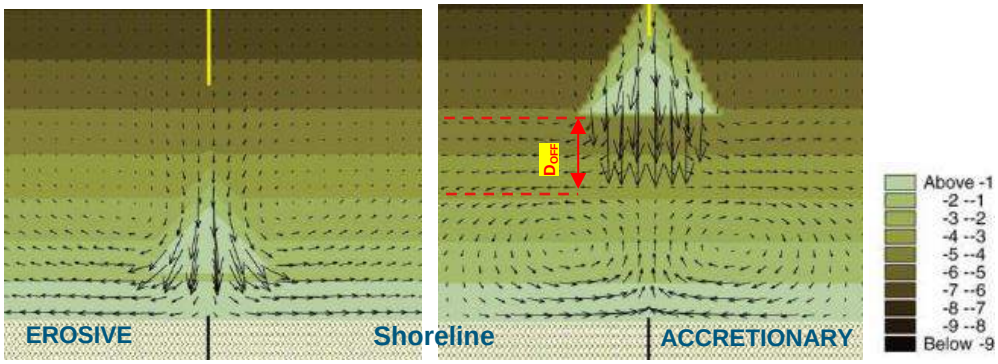


Figure 15 MIKE 21 HD output of circulation patterns around submerged structures (Ranasinghe and Turner, 2006)

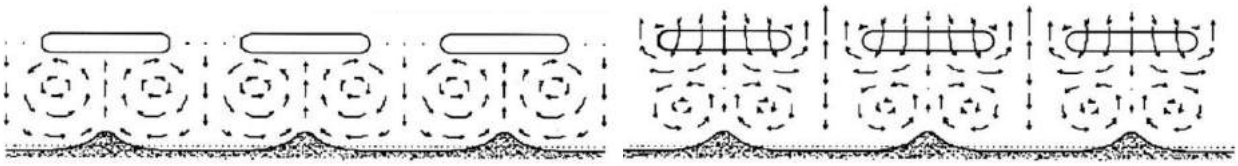


Figure 16 Representative circulation patterns around a submerged structure (Burcharth et al., 2007)

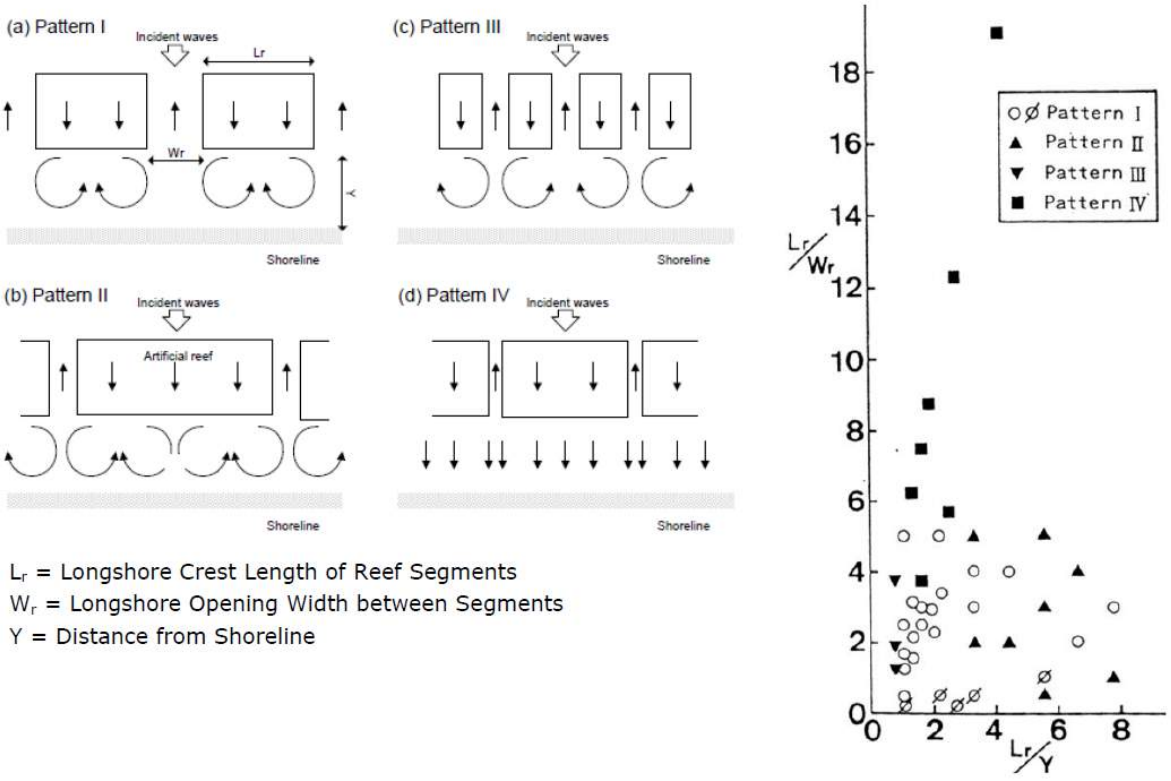


Figure 17 Current patterns and reef configuration (Yoshioka et al., 1993)

In each of their findings, it can be seen that the two-cell circulation has flows accelerating over the reef to the shoreline (creating a jet) before diverging alongshore (on the shoreline) promoting erosion. Whereas structures further offshore promote four-cell circulation patterns; here the cells closest to the shore converge in the lee of the structure promoting accretion (or a salient growth), this can be seen in Figure 15 (right). The salient growth was found to increase to some maximum value with increasing structure distance offshore. If the structure is placed further offshore from the point of maximum salient growth, the accretionary response decreases to a point where the structures influence on the shoreline is no longer measurable. The subsequent morphological response mode is detailed further in the next section.

2.4.3 Morphological Response

Andrews (1997) and Black and Andrews (2001) undertook research into possible morphological response to offshore structures based on observations of islands and natural submerged reefs and subsequent shoreline position in their lee. These investigations showed that shoreline response landward of these offshore structures resulted in either a salient growth (transient at times) or an attached tombolo, as seen in Figure 18.

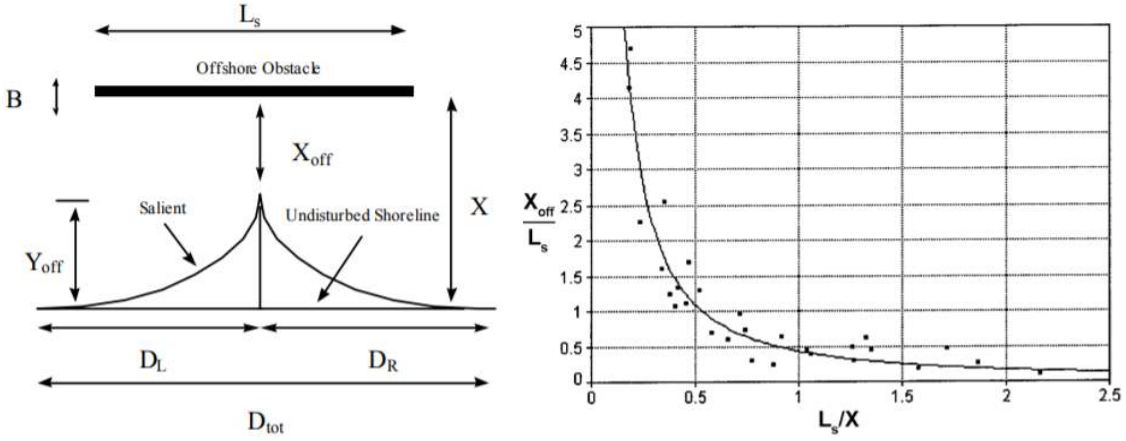


Figure 18 Example of calculated salient response for reefs (left) and field observations of salient response to approximate structure dimension. (source: Andrews (1997).

As observations of these natural structures were not made from the structures initial formation (or artificial introduction to an environment) it is difficult to determine their actual coastal response as the timeline of change would be in the order of centuries or millennia. There was also doubt as to the survey techniques used for the studies and the simplistic formulation which make no relation to incoming wave energy (and subsequent transmission).

It could also be seen that there is no mechanism in the relationship for an erosive response, meaning that all reefs, SBW, DBW or islands would result in an accretive response; in reality, observations of past SCRs will tell us otherwise.

Following this early work, Pilarczyk (2003) as well as Blacka et al (2013) summarized empirical methods of calculating shoreline response due to submerged breakwaters (SBW). Blacka's synopsis found that while there had been considerable improvement in the understanding of the mechanisms driving shoreline response for submerged reefs, or other offshore structures, the reviewed studies had significant limitations. No single study had comprehensively tested the effects of primary structural and environmental variables on quantitative shoreline response, and the shoreline response equations presented were based on approximate observations or modelling that was not calibrated or validated.

Shoreline response to a reef structure on different beach types will also differ, with littoral drift beaches experiencing:

- asymmetry in resulting shoreline alignment;
- potential downdrift erosion as a result of the formation of a salient;
- potential updrift 'groyne' effect; and
- different rates of sediment transport dependant on grain size and wave energy (or current flow).

The inherent 3D nature of these structures (especially ASR designs) and their exposure to a range of tidal and wave conditions (both energy and direction) means that a simplistic empirical formulation to ascertain shoreline response was considered impossible. Blacka (2013) stated that the findings of his investigation suggested that the available empirical techniques for assessing shoreline response are suitable only for preliminary engineering calculation and not detailed design.

With this in mind, the AASR Feasibility Study (RHDHV, 2015) sought to undertake a first pass investigation into possible layout, location and configuration and material specification for an ASR at Middleton Beach.

It was necessary to assess possible placement locations and depths of the AASR along Middleton Beach utilising an empirical formulation which included as many of the eight primary variables found to control shoreline response to an offshore breakwater (Hanson and Kraus, 1989, 1990, 1991):

1. distance offshore;
2. length of the structure;
3. transmission characteristics of the structure;
4. beach slope and/or depth at the structure (controlled in part by the sand grain size);
5. mean wave height;
6. mean wave period;
7. orientation angle of the structure; and
8. Predominant wave direction.

Ranasinghe (2010) formed an empirical relationship between the distance to shore of a submerged structure, its dimensions, the incoming wave conditions and mode of shoreline response based on both numerical and physical model testing. The numerical models were calibrated from the results of a matrix of physical model tests varying structure dimensions, distance to shore and wave height. The results formed a clear delineation between accretion and erosion modes of the leeward shore, these results can be seen in Figure 19.

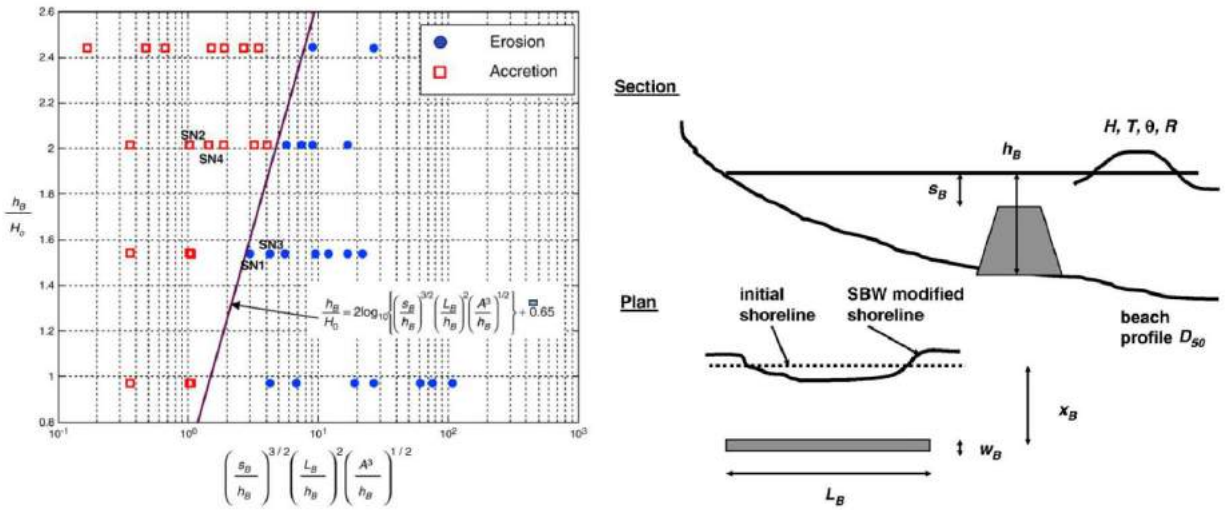


Figure 19 Mode of shoreline response to SBW based on physical model tests (left), structure dimension and model parameters (right) source: (Ranasinghe et al., 2010)

The Feasibility report (RHDHV, 2015) used the empirical relationship found in the Ranasinghe (2010) study (below) to make a first pass assumption for the distance from shore at which to place the ASR based on a non-responsive shoreline mode (neither accreting or eroding).

$$\frac{h_B}{H_0} = 2log_{10}\left\{\left(\frac{S_B}{h_B}\right)^{3/2}\left(\frac{L_B}{h_B}\right)^{2}\left(\frac{A^3}{h_B}\right)^{1/2}\right\} + 0.65$$

where: H_0 is offshore significant wave height; h_B is water depth at toe of structure; L_B is Crest level; S_B is structure length (shore parallel projection); and $A = 0.21 * D_{50}^{0.48}$ is the Bruun Parameter for a simplified shore profile (Hallemeier, 1981).

Based on the initial wave modelling, desired wave breaking conditions and concept design of the structure (RHDHV, 2015), the offshore placement of the reef was determined using:

1. the relation seen in Figure 20; a conservative approach of placing the structure at the neutral juncture between erosive and
2. An accretionary shoreline mode from the empirical Ranasinghe (2010) formula for 10 -100yr ARI wave heights.

This neutral point can be seen as D_{OFF} in Figure 15 (right), at this location the wave driven currents that sped up over the reef structure are significantly dissipated.

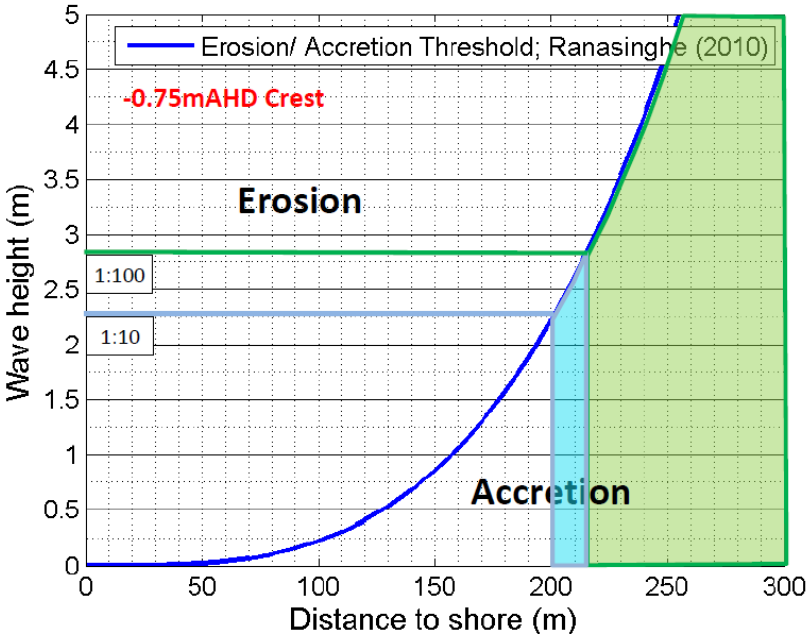


Figure 20 Predicted shoreline response in terms of wave height and distance to shore of proposed ASR at Middleton Beach based on (Ranasinghe, 2010) for -0.75mAHD crest level.

> ***Note:*** *The lower bound of the green field is defined by (initial calculation of) 100 year ARI wave height (1:100) of ~2.8m, the lower bound of the blue field by the (initial calculation of) 10 year ARI wave height (1:10) of ~2.3m. A mean D50 = 0.21mm and approximate effective shore parallel width = 100m.*

The following section details the envelope of expected shoreline response based on the four ASR layouts considered in this study (see Figure 1). It utilises the empirical formulation described above as well as the additional numerical modelling undertaken in RHDHV (2017).

The initial offshore placement of the AASR was based on four key criteria:

- having a net neutral impact on shoreline position;
- structural integrity including settlement, burial and damage to design components;
- total size of the structure and hence construction and material cost; and
- Surfer safety and amenity (i.e. minimising the offshore distance).

The efficiency in providing shoreline protection is governed by the wave driven current circulation in the lee of the structure; if located too close to shore a divergent wave driven current pattern develops which may cause beach erosion. Increasing the distance to shore allows the wave driven currents to develop a 4-cell convergent circulation pattern which would ultimately result in the development of a salient as described above and further elaborated in (Ranasinghe, 2006). On the other hand, if the structure is placed even further from shore no salient will form.

Through similar investigations at Palm Beach, Queensland (RHDHV, 2018) it was determined that the most appropriate balance between offshore placement of the structure for both surfer safety and coastal response should be based upon a 1-year ARI wave condition. At Palm Beach the proposed reef is located on a coastline where there is a strong alongshore drift and would therefore be expected to develop a salient like feature that would act as a buffer against beach erosion in the lee of the structure. On this

basis some erosion during these extreme ‘annual’ events would be acceptable. Moreover, it would also be expected that during such events, the entirety of the shoreline would undergo some extent of erosion.

The selection of distance to shore is a critical parameter in the design of an ASR and should be further assessed for the Middleton Beach setting in the physical modelling stage of the project.

An updated Extreme Value Analysis (EVA) was undertaken based upon the extracted RHDHV dataset from the 38 year hindcast model (RHDHV, 2017). A Peak over Threshold (PoT) analysis was undertaken on the dataset for the 99.5th percentile Hs at the RHDHV site for the 38 year hindcast period, as seen in Figure 21 (top). This data was used to fit to a Weibull Maximum Likelihood Estimation to find ARI wave heights at the site, Figure 21 (bottom).

Table 3 presents the results of the updated EVA.

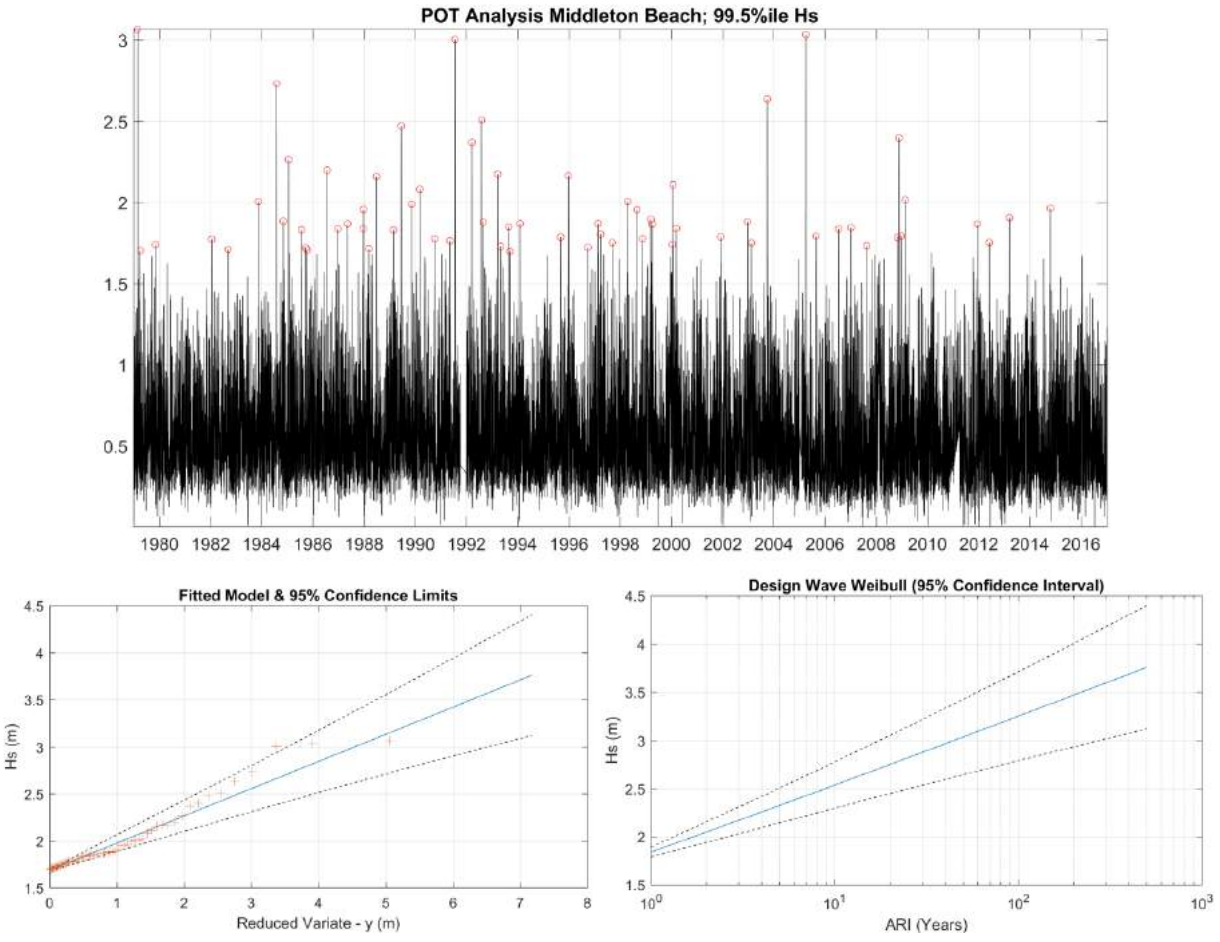


Figure 21 Peak Over Threshold (POT) analysis of the 99.4th percentile significant wave height data extracted from the 38 year hindcast model at the RHDHV AWAC location, offshore of ‘Surfers’, Middleton Beach (top). Weibull Maximum Likelihood Estimation of wave heights (bottom)

Table 3 Design wave heights at 'Surfers' location.

ARI (years)	Hs (m)
1	1.85
5	2.3
10	2.55
50	3.05
100	3.25

In order to update the Ranasinghe formulation, it was necessary to determine structure dimensions for each of the reef configurations supplied to the EPA in the approval footprint (see Figure 1). As the Ranasinghe formula was developed for a simplified 2D shape, an appropriate effective width needs to be determined for its use in the calculation. To determine the effective crest length for the more complex 3D geometry of the AASR the 1.3m contour was used. This was defined as the contour where significant wave breaking would occur based on a 1year ARI wave height of 1.85m and a corresponding wave period of 10sec. This was calculated using linear wave theory as per Goda (2000) following the localised shoaling caused by the 1:12 offshore toe of the structure. A representation of the -1.3m contour of each reef layout can be seen in Figure 22 along with structure dimensions for each configuration in the table below.

Table 4 Basic ASR dimensions for the four layouts considered in this study.

ASR Layout Option	Centre depth	Effective crest length (-1.3m contour)	Distance of shoreline to landward extent of crest	Structure orientation (°N)
Nth Inshore (NW)	-5.5m	114m	120m	125 °N
Nth Offshore (NE)	-8.0m	114m	260m	125 °N
Sth Offshore (SE)	-7.8m	114m	280m	125 °N
Sth Inshore (SW)	-5.5m	114m	120m	175 °N

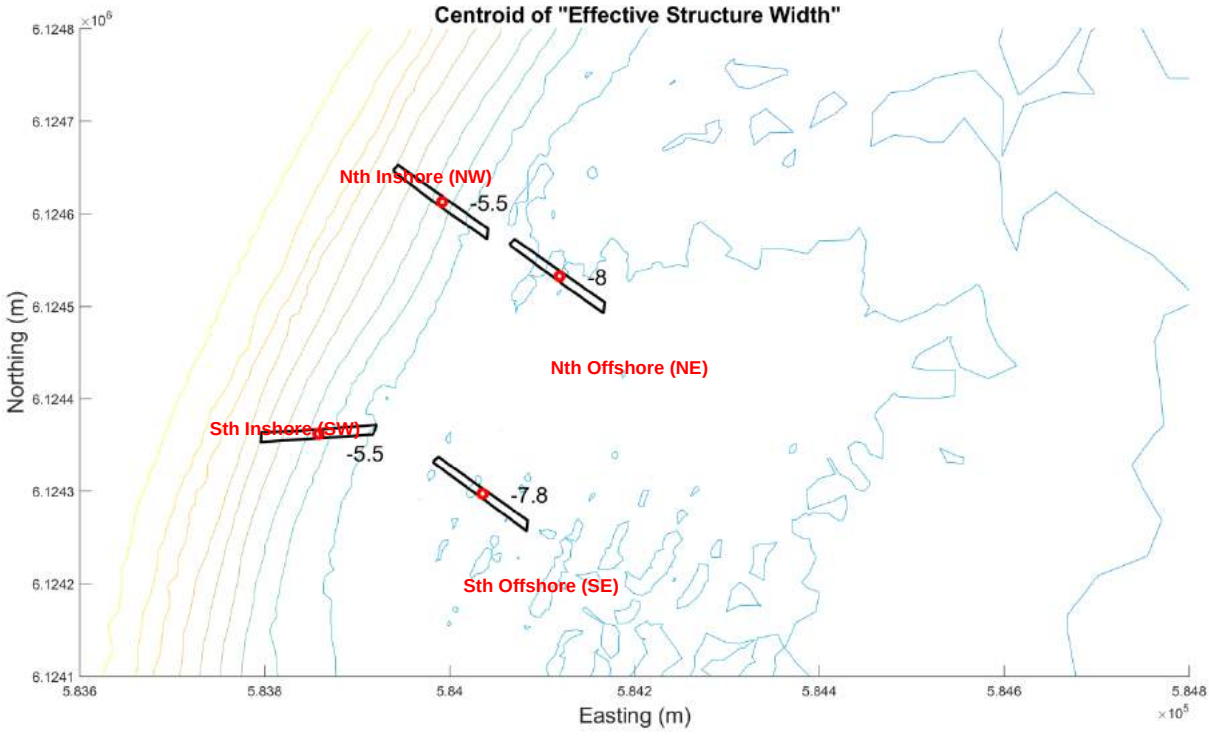


Figure 22 Representation of Effective Structure Width and other key structure dimensions for each of the four AASR layouts presented to EPA.

The Ranasinghe formulation was updated using the structure dimensions and wave parameters for a 1yr ARI event (Hs = 1.85m) and also under ambient wave conditions (Hs = 0.6m) to determine approximate distance to shore based on a null impact to shoreline response, Figure 23.

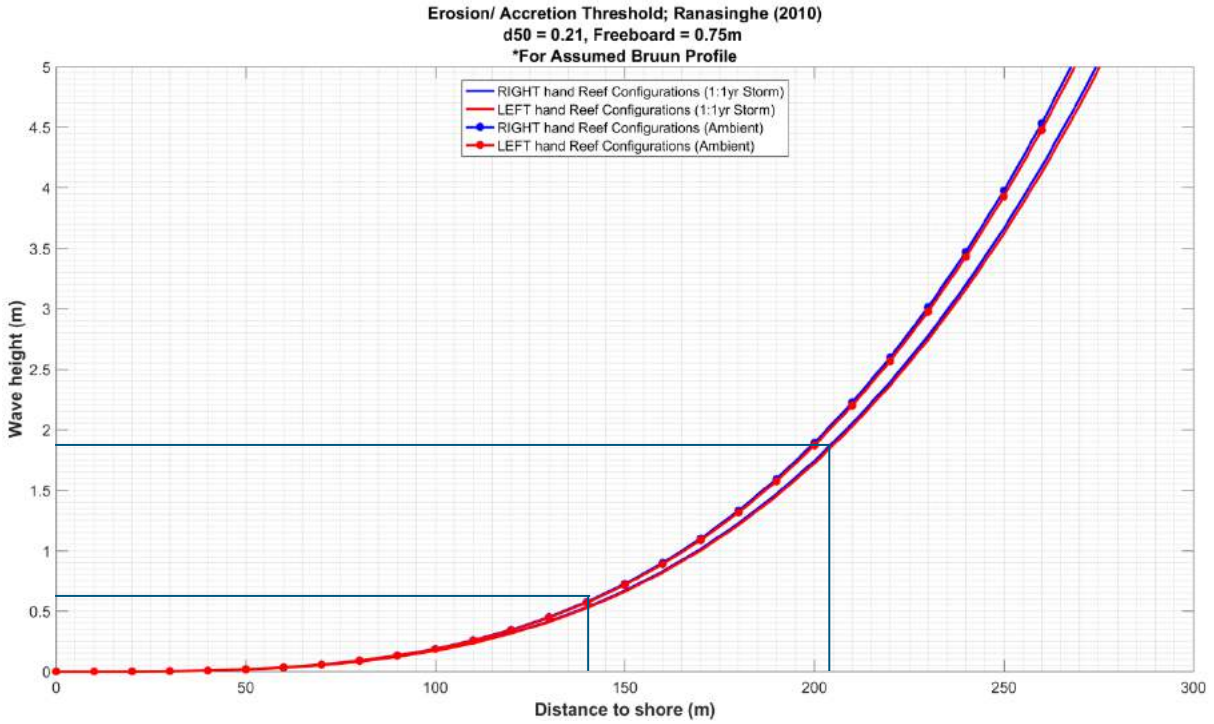


Figure 23 Updated Ranasinghe (2010) formulation to determine the minimum offshore placement of the AASR based upon the updated metocean conditions at the site. The graph shows D_{OFF} for both ambient and 1yr ARI conditions at the site.

It can be seen from Figure 23 that the minimum offshore placement distance of the AASR for a null shoreline response in its lee ($\mathbf{D_{OFF}}$) is approximately 205m and 140m from the landward extent of the crest for 1yr ARI or ambient wave condition, respectively.

2.5 Numerical Modelling

2.5.1 One-Dimensional Sediment Transport Modelling

DHI's numerical modelling suite LITPACK was used to determine the effects of a submerged structure on the coastal profile at Middleton Beach. In particular, the LITPROF module was utilised to describe cross-shore profile changes based on a time series of wave events. The model is based on the assumption that longshore gradients in hydrodynamic and sediment conditions are negligible and that the depth contours are parallel to the coastline. Thus the coastal morphology is solely described by a basic 1D cross-shore profile. A simplified representation of the AASR was used for the cross-shore profile over the reef structure. The simplified representation was based on the envelope of possible structure locations and crest heights.

In order to investigate the effect of the AASR on longshore sediment transport rates, the LITDRIFT model developed as part of RHDHV (2017) was used to compare rates with and without the AASR on a representative profile.

Cross-shore Transport

Model Setup

LITPROF calculates sediment transport from an intra-wave hydrodynamic model where the time evolution of the bed boundary layer is resolved. It provides empirical transformation of the deterministic description of transport rates and distribution the sediment across a given beach profile.

Data input into the LITPROF model for this study has been summarised in the following:

Beach Profile Data - Between October 2013 to December 2016 CoA has undertaken regular beach profile surveys at various locations along Middleton Beach. The extent and orientation for each profile can be seen in Figure 24. The most current beach profile survey at the time of the study (December 2016) was extracted and used as input into the LITPROF model for the cross-shore profile MB02 as it could be extended further than the MB02 profile from the high-res MBES survey. The MB02 profile is the closest in proximity to the proposed AASR site. The MB02 profile was extracted perpendicular to the coastline and interpolated to provide a profile chainage resolution of 10m.



Figure 24 Location of CoA beach survey transects (JKA, 2014)

Sedimentological Data - Sediment samples were taken by Geoff Bastyan and Associates (GBA) and CoA at each of the profile transects in 2013. Sediment samples were taken at five locations spread along the length of each profile. As part of this sampling campaign each sample was analysed for their particle size distribution (PSD). Samples were taken from both the sub aerial beach and intertidal zone in order to best characterise beach sediments across the profile. Sediment data collected as part of these works has been used as input into the LITPROF model.

Wave Data - A single year time series of modelled wave data has been used to estimate the annual cross-shore sediment transport rates at each of the modelled profiles. Wave data generated as part of the 38-year spectral wave hindcast exercise was analysed to identify average wave conditions expected under a typical year. From the wave power and Long Term Average (LTA) wave conditions it was determined that wave conditions modelled for 2015 represented a typical year (in terms of wave energy), for further details see (RHDHV, 2017). As such, the 2015 modelled wave data time series (3-hour resolution) was extracted at the location of the seaward extent of MB02 and used as input into the LITPROF model. The LITPROF model applies the wave time series at the MB02 seaward boundary and transforms the wave conditions to the shoreline with respect to the shape of the cross-shore profile.

Model Runs

Tide and wind induced flows are relatively weak along Middleton Beach, and sediment transport is primarily driven by waves. As such no current, tide or wind influences were included in the simulations, see RHDHV (2017) for more details.

Scenario Testing

As part of the LITPROF model development a high level model validation process was undertaken to ensure the model is generally representative of what may be inferred from the introduction of a submerged structure like the AASR. Complex 3D wave-wave interactions and wave transmission over the structure are beyond the capabilities of LITPROF. As such, the model and its results should be used as a tool and requires interpretation.

Representation of the structure in the MB02 profile was undertaken using LITPACK's Profile Evolution toolbox, which restricts the definition of the structure to the offshore distance to the centre of the structure, the crest width and depth. Detailed changes in structure slope and orientation cannot be represented; the structure is thus defined as a simple 1D non-erodible profile overlying the MB02 profile. Prior to the introduction of the structure onto the MB02 profile, sensitivity testing was undertaken to determine the most appropriate model parameters;

- Wave theory (Cnoidal, Stokes I, III, V, etc.);
- inclusion of bed ripples; and
- The morphological scaling factor which influences the width of the berm.

Following the selection of appropriate model parameters, a matrix of simulations was created to determine possible (cross-shore) profile evolution to the simplified structures. These are shown in Table 5. The matrix represents the furthest offshore and onshore placement locations as well as varying structure crest depth, initially discussed in the Feasibility Report (RHDHV, 2015).

Table 5 LITPROF Simulation Matrix

Scenario number	Scenario name	Toe distance to shoreline (m)	Crest depth (m)
90.1	MB02waves2015reef90m2mDepth	90	2.00
90.2	MB02waves2015reef90m1.5mDepth	90	1.50
90.3	MB02waves2015reef90m1mDepth	90	1.00
90.4	MB02waves2015reef90m0.75mDepth	90	0.75
240.1	MB02waves2015reef240m2mDepth	240	2.00
240.2	MB02waves2015reef240m1.5mDepth	240	1.50
240.3	MB02waves2015reef240m1mDepth	240	1.00
240.4	MB02waves2015reef240m0.75mDepth	240	0.75

Results

The results of the LITPROF simulation matrix are presented in Figure 25 and Figure 26.

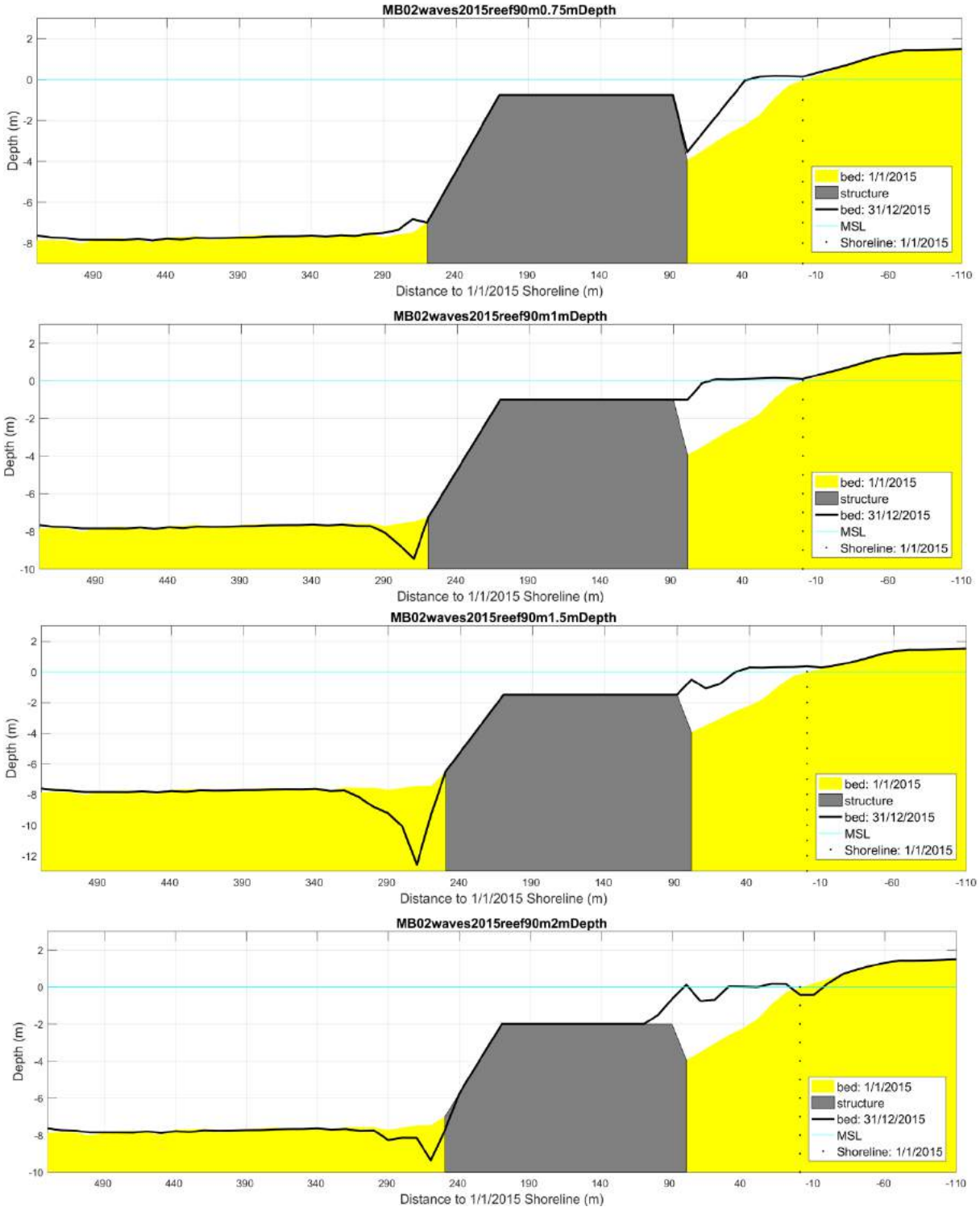


Figure 25 Scenarios 90.1-90.4; AASR with seaward toe 90m from shoreline with (top to bottom) 0.75, 1.0, 1.5, 2.0m crest depth.

Note: In this figure the yellow is the seabed at the start of the simulation (i.e. based on MB02) and the black line is the seabed at the end of the simulation (i.e. after one year).

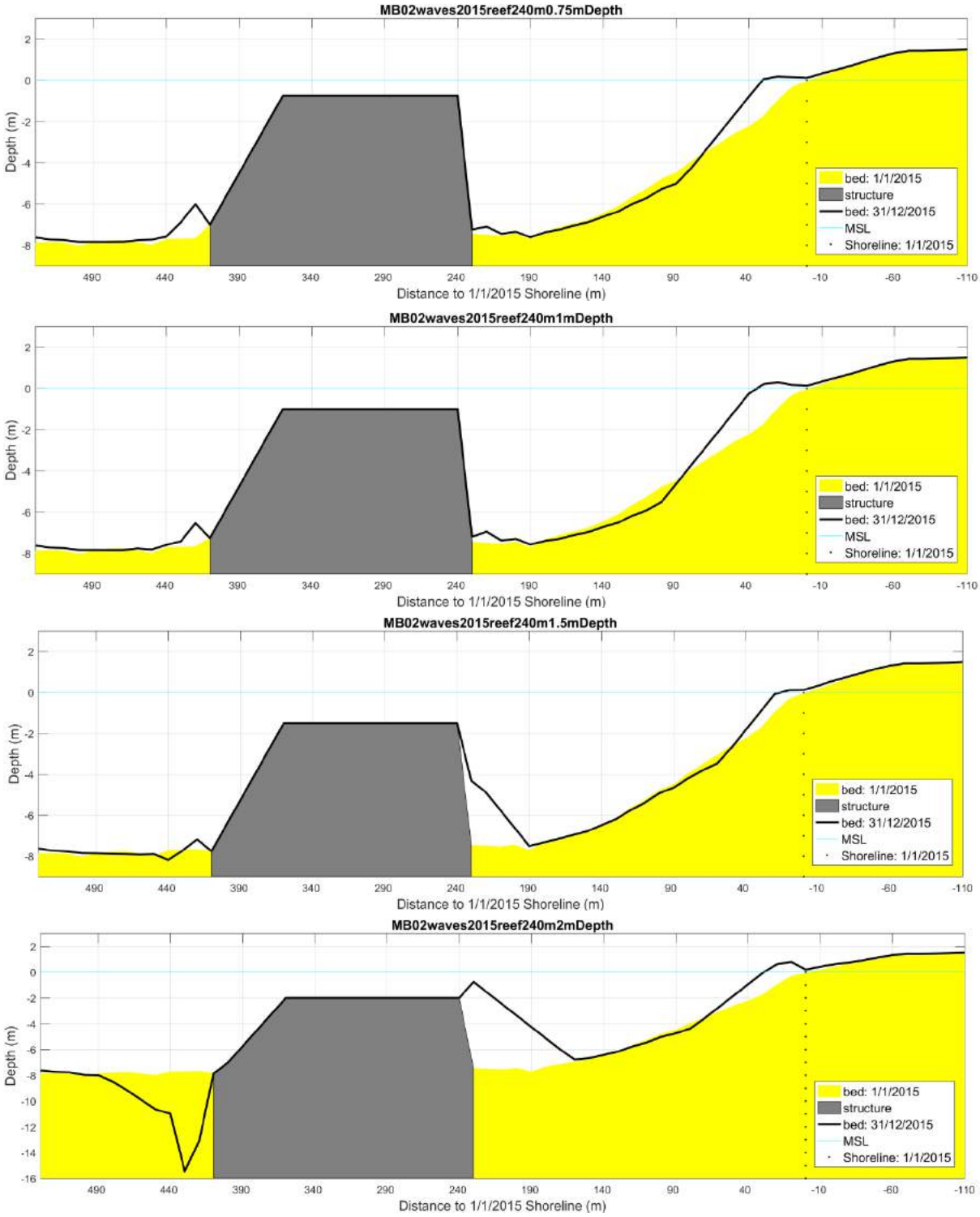


Figure 26 Scenarios 240.1 - 240.4; AASR with seaward toe 240m from shoreline with (top to bottom) 0.75, 1.0, 1.5, 2.0m crest depth.

Note: In this figure the yellow is the seabed at the start of the simulation (i.e. based on MB02) and the black line is the seabed at the end of the simulation (i.e. after one year).

Longshore Transport

Model Setup

LITDRIFT is a deterministic numerical model; the sediment transport is modelled as a function of a cross-shore profile (including shape and sediment properties) and wave climate. The main input data required for LITDRIFT include:

- beach bathymetric profile data;
- sedimentological data;
- wave data; and,
- Water level data.

In general, these parameters and the MB02 profile were kept constant from that used in the LITDRIFT modelling as part of (RHDHV, 2017). The south-west corner of Middleton Beach represents a crenulated bay shape, whereby refraction around the adjacent headland results in the incident waves being perpendicular to the shore. This orientation would result in the net longshore transport in this region to be near zero. The LITDRIFT engine was primarily developed to model sediment transport along long open stretches of exposed shorelines and is therefore not suitable for modelling longshore transport within crenulated bays. As such Profile MB02 used in the LITPROF model was used as input into the LITDRIFT model for continuity and for the relatively straight section of beach at this location. As tide and wind induced flows are expected to be relatively weak along Middleton Beach, with alongshore sediment transport primarily dominated by waves, no current or wind influences were included in the simulations.

Model Run

LITDRIFT was run for profile MB02 with and without the simplified AASR representation (Option B) for a period of one year based on the time series of waves extracted from the 38 year hindcast for 2015.

Sensitivity Testing and Model Validation

As part of the LITDRIFT model development a high level model validation process was undertaken to ensure the model was generally representative of what may be inferred from an historic aerial imagery analysis (see RHDHV, 2017), historical bathymetric surveys and past coastal process investigations.

It is important to note that the LITDRIFT model is particularly sensitive to the difference in angle between the profile orientation and the incident wave angle. For this reason care was taken to ensure the orientation of the modelled profiles best represent the true orientation perpendicular to the shoreline (and offshore contours and median incoming wave direction). When interpreting the model results it is important to note that small changes in the orientation of the shoreline of even 0.5 - 1 degree has the potential to result in significant changes to the modelled net sediment transport rates.

Results

The results of the two LITDRIFT simulations can be seen in Figure 29 and Figure 28.

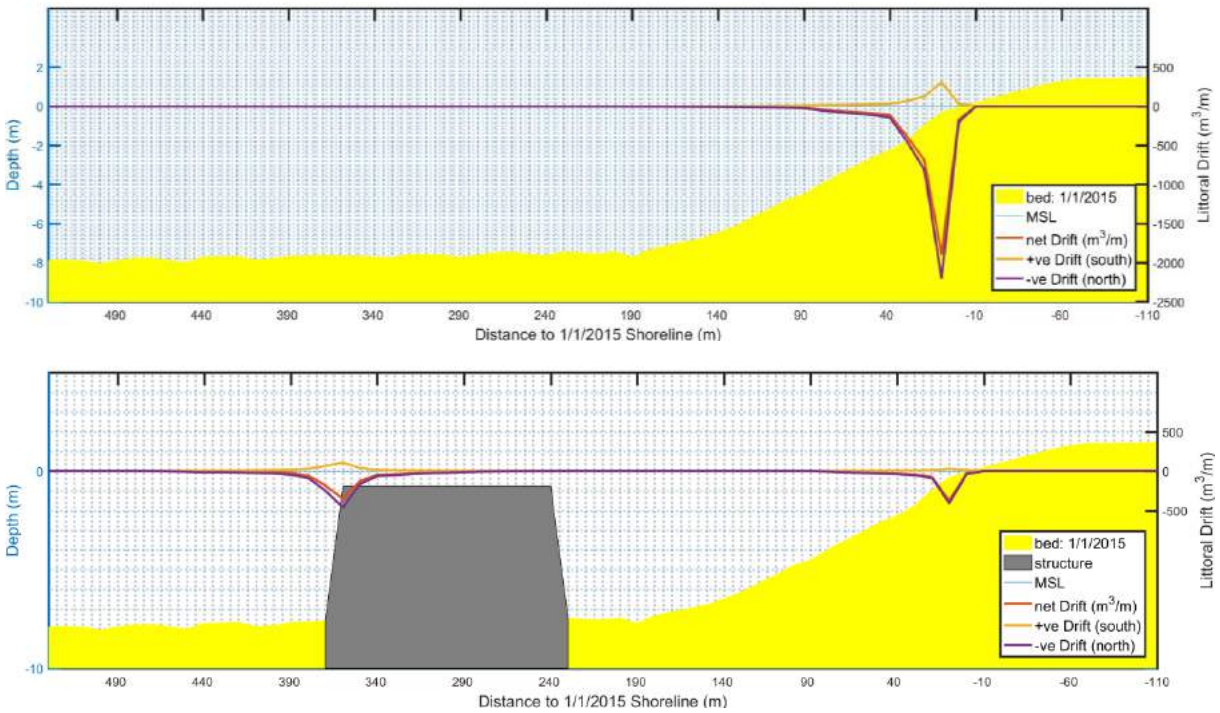


Figure 27 Longshore littoral drift (m^3/m) along the profile M02 from the LITDRIFT simulations for (top) base case (no reef) and (bottom) inclusion of reef structure at 240m offshore.

> ***Note:*** *The positive (southward), negative (northward) and net littoral drift along the profile are represented by orange, purple and red lines respectively.*

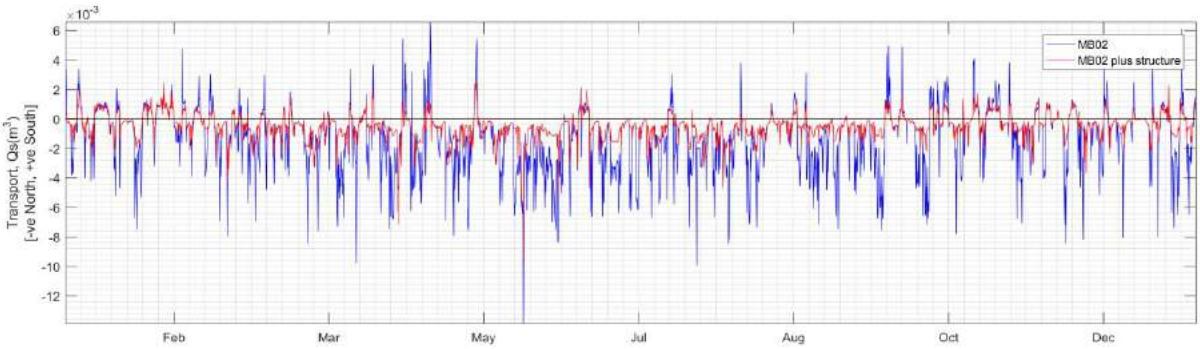


Figure 28 Annual longshore littoral transport, Qs (m^3) for the profile M02 from the LITDRIFT simulations for (red) base case (no reef) and (blue) inclusion of reef structure at 240m offshore.

2.5.2 Two–Dimensional Hydrodynamic Modelling

As part of the Emu Point to Middleton Beach Coastal Adaptation Study (RHDHV, 2017), a coupled hydrodynamic (HD) and spectral wave (SW) model was setup for a regional Albany domain and was calibrated to long term wave and current measurements made by RHDHV and DoT for locations offshore of Middleton Beach and Emu Point (respectively). In order to determine possible effects to the local hydrodynamic regime of the study site, this calibrated model was used to incorporate a range of reef bathymetries that encompass the approvals footprint described to DWER, as seen in Figure 1 (BMT, 2017).

Following RHDHVs work on the Palm Beach Shoreline Project (RHDHV, 2018); differences were found between the SW model results and the measured wave transmission over the SCR collected during laboratory physical modelling. These were not greatly improved by model calibration. Wave transformation over this type of structure is not described accurately using linear wave theory, which is adopted in the SW wave model. Further, more sophisticated, numerical modelling using a non-linear, phase-resolving wave model was, however, able to suitably represent the wave transformation over the structure. Due to the inherent 3D nature of these structures; reflections, refraction and diffraction is not adequately represented in phase-averaging SW approach. As such, the results of the numerical modelling undertaken herein should be viewed qualitatively and used for comparative purposes only and be seen as an initial step to a more detailed modelling which is recommended during later stages of the AASR Project.

The purpose of the modelling undertaken in this stage of work is to determine the envelope of possible coastal response due to the introduction of the AASR. For this purpose, the models used herein are considered suitable.

Model Description

The MIKE 21 software package has been adopted for use in this study. MIKE 21 is a numerical modelling suite that simulates flows, waves, sediment transport and ecology in rivers, lakes, estuaries, bays, coastal areas and seas in two dimensions. MIKE is developed by the Danish Hydraulic Institute (DHI).

The Flexible Mesh (FM) version of MIKE 21 has been adopted as it allows the spatial resolution of the computational grid to be locally increased in areas of interest, i.e. at the AASR project site, while the resolution in other areas can be coarser to help maintain acceptable model run times. The spatial discretisation of the equations in MIKE 21 FM is performed using a cell-centred finite volume method.

The MIKE 21 FM Hydrodynamic (HD) and spectral wave (SW) modules have been used in this study (DHI, 2017a). The HD module system is based on the numerical solution of the two-dimensional shallow water equations - the depth-integrated incompressible Reynolds averaged Navier-Stokes equations. Thus, the model consists of continuity, momentum, temperature, salinity and density equations. The SW module calculates the integrated wave parameters; significant wave height (Hs), peak wave period (Tp), mean wave direction (MWD) and directional spreading (DSD) by solving the wave action equation for the 1st and 2nd spectral moment of the wave energy frequency spectrum.

Modelling Approach

Utilising the 38 year hindcast data (RHDHV, 2017), both average and extreme wave conditions were selected based on the long term statistics as well as a ‘low energy’ event based on the 20th percentile wave heights from the Long Term Average. A search was then made of the 38 year dataset to find actual representative ‘low energy’, ‘average’ and ‘extreme’ events. Boundary wave conditions of each of these events were sourced from the 38 years of results produced as part of (RHDHV, 2017).

The SW model was run for each of these wave conditions (‘low energy’, ‘average’, ‘extreme’) and each of the bathymetric configurations:

1. Base case (2016 survey – no reef);
2. Offshore NE Right;
3. Inshore NW Right,
4. Offshore SE Right and;
5. Inshore SW Left.

Each of these configurations can be seen at the outer extents of the approvals footprint (Figure 1) and in the model meshes (Figure 12). As the SW model was used to model a single wave condition over a relatively small domain, it was run using a directionally de-coupled, quasi-stationary formulation. This approach also reduced computational run times, allowing for more sensitivity testing within the project schedule.

The SW model produced two-dimensional wave radiation stress maps of the domain outputting the following parameters; sxx, sxy and syy (m^3/s). These were then used to force the HD model. As the area of interest has been noted as having little longshore sediment transport (RHDHV, 2017) and the Middleton Beach area is micro-tidal, the HD model was run without any other external hydrodynamic forcing, i.e.; water level fluctuation or flux at any of the boundaries. This was to directly compare the effects of incorporating the reef structures within the model domain.

Model Setup

The MIKE 21 FM domain, computational mesh and interpolated model bathymetry for each model layout are presented in Figure 12. The model mesh has been refined following a number of sensitivity tests in order to create efficiency with run time, bathymetric representation of each reef structure and extent. The mesh configuration was kept constant for each ASR layout (as well as the base case) to be consistent for later comparison of results.

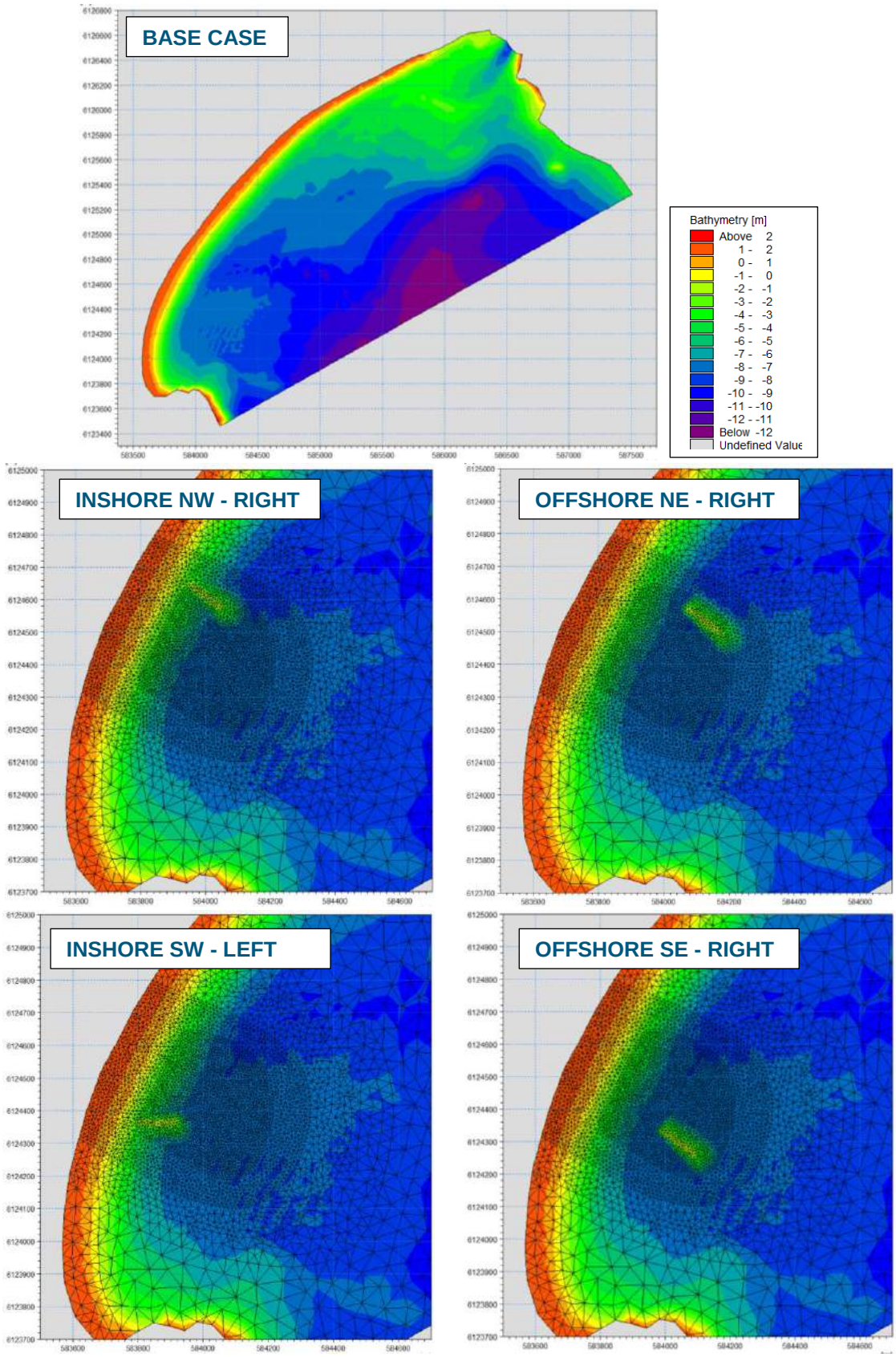


Figure 29 Interpolated bathymetry for the BASE case (top) and close up of computational mesh and bathymetry for (clockwise from middle) NW right, NE right, SW left, SE right

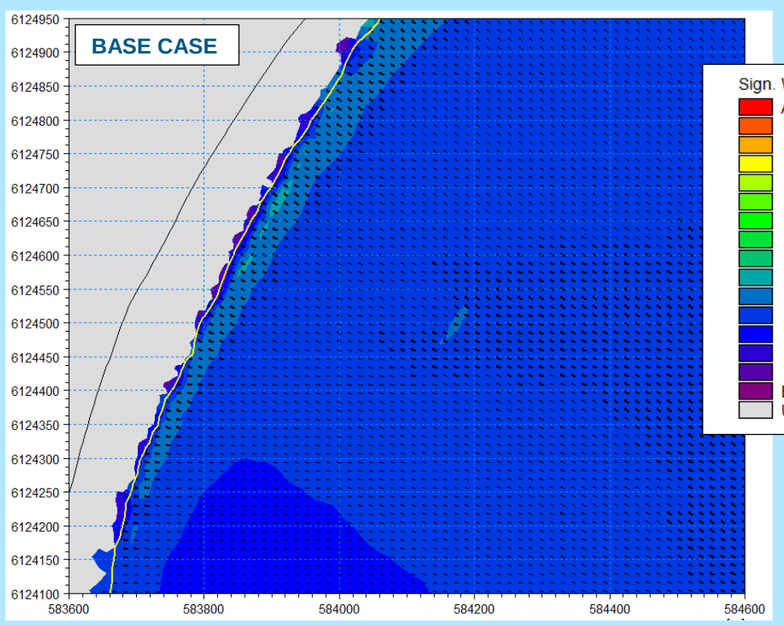
Hydraulic roughness and bed resistance were applied to both the HD and SW models respectively. These values were based on recent physical model tests undertaken as part of the Palm Beach Shoreline Project (RHDHV, 2018) where wave transmission was measured over a submerged rock structure on a mobile sand bed in a 3D wave tank. The manning's (M, m1/3/s) and Nikuradse roughness (kn, m) were altered accordingly.

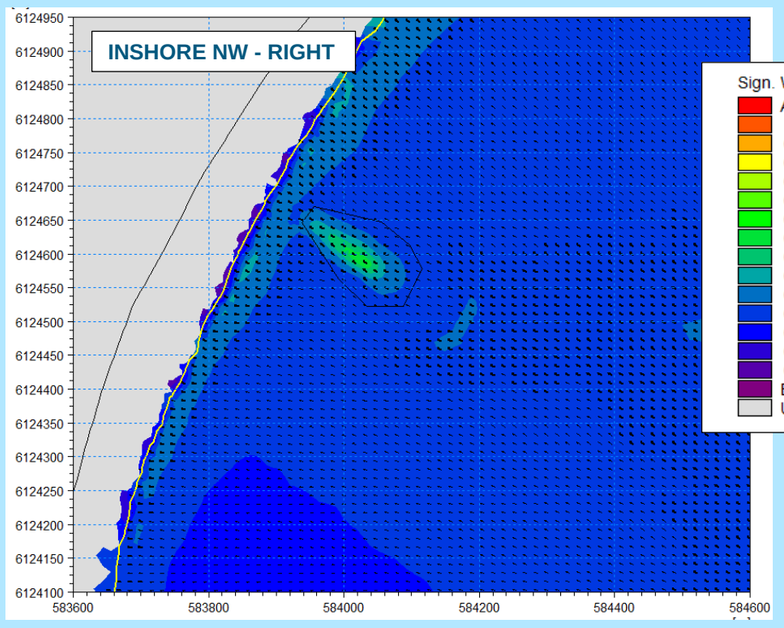
Results

Results for both the SW and HD runs are presented below as 2D significant wave height, Hs (m) maps for the SW and 2D current speed (m/s) maps for the HD runs. Current speed difference plots have also been produced for each reef layout (in comparison) to the base (no structure) case. The difference plots show increases in current speed in red and areas where current speeds have been reduced in blue. The vectors represent current directions from the scenario (reef layout) conditions.

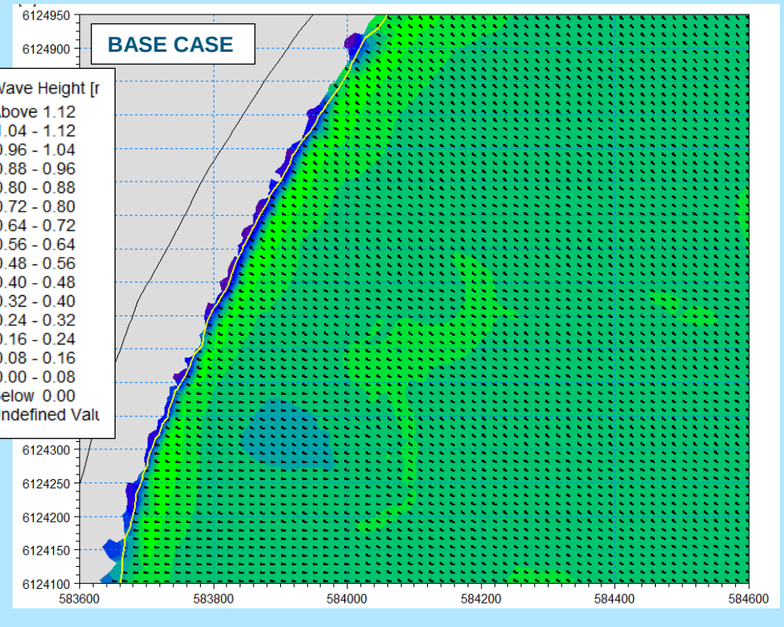
SW

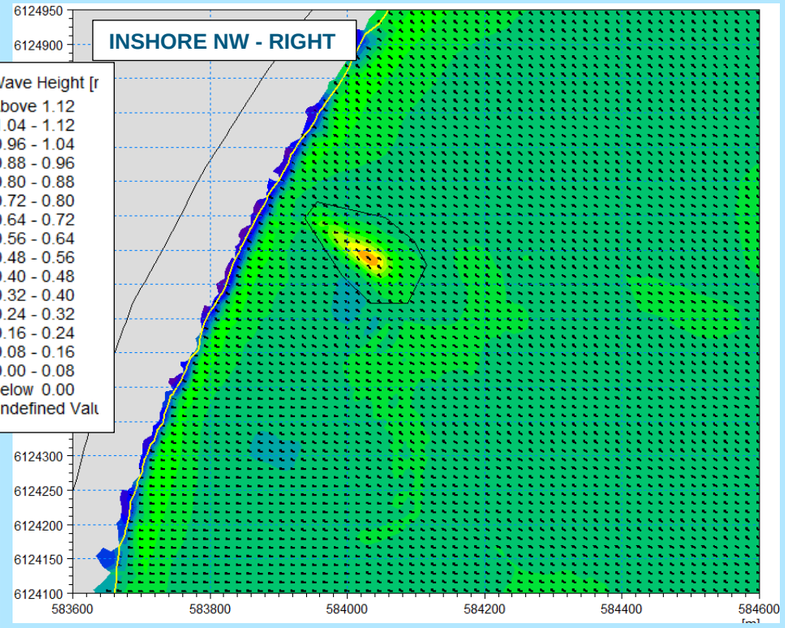
Low Energy Conditions: 14/12/2012 13:30
Offshore: Hs = 0.33m, Tp = 12.7sec, Dp = 121°



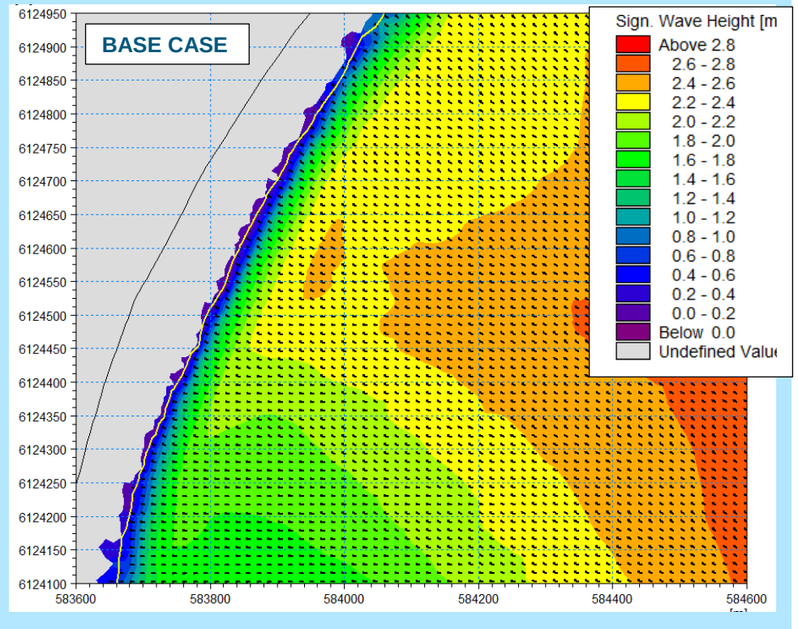


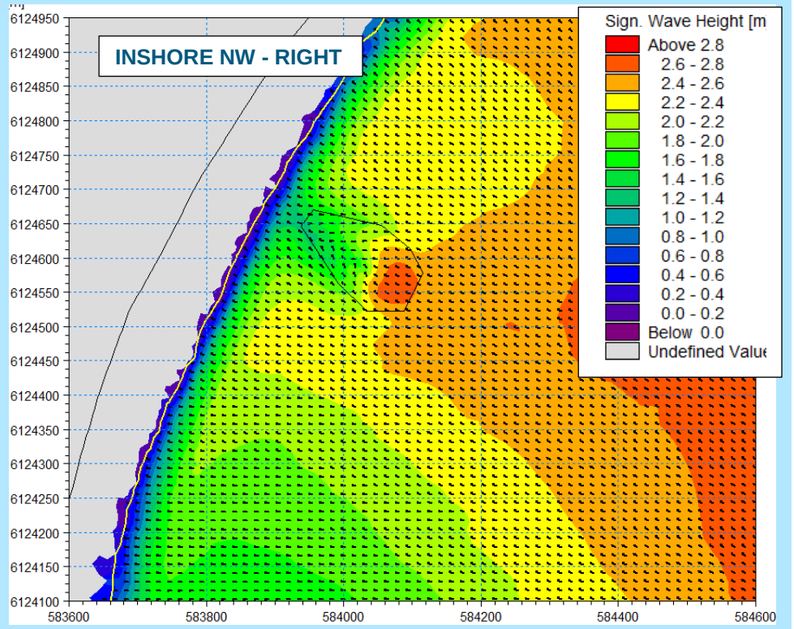
Average Conditions: 17/12/2015 13:30
Offshore: Hs = 0.6m, Tp = 13.4sec, Dp = 121°

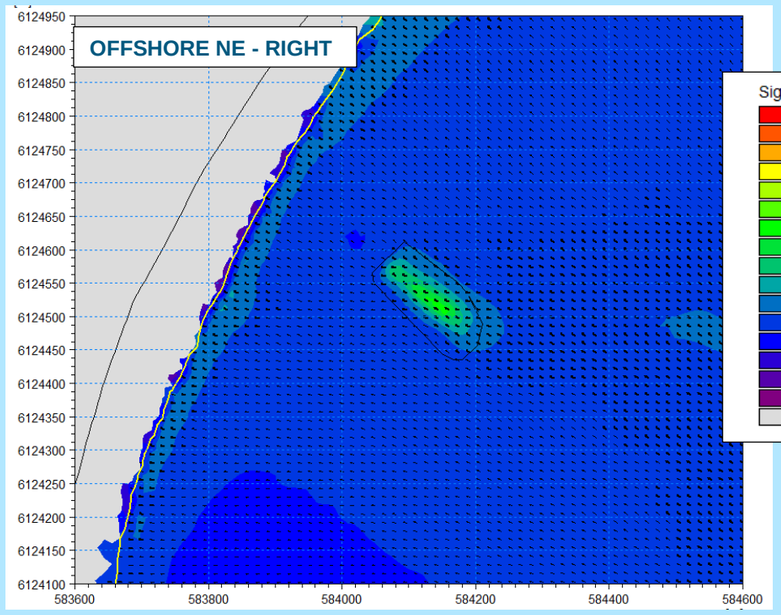


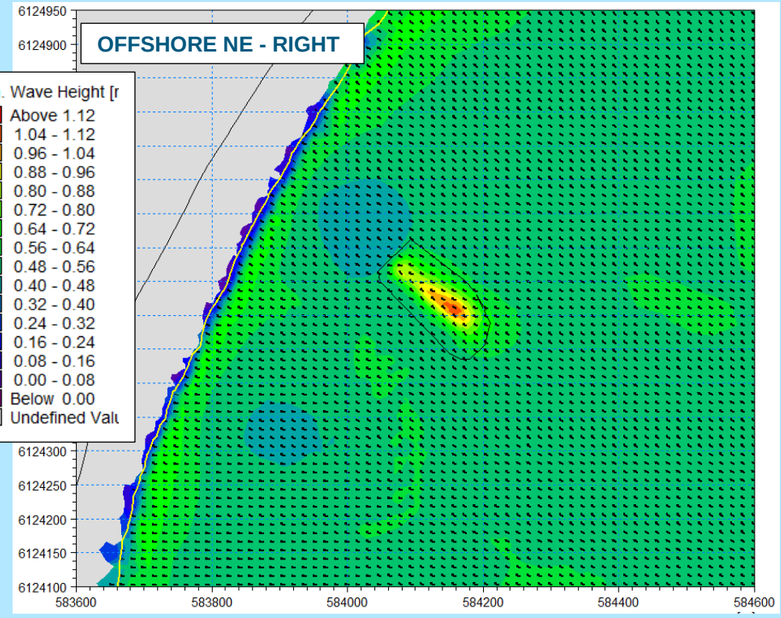


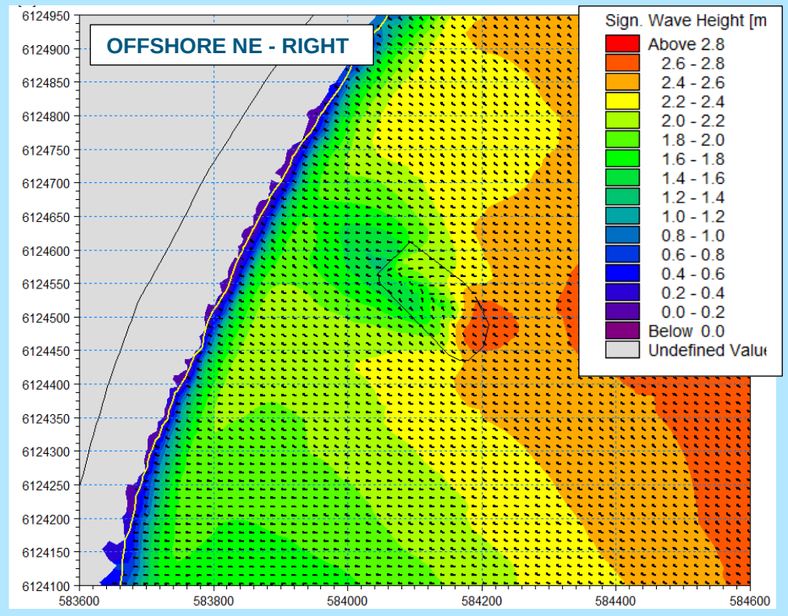
Extreme Event : 02/08/1984 22:30
Offshore: Hs = 2.7m, Tp = 11.5sec, 126°

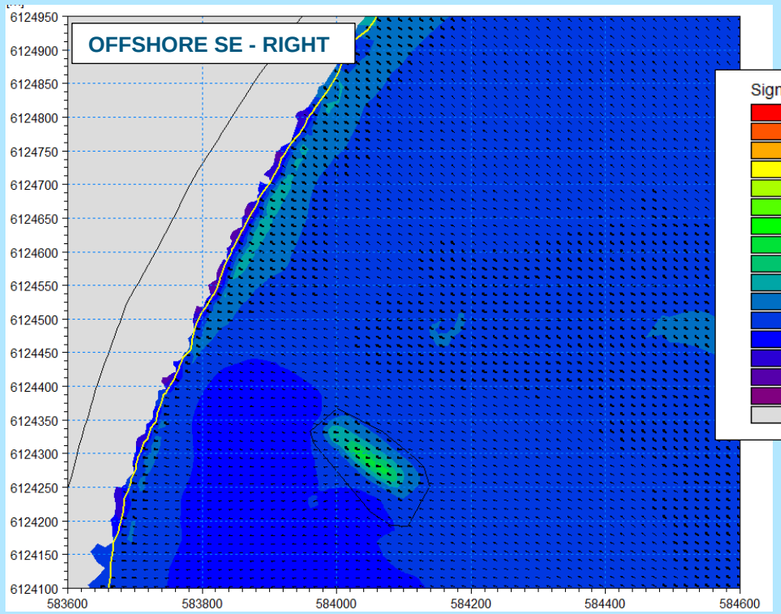


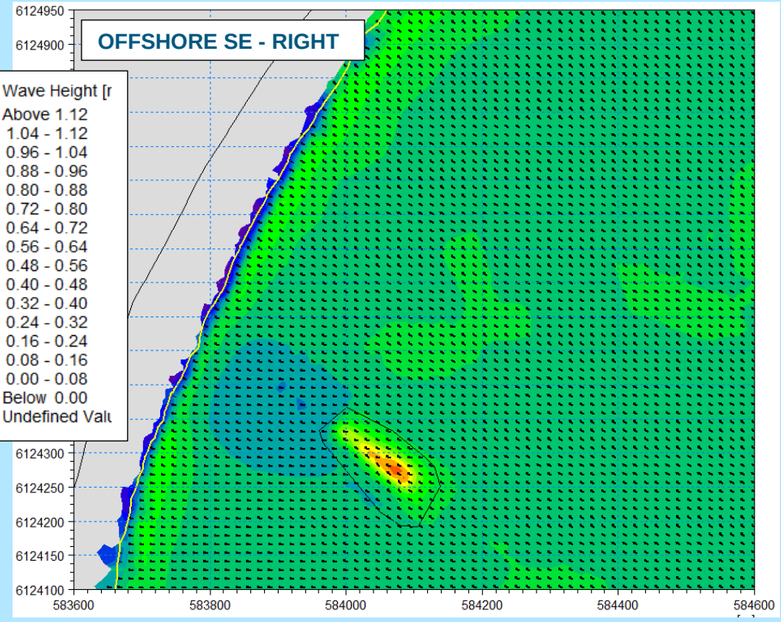


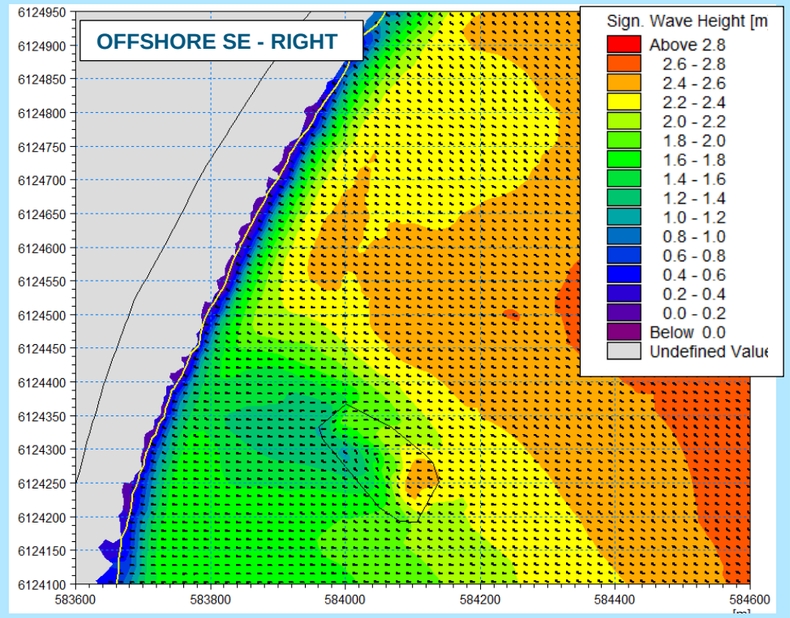

OFFSHORE NE - RIGHT
Sign. Wave Height [r
Above 1.12
1.04 - 1.12
0.96 - 1.04
0.88 - 0.96
0.80 - 0.88
0.72 - 0.80
0.64 - 0.72
0.56 - 0.64
0.48 - 0.56
0.40 - 0.48
0.32 - 0.40
0.24 - 0.32
0.16 - 0.24
0.08 - 0.16
0.00 - 0.08
Below 0.00
Undefined Valu
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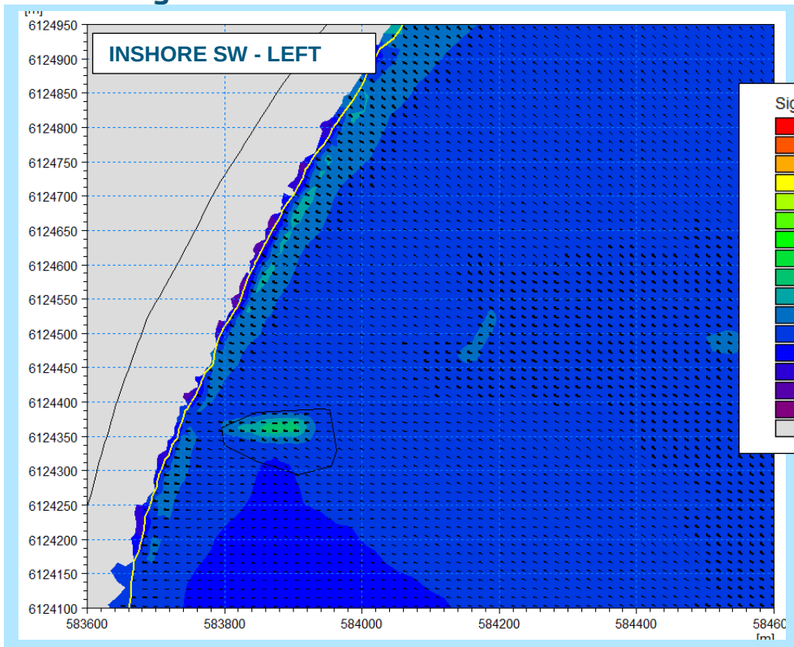

OFFSHORE NE - RIGHT
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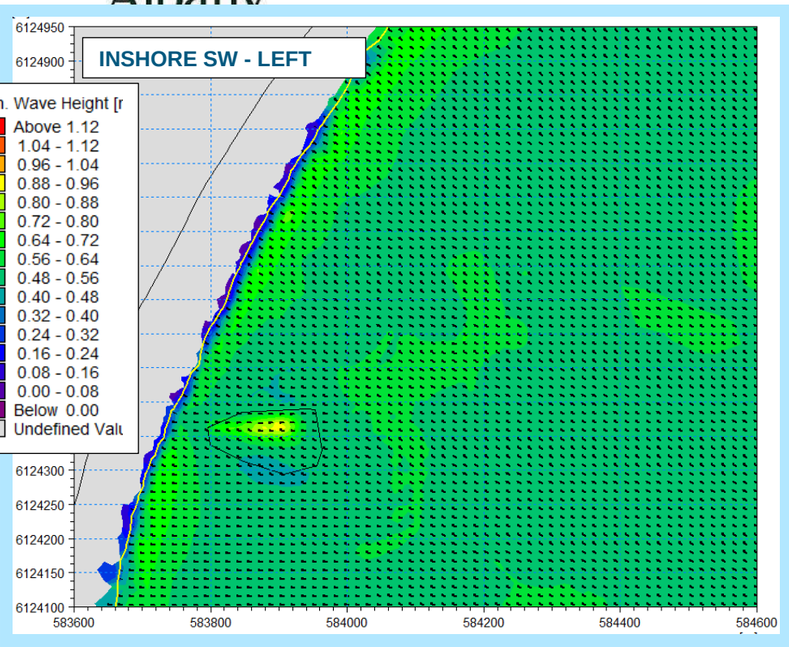

OFFSHORE NE - RIGHT
Sign. Wave Height [m
Above 2.8
2.6 - 2.8
2.4 - 2.6
2.2 - 2.4
2.0 - 2.2
1.8 - 2.0
1.6 - 1.8
1.4 - 1.6
1.2 - 1.4
1.0 - 1.2
0.8 - 1.0
0.6 - 0.8
0.4 - 0.6
0.2 - 0.4
0.0 - 0.2
Below 0.0
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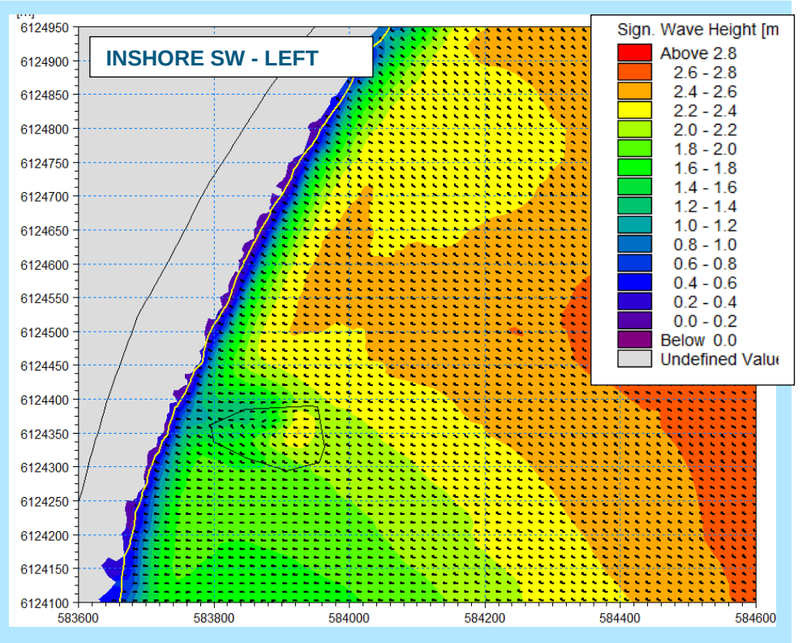

OFFSHORE SE - RIGHT
Sign. Wave Height [r
Above 1.12
1.04 - 1.12
0.96 - 1.04
0.88 - 0.96
0.80 - 0.88
0.72 - 0.80
0.64 - 0.72
0.56 - 0.64
0.48 - 0.56
0.40 - 0.48
0.32 - 0.40
0.24 - 0.32
0.16 - 0.24
0.08 - 0.16
0.00 - 0.08
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OFFSHORE SE - RIGHT
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OFFSHORE SE - RIGHT
Sign. Wave Height [m
Above 2.8
2.6 - 2.8
2.4 - 2.6
2.2 - 2.4
2.0 - 2.2
1.8 - 2.0
1.6 - 1.8
1.4 - 1.6
1.2 - 1.4
1.0 - 1.2
0.8 - 1.0
0.6 - 0.8
0.4 - 0.6
0.2 - 0.4
0.0 - 0.2
Below 0.0
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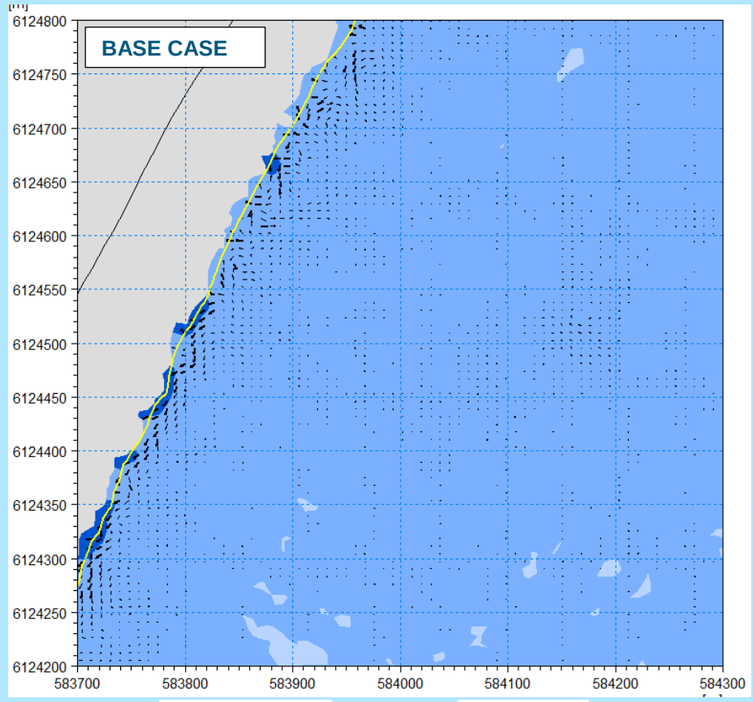


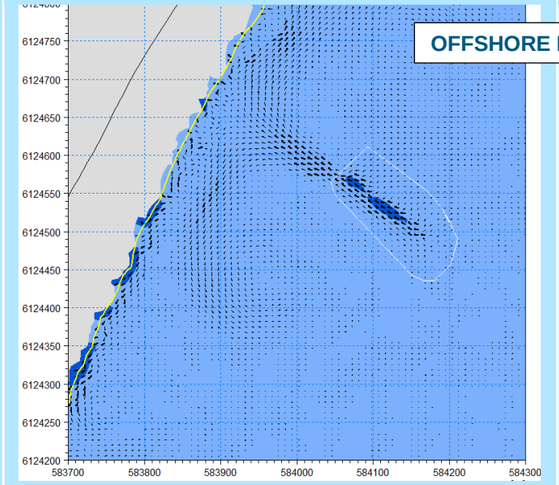


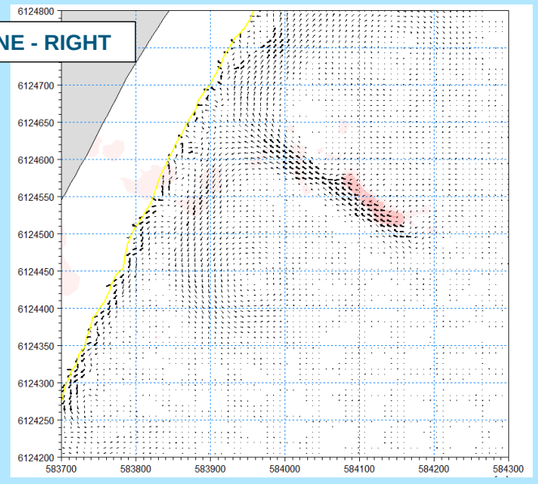


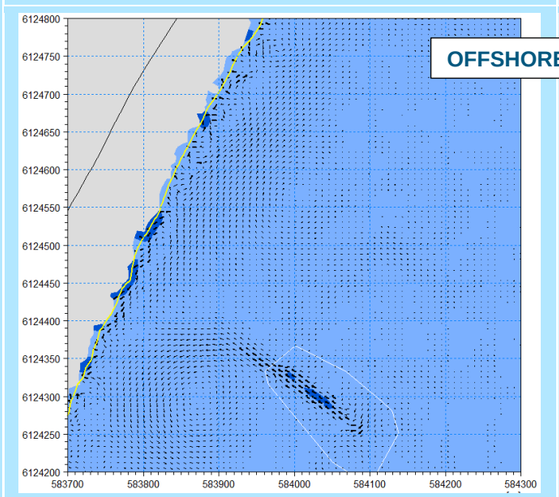
**Note – the Low Energy & Average and Extreme Wave Condition plots have differing scales*

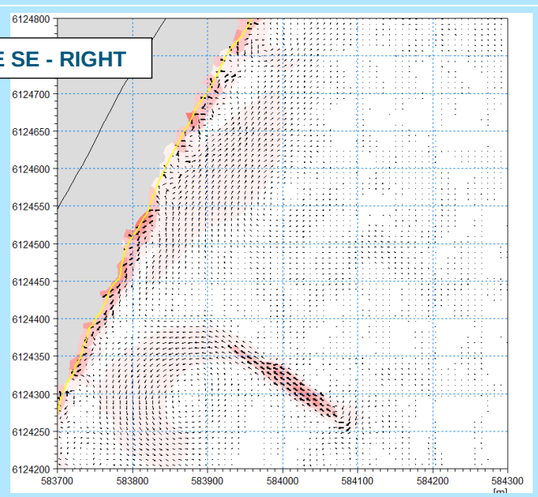
Low Energy Conditions: 14/12/2012 13:30 Offshore: Hs = 0.33m, Tp = 12.7sec, Dp = 121°

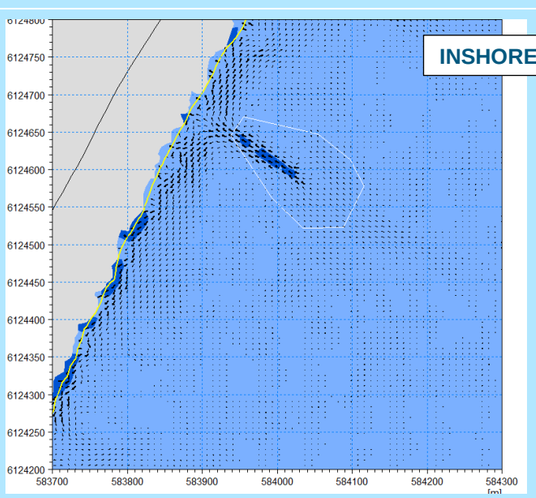


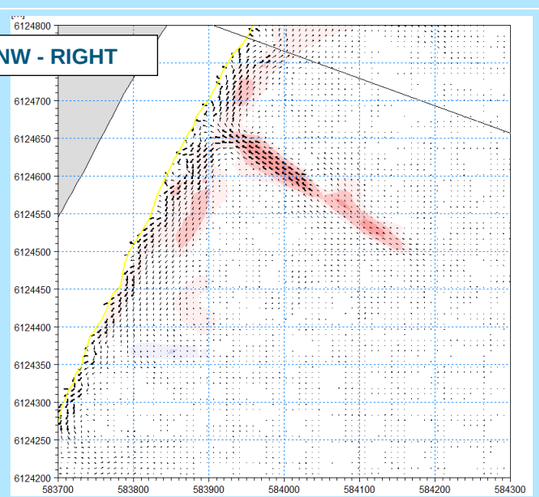


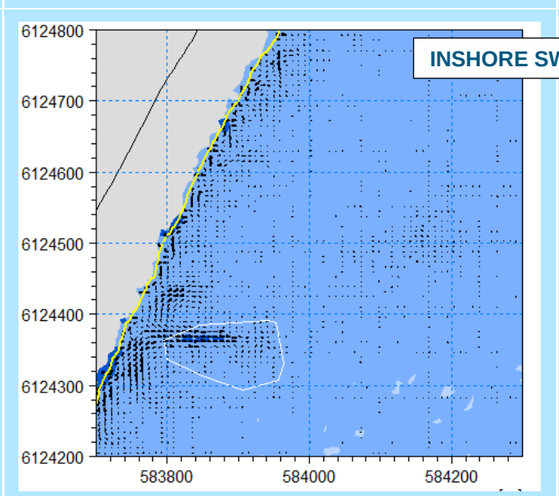


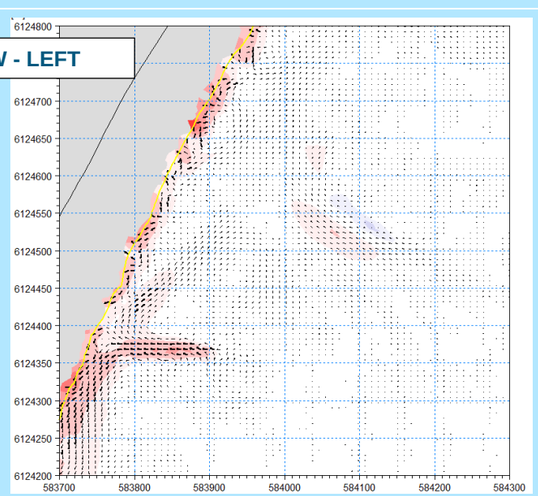




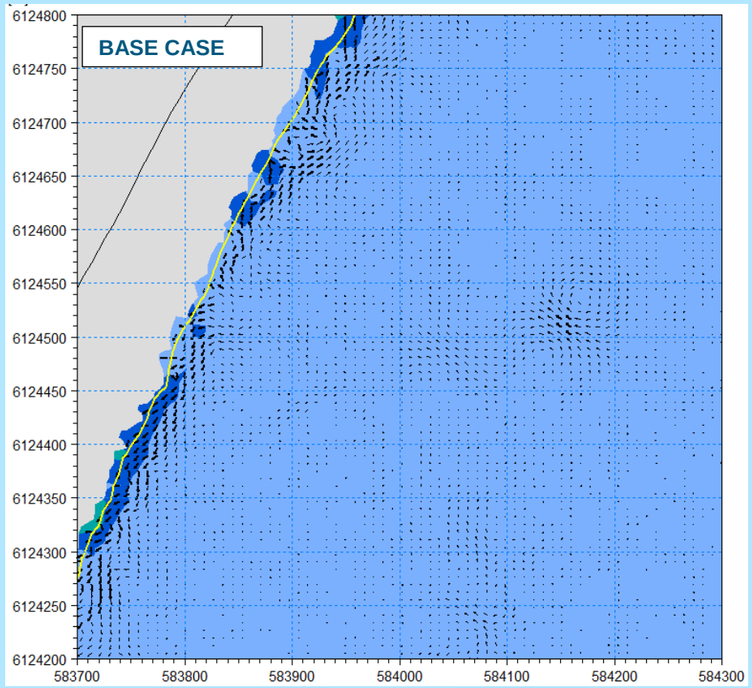


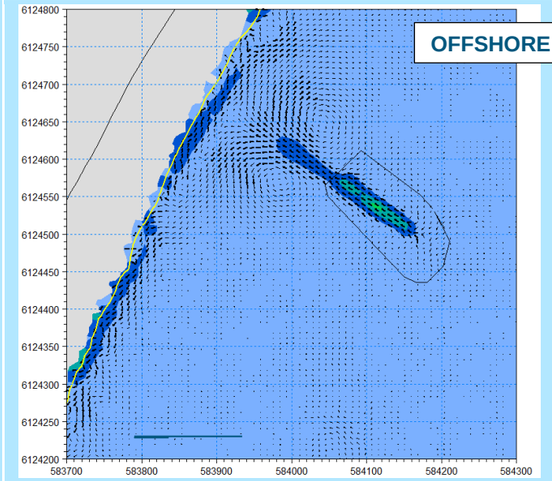


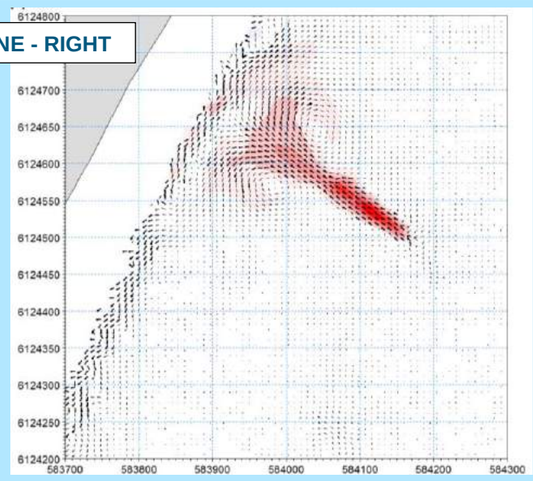


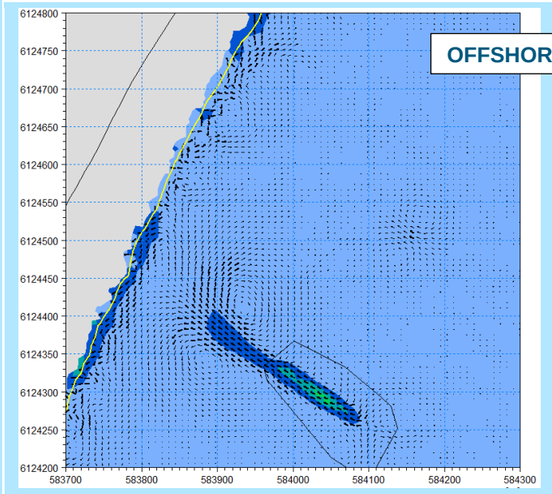


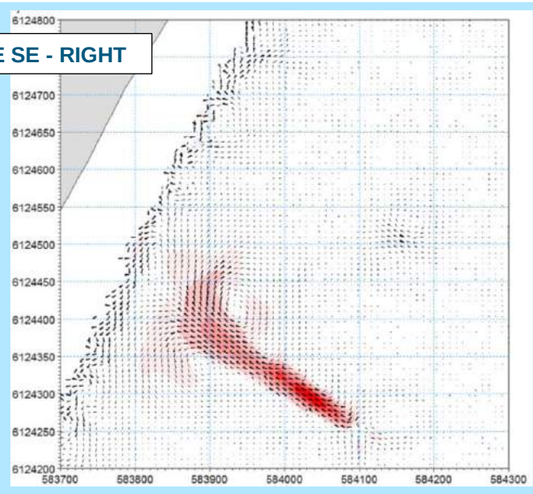
Average Conditions: 17/12/2015 13:30; Offshore: Hs = 0.6m, Tp = 13.4sec, Dp = 121°

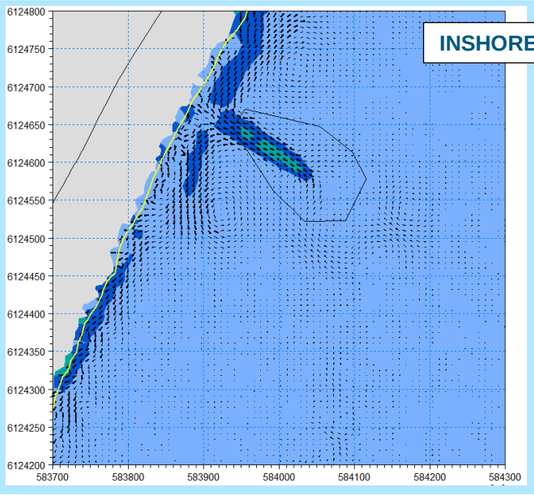


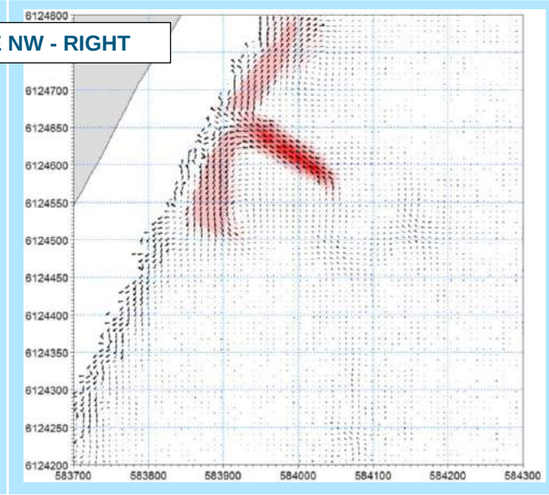


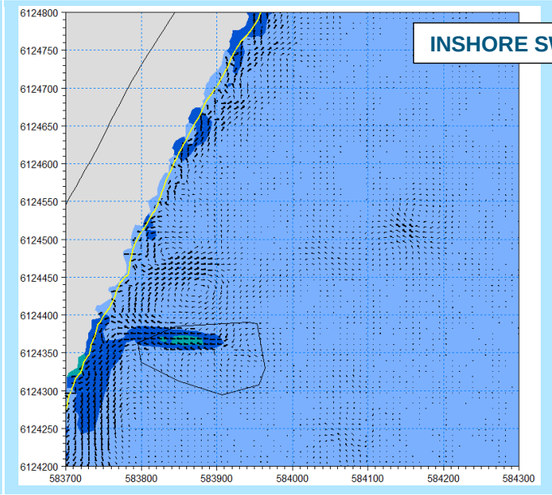


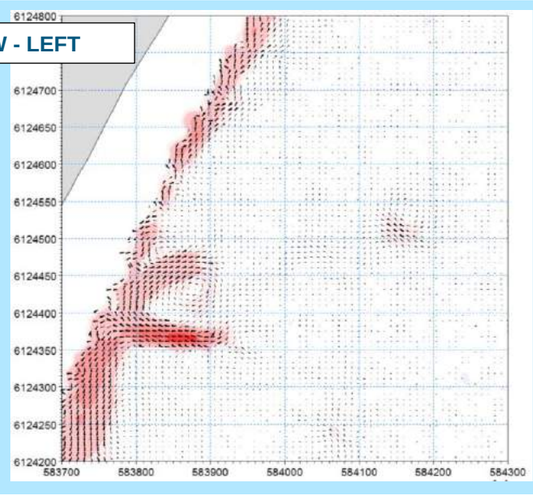




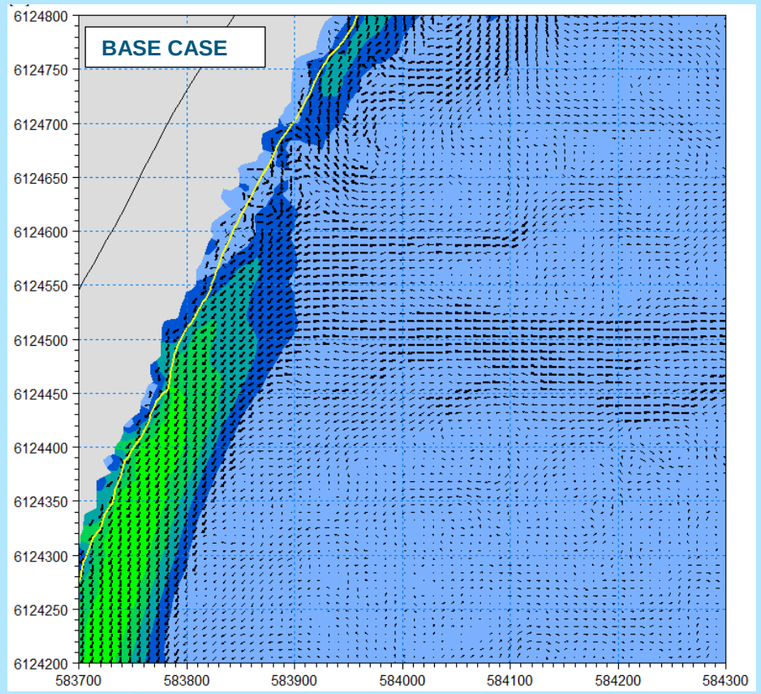


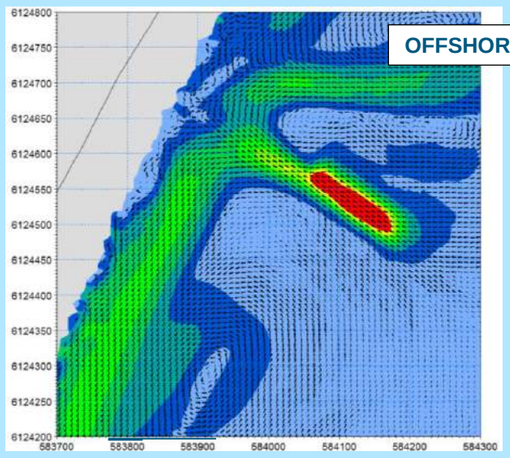


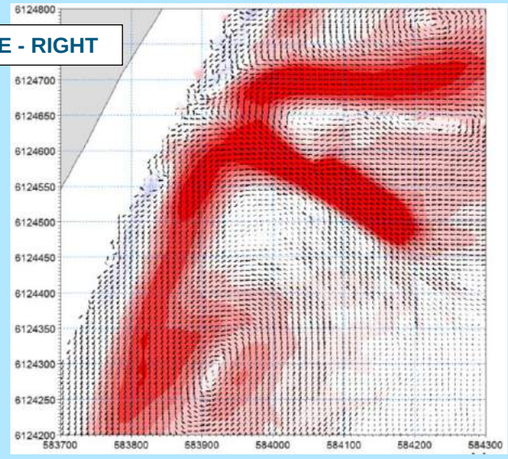


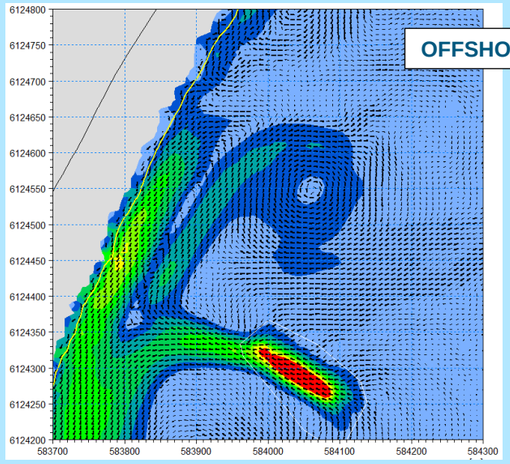


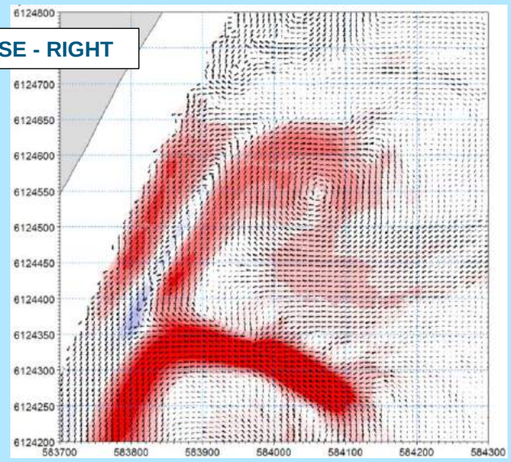
Extreme Event : 02/08/1984 22:30 Offshore: Hs = 2.7m, Tp = 11.5sec, 126°

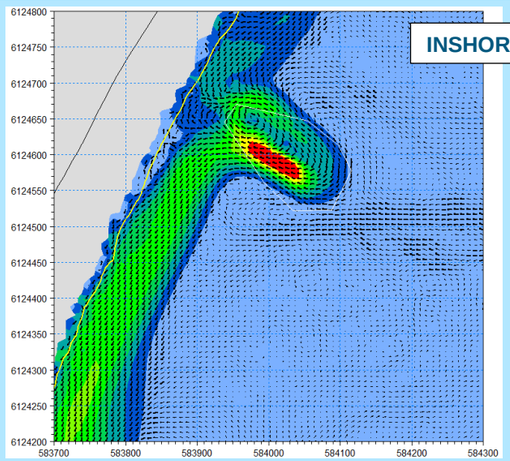


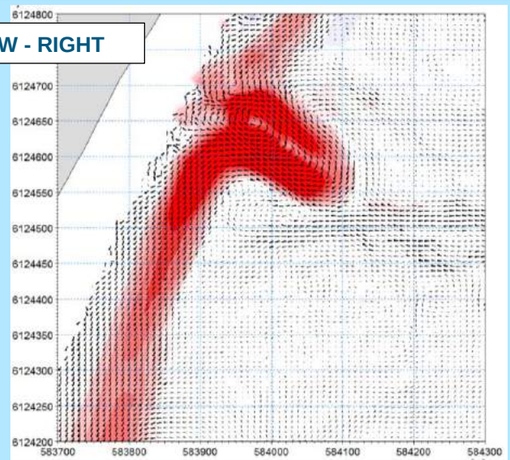


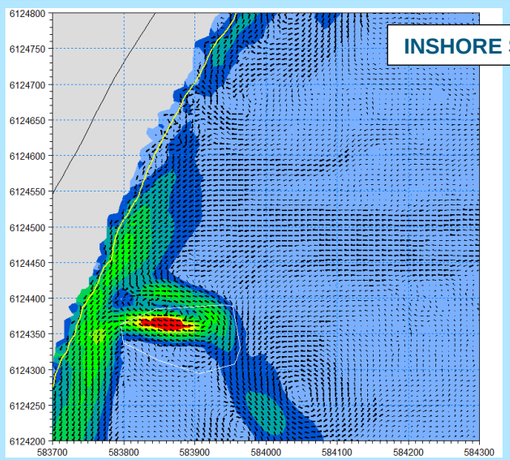


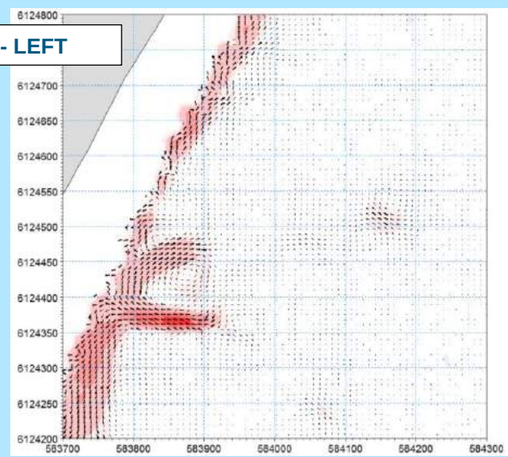












3 Discussion

The key finding from the literature review and the empirical formulation undertaking is that the morphological response in the lee of a submerged structure is directly linked to the proximity of the increased current flow over the structure to the shoreline. This accelerated jet is caused by gradients in wave radiation stress as waves encounter the structure. There are four main modes of shoreline response to this local hydrodynamic regime;

1. If the structure is too close to the shoreline, the accelerated jet diverges as it interacts with the shoreline promoting erosion.
2. If the structure is placed further offshore, the accelerated jet has room to dissipate and diverge in the lee of the structure, this divergence causes a counter-current which produces outward flow from the shoreline, promoting accretion.
3. If the structure is placed at the null point between these two zones, the shoreline will remain unchanged, under constant conditions.
4. If the structure is placed at some distance from shore, the influence of the current regime is negligible and the shoreline will remain unchanged.

These modes have been observed in the 2D hydrodynamic modelling in Section 2.5.2. One of the key questions to be answered through this exercise is the appropriate offshore distance of the reef. As can be seen from Ranasinghe's empirical relationship, wave height is directly related to the distance from shore of the null point in shoreline response; higher wave height means further offshore placement of the reef and vice versa. Therefore, the determination of the design wave height of the structure is one of the key design parameters that need to be determined prior to the detailed design phase. The results of the modelling of the three wave conditions showed the following;

Low Energy Wave Condition: The 2D hydrodynamic simulations showed generally very low current speeds (~0.2m/s) for all reef layouts. The two most northerly layouts (Inshore NW and Offshore NE) appear to cause no increase in current speed along the shoreline, indicating an accretionary mode (or at least no erosion) in the lee of these structures. The most nearshore layout (NW) appeared to be at its nearshore limit, i.e. if moved further shoreward the divergence from the accelerated jet would intersect the shoreline promoting erosion.

It can be seen in the base cases for the higher energy events that this northerly location appears to be at the divergence point of the natural longshore current, as seen in Figure 30. This is an important point to note as disruption of this current (or intensification) may have impacts on up/downdrift locations along Middleton Beach.

The placement of the AASR in this more northerly location tends to align with this general pattern. The accelerated current jet over the reef appears to flow directly into this natural divergence zone. The offshore location (NE) appears to create a moderate four cell circulation pattern, implying accretion in its lee.

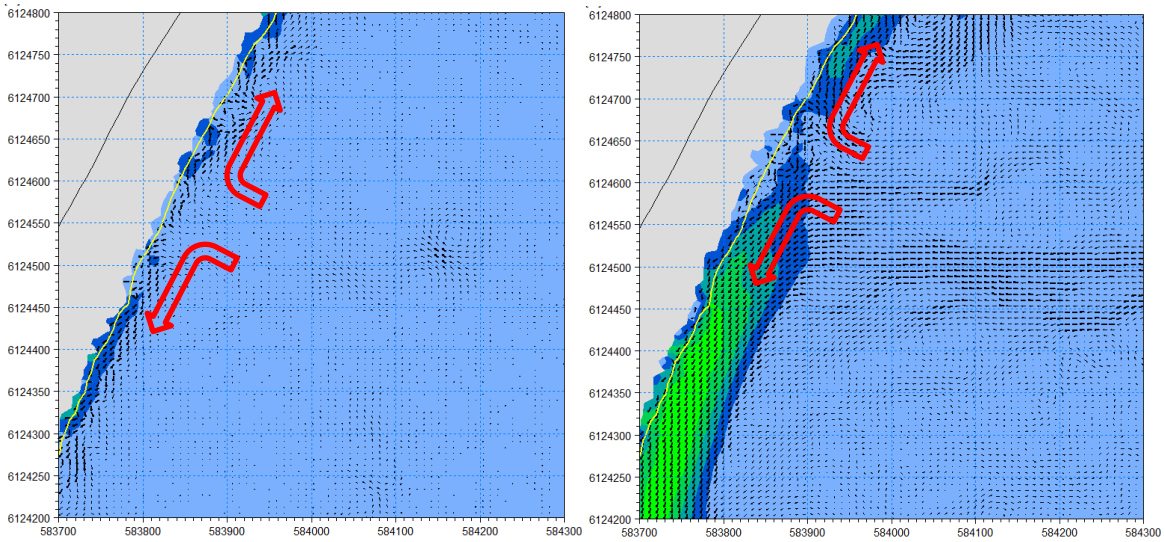


Figure 30 Divergence in the longshore current direction seen in the base case (no reef) simulations for the average wave conditions (left) and extreme wave conditions (right).

Moving the reef layouts to the south of this natural divergence pattern, (as seen in Offshore SE and Inshore SW layouts) appears to shift this divergence feature further to the south. The current difference plots for these layouts both appear to have significantly increased current speeds along the shoreline to their north. This disruption may result in localised erosion to the north of the structure and may even cause changes further up/downdrift. As with the Offshore NE layout, the Offshore SE layout appears to result in a mild four cell circulation pattern.

Further, it should also be noted that for the offshore layouts there appeared to be no increase in wave height along the shoreline (wave focussing) occurring in the lee of these structures (noting that the SW model does not sufficiently resolve 3D wave interactions). This leads us to believe that even for this benign wave condition, waves were breaking on each reef layout, or that wave focussing was contained to the structure footprint, creating a wave shadow (or reduction) in their lee. However, given the limitation of the SW model, this requires verification in later project stages.

The results would infer that designing to this low energy condition would mean that an offshore structure distance of 90m (to land ward toe) would result in a leeward current pattern that would be neutral to shoreline change (as per the findings of Section 2.4 for the Inshore NW layout. Further, detailed investigation is required to determine the up/downdrift impacts of moving the structure to the location of the southerly layouts (Offshore SE/ Inshore SW).

Average Condition: The average wave conditions had very much the same results as the low energy wave conditions and the same expected outcomes; both the offshore reef layouts (NE, SE) appeared to induce accretionary beach modes (4-cell circulation). The northerly inshore layout (NW) still appeared to be at the null point of shoreline response (although current speeds were much higher). Moving the layouts to the south (SE/SW) caused higher current speed difference to the shoreline north of these structures.

The results of the spectral wave modelling showed significant wave reduction in the lee of all reef layouts inferring that under average wave conditions, waves will break on the reef structures, significantly reducing the wave heights in the lee. However, as stated above this requires verification in later project stages.

Extreme Conditions: The simulated extreme wave event of 1984 showed very high current speeds along the shoreline in all scenarios; the base case shows current speeds in the order

of >1m/s directly on the shoreline (except in the divergence zone described above). These results (and photographic evidence from the actual 1984 event) show that the entirety of Middleton Beach underwent some sort of erosion during this event. The inclusion of a reef structure is expected to have little impact on coastal protection during such a large event. Interestingly enough, it can be seen that the northerly placed structures have little impact on the current speeds encountered along the shoreline. That is to say they, do not add to the magnitude of these currents (this is seen in the white region along the shoreline for the difference plots of the NW, NE simulations). The southerly placed structures (SE, SW) appear to induce a greater disruption to the current regime than that seen in the base case, in all cases increasing current speeds (regions of red in the difference plots).

Both offshore layouts (NE, SE) appear to either create a small four cell pattern or be at the null point between the 2 and 4 cell configurations. Both of the inshore cases (SW, NW) have leeward current patterns diverging directly into the shoreline.

The inflection point identified in Figure 30 is an important feature that warrants further study into the future. It is anticipated that constructing the reef with this feature in mind will reduce any possible impacts on the greater Middleton Beach. A worst-case scenario of severe erosion or accretion in the lee of the structure would, at this alongshore position, have negligible impact on both Ellen Cove and Emu Point.

Severe erosion at this location may reduce beach amenity or the erosive buffer available in the upper dune volume for larger storms. As there is very little infrastructure landward of this location, it is expected that this risk is quite low. Remediation options undertaken by the city may include; reduction of the crest height of the reef, placement of sand on the upper beach in the effected zone however it is anticipated that any shoreline response will be very subtle and transient.

1D Sediment Transport Modelling

The simplified sediment transport modelling undertaken herein gives an insight into the appropriate long and cross-shore placement of the structure. The results of the LITDRIFT model reiterate the location of the divergent current pattern in the vicinity of the MB02 transect. The 2015 littoral drift curves show that there is a net weak littoral drift to the north at that location. There are interim patterns of southward movement to Ellen Cove; these are more prevalent during the summer months when there is a higher frequency of northerly winds and higher standard deviation in the usually unidirectional waves.

The sediment transport equation used in the LITDRIFT engine is mainly dependant on the incoming wave angle to the alignment of the cross-shore profile. As seen in the long term wave statistics at the offshore extent of MB02, by the time waves have been transformed through King George Sound, they are fairly unidirectional having aligned with the uniform bathymetric contours offshore of MB02. Any deviation from this alignment drives a longshore transport in that direction. The slight net northerly transport at this location means that MB02 is located just south of the divergent pattern explained above, where it is expected there would be a net zero (or very small) transport.

Most of the transport along the MB02 profile can be seen to occur along the shoreline (or shallow areas of the bar) this is where wave radiation stresses are expected to be the highest. It can be seen that the introduction of the reef, greatly reduces the longshore transport along the profile, Figure 27 and Figure 28. This is predominantly due to the reduction in wave energy over the structure (causing a wave shadow in the lee, mentioned above). As such, wave energy is significantly dissipated before interacting with the shoreline, driving a smaller transport rate. It can also be seen that there is a small transport driven on the offshore toe of the structure. This is the assumed location where wave breaking is initiated. As LITDRIFT is unable to resolve the 3D shape of the reef structure and the porosity and impermeable nature of the rocks used for its construction, it is unable to determine the

amount of dissipation that would occur here, generally reducing any transport (at the edge of the structure) at this location to near zero.

Due to the generally ambient nature of the modelled year (2015) the LITPROF modelling did not show much cross-shore variation in profile. This was seen in the base case (no structure) sensitivity tests prior to the scenario testing. In general, there was little variation in offshore bar movement during this simulation and the final profile (31/12/2015) showed an accretive shoreline, this correlated well to the general accretionary pattern described for Middleton Beach in the conceptual model in (RHDVH, 2017).

In all the simulations tested, there appeared to be a widening of the shoreline. As with the LITDRIFT modelling, LITPROF was unable to resolve the 3D nature of the structure and there can be seen scour at the offshore toe of the layouts tested. The engine is unable to resolve localised reduction in current magnitude at the toe due to the permeability of the rock structure and generally gentle slope of the toe. A key finding of the LITPROF modelling is the generally benign, accretionary cross-shore sediment transport regime at the MB02 location.

4 Conclusion and Recommendations

Although a simplified approach, the modelling undertaken herein is a warranted initial step for the prediction of shoreline response from the introduction of the AASR along Middleton Beach. The results have shown that an artificial surfing reef can be constructed within the approvals footprint given to the EPA whilst not having a significant impact on coastal processes at Middleton Beach. The initial modelling provides confidence that the key project objective of limiting any impacts on Middleton Beach can be maintained through the detailed design stage.

The modelling and empirical review undertaken herein has shown that the minimum offshore placement distance of the AASR for a null shoreline response is approximately 205m for a 1yr ARI wave event from the landward extent of the structure or 140m under ambient wave conditions.

A detailed literature review of constructed artificial reefs was undertaken focussing on projects with similar characteristics to that of the AASR. Cables Station Reef in Perth, constructed in mid-1999 was found to be the most comparable in terms of geographic location and metocean conditions. However, local geomorphology (limestone reef) means that the empirical formulations that are most applicable to the assessment of the AASR are not directly applicable to Cables Station Reef. These empirical formulations are relevant for sandy beaches such as that of Middleton Beach.

Wave and current modelling was undertaken for a range of reef configurations and offshore positions under a variety of event-based wave conditions. The modelling showed, similarly to the empirical calculations, that the AASR layouts either produced a 2 or 4-cell current circulation pattern in their lee. In general, the more inshore reef placements (approx. 120m from shoreline) showed circulation patterns that would be expected to promote erosion, even under average conditions. However, the more offshore reef configurations (approx. 280m from shoreline) showed local circulation patterns which would be expected to produce no significant impact on the shoreline.

A phase-averaging, linear wave theory approach was adopted for the wave and current modelling, which has been previously shown to have some limitation in fully representing the wave transformation over the structure. A more sophisticated, non-linear, phase-resolving wave model is recommended for later project stages. However, the results attained through this present undertaking have given qualitative results that can be interpreted for comparative purposes.

One dimensional longshore and cross-shore sediment transport modelling was undertaken of an idealised beach profile at the Surfers location at Middleton Beach, both with and without the AASR in place at the offshore location for a representative year. The introduction of the reef was seen to dissipate the current along the shoreline (and sediment transport) in this location. The results of the one dimensional cross-shore modelling showed a general accretionary trend with sediment being deposited in the lee of the structure, appearing to be transported from the offshore toe.

Although a simplistic modelling approach, the results generally followed the findings of the recent Emu Point to Middleton Beach Coastal Adaptation Study (RHDHV, 2017); longshore sediment transport at the site was generally very low with a net annual longshore transport in the order of 10,000m^3 to the north (with the majority of this transport occurring along the shoreline) and that this section of Middleton Beach shows a long-term trend of accretion.

An interesting outcome of the current pattern modelling was that the location chosen for the AASR, was in the vicinity of what appeared to be an inflection point along Middleton Beach. At this location, northerly travelling currents from Ellen Cove converged with the Southerly directed currents from Emu Point as they changed direction and headed offshore. Although these currents are generally very low

in magnitude, it is an important feature to note when continuing with the later design phases especially when considering alongshore placement of the structure and possible disruption of coastal processes.

Following the modelling undertaken herein, the original recommended design (option B) remains the recommended design option to take forward. The longshore and cross-shore placement of this structure is seen to have the least impact to coastal processes. However, the depth, location, shape, orientation and structure volume are still design parameters that require further investigate and optimisation in order to meet the overall project objectives such as budget, surf amenity and the recommendation of the ASG.

The work has also introduced important design considerations to be further explored in the detailed design phase, such as;

- Design wave conditions - is it more appropriate to design to a 1 year ARI scenario, an operational wave height or a larger design wave height. This question will aid to prioritise project objectives of usability, safety, structural stability, maintenance and overall amount of surfable days. This study showed the dependency on offshore distance to this parameter.
- Crest height - this is also related to the design wave conditions above but also touches on more complex design issues of surfability and structural stability as well as safety.
- Morphological response - to what extent is intermittent morphological change in the lee of the structure acceptable? Is there a tipping point between coastal protection objectives, surf amenity and budget?

These questions should be considered holistic moving through the approvals and detailed design phases of this project as they will help to define and finalise the overall key project objectives. It is recommended that once these key design parameters and project objectives are finalised, a more detailed numerical modelling investigation be undertaken. This should be undertaken in conjunction with a sufficiently robust physical modelling program to ensure uncertainty is reduced prior to construction phase of the project.

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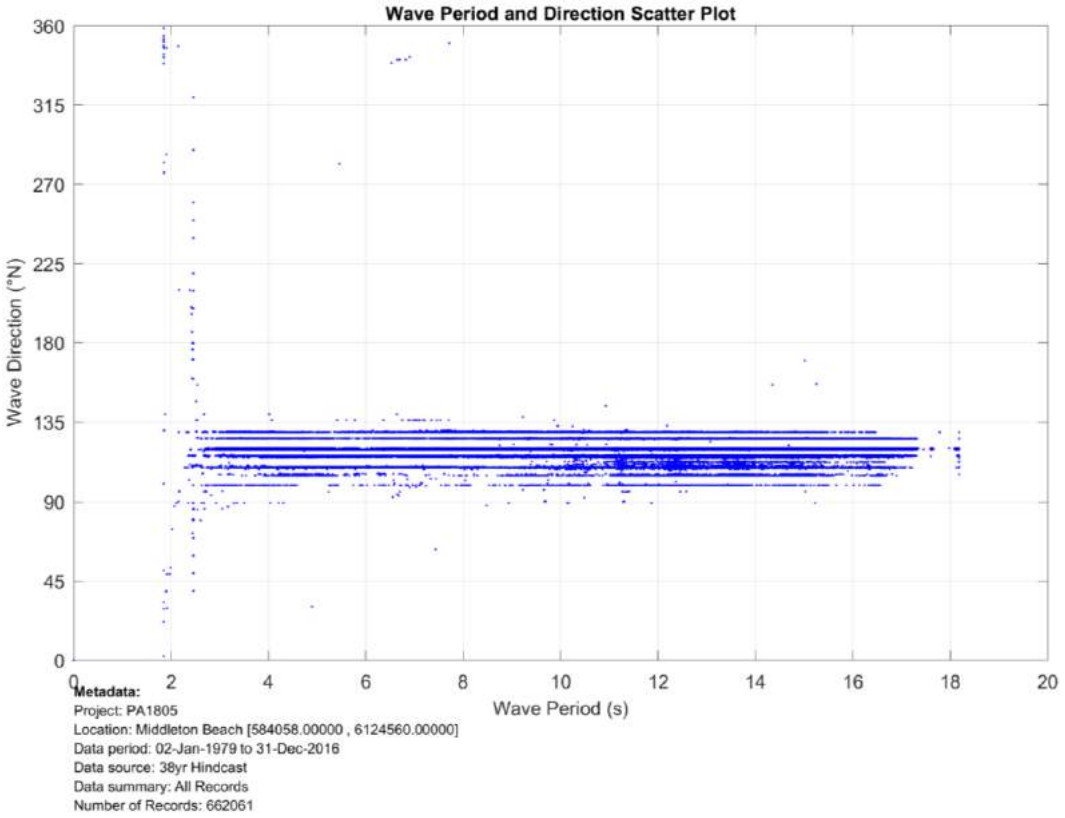
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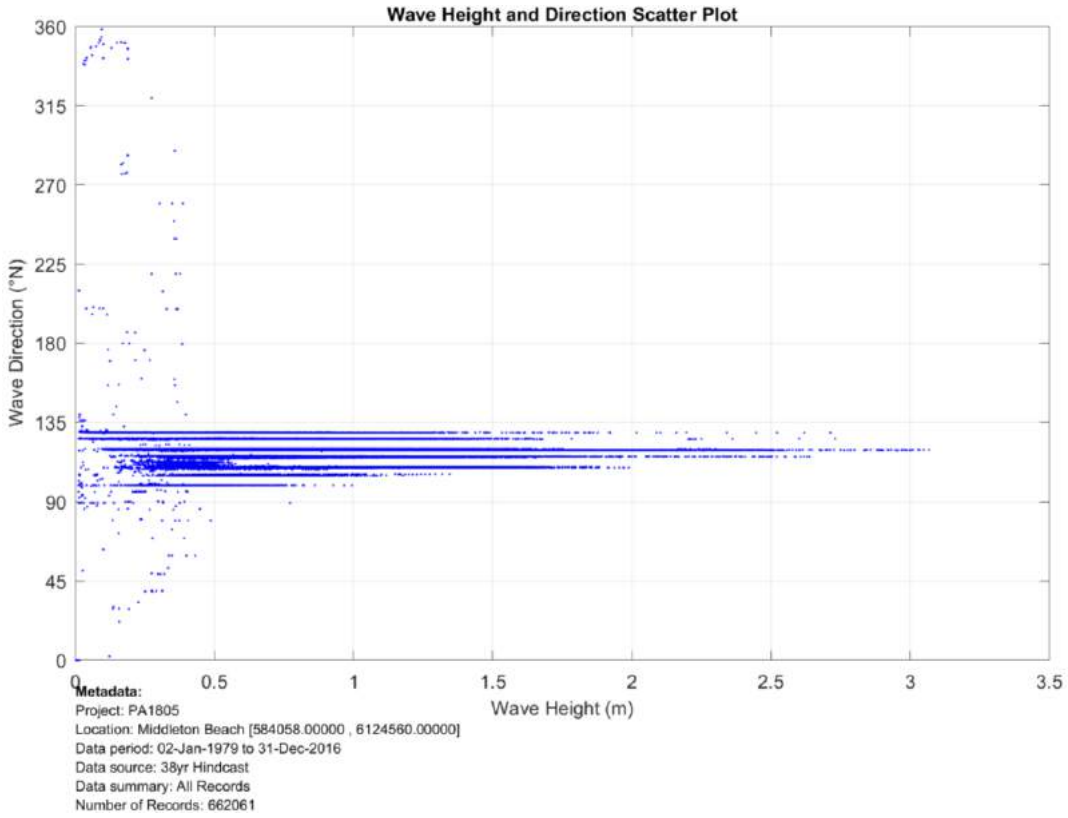
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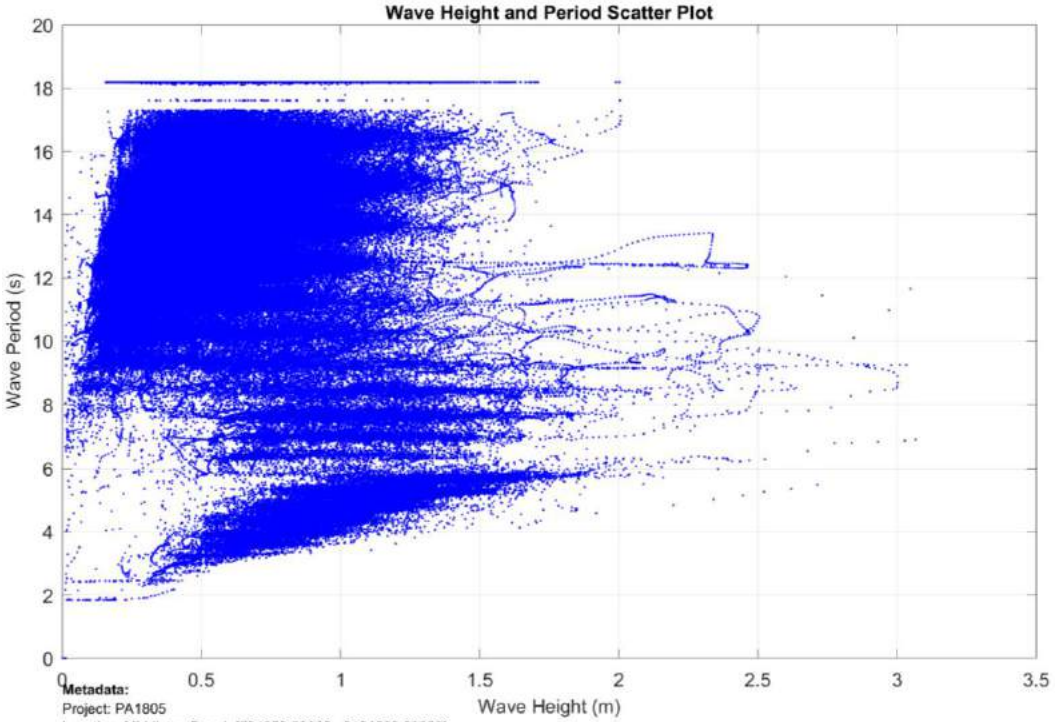
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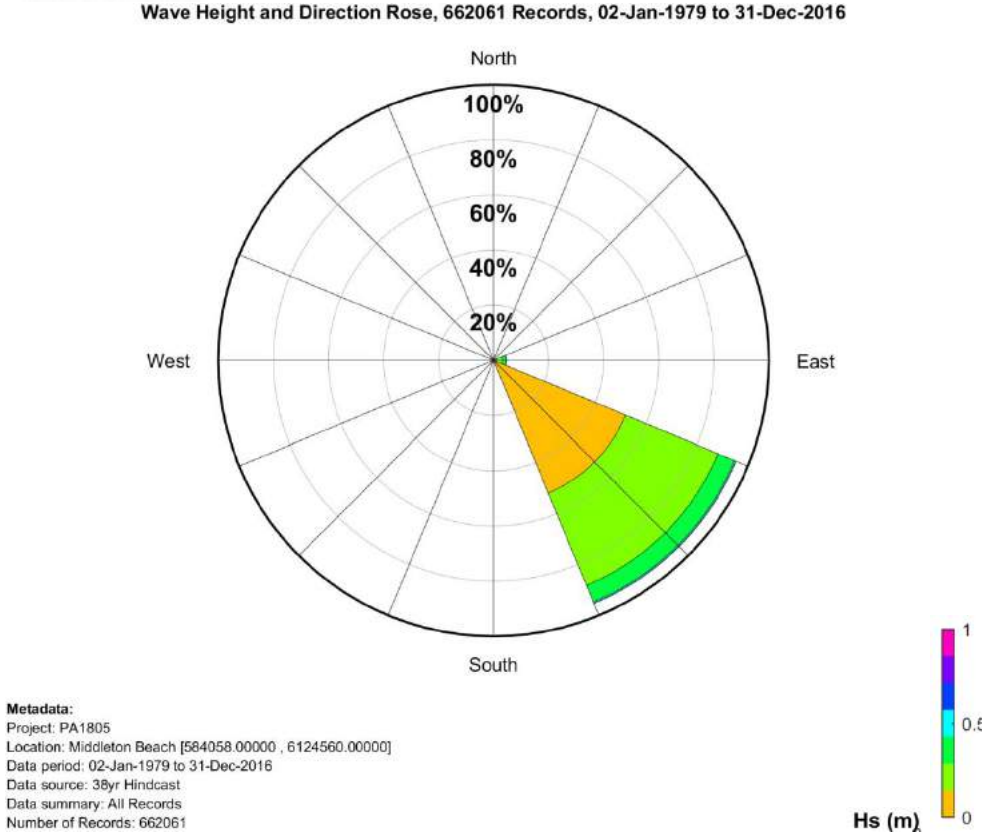
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Appendix A – Joint frequency Analysis of RHDHV AWAC Site from 38 year hindcast

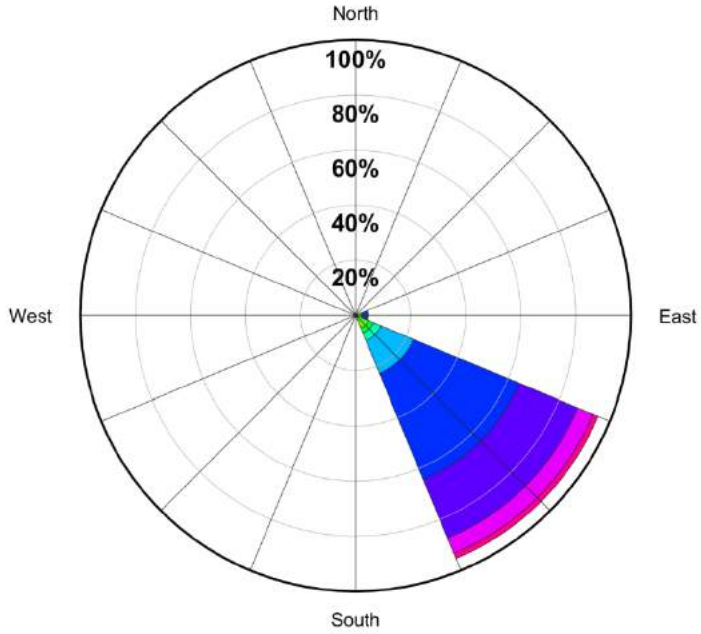


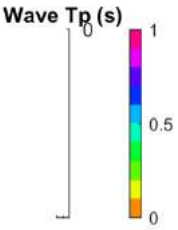



Wave Height and Period Scatter Plot
Wave Period (s)
Wave Height (m)
0
2
4
6
8
10
12
14
16
18
20
0
0.5
1
1.5
2
2.5
3
3.5
Metadata:
Project: PA1805
Location: Middleton Beach [584058.00000 , 6124560.00000]
Data period: 02-Jan-1979 to 31-Dec-2016
Data source: 38yr Hindcast
Data summary: All Records
Number of Records: 662061


Wave Height and Direction Rose, 662061 Records, 02-Jan-1979 to 31-Dec-2016
North
West
East
South
100%
80%
60%
40%
20%
Hs (m)
1
0
Metadata:
Project: PA1805
Location: Middleton Beach [584058.00000 , 6124560.00000]
Data period: 02-Jan-1979 to 31-Dec-2016
Data source: 38yr Hindcast
Data summary: All Records
Number of Records: 662061

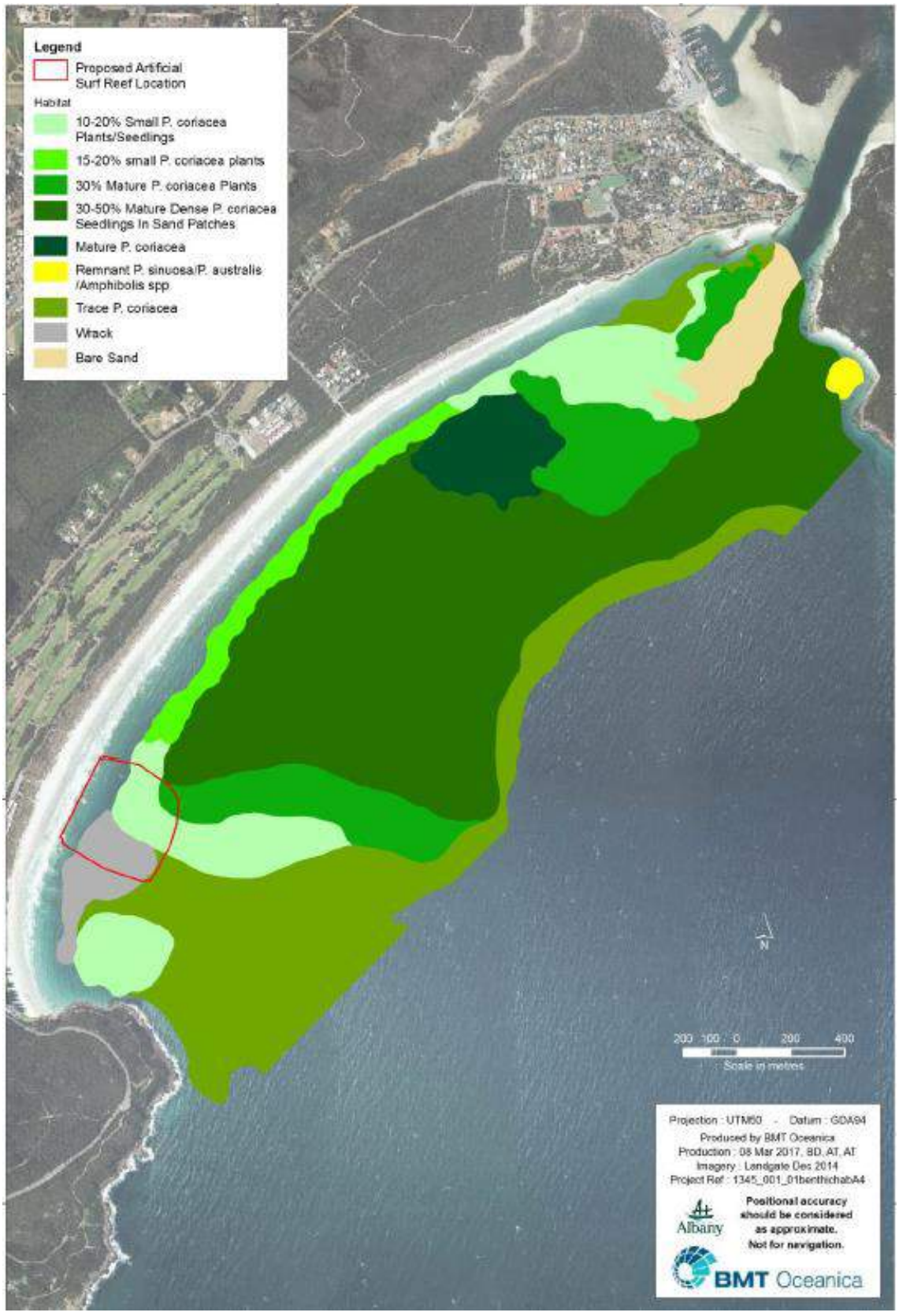
Wave Period and Direction Rose, 662061 Records, 02-Jan-1979 to 31-Dec-2016





Metadata:
Project: PA1805
Location: Middleton Beach [584058.00000 , 6124560.00000]
Data period: 02-Jan-1979 to 31-Dec-2016
Data source: 38yr Hindcast
Data summary: All Records
Number of Records: 662061

Appendix B – benthic fauna habitats (BMT, 2017)



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